INTERVIEW-PREP PACKET 1 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS · V2 — SEVEN FLEXPORT PRIMARIES

Amazon NACF Supply Chain loop — Packet 1: Snigdha Tripathy

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Snigdha Tripathy

Senior Manager, Supply Chain — NACF Supply Chain · HIRING MANAGER

Wednesday, October 29, 2026 · 10:00 am PDT · 55-minute video interview

Strategy in one line. The hiring manager judges fit for the actual work — and her public profile (EU inventory-placement strategy across the crossdock network; readiness for 50+ FC launches; a Kaizen / Lean trainer; teams built from scratch — public-profile inference in "Read me first") says she will test whether the mechanisms transfer to her world. Open with S11 (a boarding process behind published gates, run from a weekly input/output review) and hold S12 (a committed roadmap re-sequenced on cost-to-serve evidence). v2 adds five distinct Flexport primaries — S11-A the Monday review as an analysis, S11-B launch readiness as standard work, S11-C retiring an acquired product's book, S12-A the Asia latency investigation, S16-A the data model under the review — so every Leadership Principle has a primary and a backup from one company. Lead every answer with the mechanism left behind, the priced trade-off, and the team that grew to own it. Systems are enablers; the process redesign and its business results are the story.

LP focus — my guess, not confirmed: Deliver Results · Hire & Develop / Earth's Best Employer · Dive Deep. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and her public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Think Big / Frugality and Ownership / Earn Trust remain live alternates carried by the twist maps.

Also expect: "tell me about yourself" (75-second version, once, here) and "why Amazon" (hiring managers usually ask it); one functional question on how I would approach the NACF process space.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

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Every new story is tagged NEW and carries five labeled facets (data analysis · solution design · math & data logic · implementation · probe questions), four `Before / Fix / After / Dollars` tables, a four-question behavioral pack, three angle cards and three trap drills. Each primary opens with a SAFE-SAY box (the sentence for its most attackable number and for "what is your source for that?"). Yellow VERIFY-WITH-RAJESH tags mark only the specifics that are in none of the source files; each is repeated on the VERIFY page with the safe sentence to say instead.

5-Minute Cram Sheet — Snigdha Tripathy · Wed Oct 29 · 10:00 am PDT NEW

HER HOOKS → MY PROOF (PUBLIC-PROFILE INFERENCE)

- **EU IXD placement strategy** → S11 wave machine: a selection bar published before the first node, a ~50-container calibration per node, a public hold when a node misses; S11-C: three brokers = 75.0% of containers, so the few lanes that carry the volume get bespoke handling.
- **$80MM savings** (her number — framing only) → my discipline on dollars: ~$5M avoidance said once (S12); $0.66 → $0.03; anything priced is "priced, not saved" (S16-A).
- **50+ FC launches** → S11-B readiness: four gates, 100 containers × 7 checks, six graduation criteria, and the pilot that ran later than the first testing plan, owned.
- **Kaizen / Lean trainer** → S11-A waste = 3.67 touches and standard work; S11-B jidoka (stop the line, hold wave 3); S12-A genchi genbutsu and PDCA; S16-A poka-yoke (bridge table, parity gate).

THREE OPENING SENTENCES

- *Most significant delivery.* "Cost-to-serve on Flexport's shipment-visibility product went from sixty-six cents a container to three while customers went from seven to sixty-eight — nine point seven times — with zero rollbacks, because boarding went through six published gates, not around them."
- *Changed a committed plan.* "I re-opened my manager's eighteen-month roadmap privately, with numbers — partners first, operators on the same chassis — and value landed in month six with sixty-six percent less effort."
- *Dive into data.* "A twenty-two-fold cost drop decomposes into two levers — containers needing a human, and touches on each — and which one moved decides what you do next."

FIVE NUMBERS I MUST NOT GET WRONG

- **7 → 68 = 9.7×** — never rounded up; no other growth multiple is ever said.
- **$0.66 → $0.03, about 95%**; direct dollars ≈ $50K a year *modeled*; the ~$1.5M pre-production estimate is a user anchor, separate, never added.
- **~$5M** is avoidance, said once — never annual; run-rate ≈ $1.5M a year is a separate modeled number.
- **S11-C:** printed $185,892 = six of nine segments; nine rows = $254,952; lead with containers (294,581).
- **S12-A:** never quote 81%; like-for-like is −44.2% overall (Apr → Sep). **S16-A:** $103,680 and $100,360 are prices, not savings.

SEVEN STORIES — TELLING ORDER AND THE ONE NUMBER

1	**S11**	**7 → 68 customers (9.7×); $0.66 → $0.03**	Deliver Results · Ownership
2	**S11-B**	**100 containers × 7 checks; 41 of 59 failures at two hops**	Highest Standards · Customer Obsession
3	**S12**	**month 18 → 6; −66% effort; ~$5M avoidance (record), said once**	Think Big · Backbone · Frugality
4	**S11-A**	**22× = 6.0× (coverage) × 3.67× (touches)**	Dive Deep · Earn Trust
5	**S12-A**	**Asia 21.19 s vs US 3.3 s = 6.42× on 45.3% of views**	Dive Deep · Customer · Bias for Action
6	**S11-C**	**79 of 79 customers placed; 3 brokers = 75.0% of containers**	Customer Obsession · Broad Responsibility
7	**S16-A**	**system accuracy 60% → 90%; over $100K priced, not saved**	Invent & Simplify · Frugality

BRIDGE IN ONE BREATH (BOOKLET FRAMEWORKS)

- Inbound capacity: dock doors × hours × unload rate; utilization knee ≈ 85%; labor by UPH; cubic fill.
- Inventory health: weeks of cover — under 2 expedite, 2–4 healthy, over 6 excess; safety stock z · σ · √LT.
- Placement: distance-to-customer priced against cost per unit moved — the edge-near-demand logic of S12-A, the concentration logic of S11-C.

ROOM RULES

- Pitch, land the metric, **stop**. Name who owned what — program-lead voice.
- Every dollar with its basis word; if a number is yellow in the text, check the last page first.
- Overlaps: S16-A, S11-C, S12-A, S11-B facts are also in other packets — see "Read me first"; fall back to the router's backup if the panel flags a clash.
- Bridge in one sentence, then hand it back to her with a question.
- If challenged on a number, read the story's SAFE-SAY box; if it is yellow, read its safe sentence on the VERIFY page — never the unconfirmed specific.

Question → Story Router — 16 Leadership Principles + 11 common prompts NEW

Each Leadership Principle row quotes a **real question from the Amazon question bank (AIQB reference guide)**, then routes it: find the LP, tell the primary (the B-card is the 90-second STAR in that story's Behavioral Pack), keep the backup for "another example". One number, then stop. Never the same story twice in one loop.

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Customer Obsession *"Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?"* AIQB: Customer Obsession Q8	S11-C · B1 — nine segments, 79 destinations	S11-B · B2 (shadow accounts) · S11 · B3	"Every one of seventy-nine customers got a destination and a date — not an end-of-life email."	**79 of 79 = 55 + 3 + 9 + 9 + 3**
Ownership *"Give me an example of an initiative you undertook because you saw that it could benefit the whole company or your customers, but wasn't within any group's individual responsibility so nothing was being done."* AIQB: Ownership Q3	S11-A · failure card — the review nobody owned, run by hand for months	S11 · B2 (the critique accepted)	"Nobody owned the measurement system, so I ran it by hand every Monday — about four hours a week at its worst — until the formulas were stable enough to automate."	**40.0% → 21.9% week, explained**
Invent and Simplify *"Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?"* AIQB: Invent and Simplify Q1	S16-A · B1 — four systems, one pipeline	S11-A · conflict card (seven metrics)	"Four systems and one person made the Monday number; I simplified it to one pipeline and seven frozen definitions."	**system accuracy 60% → 90%**
Are Right, A Lot *"Tell me about a time when you didn't have enough data to make the right decision. What did you do? What path did you take? Did the decision turn out to be the correct one?"* AIQB: Are Right, A Lot Q1	S11-A · B3 — 99.4% on three touches	S11-A · B4 (the carrier-format-change week) · S12 · failure card (Asia 21.19 s the plan did not see)	"One week printed ninety-nine point four percent — on three touches. I did not have enough data to steer on it, so I printed it as a data miss."	**N = 3 → discarded**
Learn and Be Curious *"We all have things about ourselves we'd like to improve on at work. Give me an example of something that you've worked on to improve your overall work effectiveness."* AIQB: Learn and Be Curious Q4	S11-B · B4 — the honest miss	S12-A · failure card (no regional P50 / P90)	"The pilot ran later than the first testing plan; my test estimates were optimistic and monitoring arrived late, so I size testing on hop count now."	**41 of 59 failing rows at two hops (my recount)**
Hire and Develop the Best *"Describe a time when you constructed a team to accomplish a goal. What factors did you consider in constructing the team? Did you factor diversity into your team construction?"* AIQB: Hire and Develop the Best Q9	S11 · the input owners across nine teams who sign their metric — a team-level answer; for a named person see VERIFY-WITH-RAJESH · V1	S11-A · conflict card (owners sign what they are red against)	"I grew the input owners across nine teams into people who sign and defend their own definitions; the review runs on them, not on me."	**9 teams · 7 definitions**
Insist on the Highest Standards *"Tell me about a time when you had to make a decision between standards and delivery. What tradeoffs did you have make? What was the outcome?"* AIQB: Highest Standards Q4	S11-B · B1 — UAT opens at 80%	S16-A · B4 (1% parity gate) · S11-A · B2	"UAT did not start until eighty percent of shipments showed minor-to-no issues — the gate was data quality, not the calendar."	**100 × 7 checks · 80% gate**

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Think Big *"Give me an example of how you have changed the direction or view of a specific function/department and helped them embrace a new way of thinking. Why was a change needed?"* AIQB: Think Big Q2	S12 · B1 — month 6 not 18	S11 · B1 (7 → 68 on the same ops team)	"The roadmap said eighteen months to first value; I re-opened it and landed value in month six, with sixty-six percent less effort."	**18 → 6 months; −66%**
Bias for Action *"Give me an example of a calculated risk that you have taken where speed was critical. What was the situation and how did you handle it?"* AIQB: Bias for Action Q1	S12-A · B3 — reversible changes first	S12 · reversal trigger written before the swing	"We started with changes that could be undone — a double CDN removed, a firewall on — and held the expensive one for the metrics."	**52-day plan; 4 of 8 rows done**
Frugality *"Give me an example of how you have helped save costs or eliminate waste within your role or organization. What was the situation? What was the impact?"* AIQB: Frugality Q1	S12 · B3 — ~$5M avoided	S16-A · B3 (priced, not saved)	"About five million dollars of cost avoidance — engineering never spent plus value pulled forward — and I say it once."	**~$5M avoidance (record), said once**
Earn Trust *"Give me an example of a tough or critical piece of feedback you received. What was it and what did you do about it?"* AIQB: Earn Trust Q2	S11 · B2 — the critique accepted	S11-A · B3 · S12 · B4 · S11-C · B4	"Leadership said the program wasn't showing value; I'd been reporting activity, not evidence — and I accepted that first."	**#3 → #14 of 16 → rollout unlocked**
Dive Deep *"Tell me about a problem you had to solve that required in-depth thought and analysis. How did you know you were focusing on the right things? What was the outcome?"* AIQB: Dive Deep Q3	S11-A · B1 — 22× = 6.0× × 3.67×	S12-A · B1 · S16-A · B2 · S11-B · B3	"A twenty-two-fold cost drop — and I can say which lever did it: six times fewer containers needing a human, times 3.7 fewer touches."	**22× = 6.0× × 3.67×**
Have Backbone; Disagree and Commit *"Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business. What was it and how did you handle it?"* AIQB: Have a Backbone Q1	S12 · B2 — private first, two days to break it	S11-C · B3 (the dissent on the page)	"The plan had my manager's name on it; I read it to him alone, with numbers, and gave engineering two days to break it."	**2 points argued, 1 adjusted**
Deliver Results *"Tell me about a time when you not only met a goal but considerably exceeded expectations. How were you able to do it? What challenges did you have to overcome?"* AIQB: Deliver Results Q3	S11 · B1 — both targets early	S12 (value in month 6)	"Both public targets — 68 customers and 170 monthly users — landed two months early with zero rollbacks, behind six published gates, not around them."	**7 → 68 (9.7×)**
Strive to be Earth's Best Employer *"Describe a time when you implemented a change at work aimed at improving the work experience of yourself or coworkers. What was the goal?"* AIQB: Strive to Be Earth's Best Employer Q3	S16-A · B1 — retired the Monday stitching: one engineer maintains the pipeline instead of one PM rebuilding it	S12 · B4 (operators' two quarters, priced and published)	"Every Monday I stitched four systems by hand and the hours came out of rollout time; I turned it into a pipeline so the review no longer depends on one person."	**system accuracy 60% → 90%**

LEADERSHIP PRINCIPLE · AIQB QUESTION	PRIMARY STORY · CARD	BACKUP	15-SECOND OPENER	ONE NUMBER
Success and Scale Bring Broad Responsibility *"Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization."* AIQB: Broad Responsibility Q5	S11-C · B2 — concentration	S12-A · B2 (Asia = 45.3% of views)	"Three customers were four percent of the book and three-quarters of the containers — a template would have been wrong for them."	**3.8% of customers = 75.0% of containers**

ELEVEN COMMON NON-LP PROMPTS

PROMPT	PRIMARY	BACKUP	NUMBER
Tell me about a time you failed	S11-B · B4 (the pilot ran later than the first testing plan)	S12-A failure card	**later than the first testing plan**
Disagreed with your manager	S12 · B2	S11-C · B3	**2 points argued, 1 adjusted**
Influenced without authority	S11 · conflict card (Sales held to six published criteria)	S11-B · conflict card (nine teams, one seam)	**6 criteria · wave 3 held**
Reduced cost	S12 · B3	S11 (CTS) · S16-A · B3	**~$5M avoidance once · $0.66 → $0.03 (both record)**
Handled ambiguity	S11-C (nine segments)	S11 · B2	**79 customers, 9 segments**
Simplified a process	S16-A · B1	S11-A conflict card	**4 systems → 1**
Decided with incomplete data	S12-A (P95 on 316 tail views)	S11-A (N = 3)	**P99 = 63 views vs P95 = 316**
Missed a deadline	S11-B · B4	S16-A (the two-week security-review miss — one sentence only)	**later than the first testing plan**
Mentored someone	S11 — owner growth (team-level) VERIFY-WITH-RAJESH · V1	S11-A	**owners across 9 teams signing definitions**
Delivered under a tight timeline	S12-A · B3	S11 · B1 (both targets two months early)	**52-day plan**
Changed your mind because of data	S12 (adjusted 1 of 2 points)	S11-B (re-sized testing on hop count)	**1 of 2 points**

SAME STORY ASKED TWICE — SWITCH TO A DIFFERENT ANGLE CARD (LP · HEADLINE METRIC)

STORY	DATA / ANALYTICAL CARD	CONFLICT / BACKBONE CARD	HONEST-FAILURE CARD
S11	**Dive Deep** — $0.66 / $0.18 = 3.67 → 0.17 touches	**Have Backbone; Disagree and Commit** — six criteria; wave 3 held	**Learn and Be Curious** — #3 → #14 of 16
S11-A	**Dive Deep** — 22.0× = 6.0× coverage × 3.67× touch efficiency; fair split 56% / 44% (modeled)	**Insist on the Highest Standards** — seven frozen metrics, each signed by the owner who would be red against it	**Ownership** — by hand for months, about four hours a week at its worst; no longer a single point of failure once the pipeline matched within 1%
S11-B	**Dive Deep** — 41 of 59 (69.5%) at route + milestone; 3 of 105 all-seven-clean	**Ownership** — nine teams aligned on one seam; time to localize cut about 50% (secondary source)	**Learn and Be Curious** — first plan early November → launched December 2021; hours to localize a bug at six-plus layers
S11-C	**Dive Deep** — 3.8% of customers = 75.0% of containers = 54.7% of ARR	**Have Backbone; Disagree and Commit** — four dissenting inputs; one decider; each impacted party commits	**Learn and Be Curious** — 2 open questions; break-even $15,491 / month
S12	**Dive Deep** — 339.5 min × $0.45 ≈ $153; −25% vs −50%	**Have Backbone; Disagree and Commit** — two points argued; one adjusted	**Are Right, A Lot** — 21.19 s vs 3.3 s
S12-A	**Dive Deep** — 6.42× on 6.31k of 13.92k views	**Have Backbone; Disagree and Commit** — one conditional row (Q3 or later); 4 of 8 rows done	**Learn and Be Curious** — window-mismatched 81.1% — not quoted
S16-A	**Dive Deep** — 12 / 18 = 0.667 vs 12 / 10 = 1.20 (illustrative); 4% drop (record)	**Have Backbone; Disagree and Commit** — seven frozen metrics; one bridge table	**Earn Trust** — two-week security-review miss; zero Monday reviews skipped

B-cards: each story's Behavioral Pack has four separate 90-second STARs (B1–B4), each headed by the real AIQB question it answers; the three angle cards (data · conflict · honest failure) give a different opener and metric when the same story is asked twice.

Read me first

The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

Inference from the title. The hiring manager owns the team this role joins and the outcomes it is measured on. A Senior Manager of Supply Chain in NACF is accountable for network-level processes — how inventory is planned, placed, moved and recovered across fulfillment centers — and for the operating cadence (weekly business reviews, goals, mechanisms) that keeps those processes honest. She is the one interviewer who will also judge *fit for the actual work*: can this person define and run a process across teams that do not report to them, make value legible to leadership, and change a committed plan on evidence without breaking trust.

What that means for the answers. Lead with scale-with-mechanisms (S11) and roadmap judgment under unit economics (S12); v2 adds five distinct Flexport primaries (S11-A, S11-B, S11-C, S12-A, S16-A) so every Leadership Principle has a primary and a backup from one company — see the Router. All seven are Flexport stories, told as process redesigns: a boarding process behind published gates with a weekly input/output review, a launch roadmap re-sequenced on cost-to-serve, and the analysis, readiness, retirement, latency and data-model work underneath them. She will probably also ask the openers — "tell me about yourself" and "why Amazon" — and a functional question on how I would approach the NACF process space in the first 90 days. The JD's own phrase, "designing or modeling the supply chain as opposed to working in it," is the lens to speak to: I designed the boarding process and the review; I did not run the ops desk.

What her public profile adds (public-profile inference, labeled)

The profile, as pasted (public-profile inference — directional, not verified biography). 11+ years at Amazon across operations, launch and learning: EU IXD & Placement Operations Manager, Luxembourg, 2017–21 — defined the EU inventory-placement strategy for the crossdock network (distance-to-customer, speed-vs-cost) and built the team ground-up (+$50MM savings in 2020); Senior Manager EU IXD, 2021–22 — ran the EU crossdock network with a team of six plus interns ($80MM savings in 2021; IXD grown to 60% of inbound, +46% YoY); Senior Launch Program Manager, 2014–17 — readiness for 50+ FC launches including Amazon.in; FC Learning & Development Manager — Kaizen / Lean / CI training, learning curves and skill matrices, delivered an S-Team goal; Dell training background; SP Jain PGPM. **These are her numbers from her profile, quoted for framing only — they are not canon and never enter my stories.**

What that means for the answers. Three adjustments. First, lead with the *mechanism left behind* and who owns it now — she built teams and mechanisms herself (a placement strategy, a crossdock network, an L&D program), so "what still runs without you?" is her native question. Second, *price the trade-off* the way she priced distance-to-customer against speed and cost — S11's boarding pace vs automation maturity and S12's operators' two further quarters are exactly her shape of decision. Third, expect *transferability probes* — "how would this apply to a placement / network problem?" — and expect her to spot borrowed Leadership-Principle language: answer in the story's own words (gates, denominators, reversal triggers), never in recited LP slogans.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Think Big / Frugality for S12, Ownership / Earn Trust for the critique-and-private-first threads — the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Deliver Results	The HM carries the outcome LP almost by default — she needs evidence the candidate lands public targets with the organization intact. S11's two public targets two months early with zero rollbacks is the cleanest record in the set.	S11 (open) · backup: S18 in the master packet — only if S11 is already spent
Hire & Develop / Earth's Best Employer (one of the two)	Her profile is teams built and grown: a placement team built ground-up in Luxembourg, a team of six plus interns running the EU crossdock network, and an S-Team goal delivered as an FC L&D manager on Kaizen / Lean / CI training and skill matrices (public-profile inference). She will probe who grew on my programs: S11's input owners across nine teams signing the definitions they would be red against, wave teams grown into owners; S12's delivery teams now running the burn-up, the swap rule and the re-estimation habit.	S11 (owners own their lines) · S12 (mechanisms the teams now run)
Dive Deep	She ran a crossdock network hands-on and defined placement strategy at node level (public-profile inference) — expect her to test whether my numbers survive decomposition: CTS $0.66 → $0.03 cross-walked to ≈3.7 touches at $0.18 each; S12's cost translated at the work-item grain (339.5 minutes; 2,200 interactions per departed shipment). The Ownership / Earn Trust probes she may still carry are covered in the twist maps — the critique accepted (S11) and the private-first flip (S12).	S11 (per-container cross-walk) · S12 (work-item grain)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For the HM specifically:** name the mechanism in Amazon's own vocabulary where it is honest — a weekly input/output review with frozen metric definitions, graduation criteria published before the wave, a reversal trigger written before the flip. Do not claim to have invented the WBR; say it was copied from a trusted review, which is the point.

- **Answer her native question — what did you leave behind, and who owns it now?** Every story lands on a mechanism that still runs without me (the Monday review and the graduation gates; the rebuild-vs-reuse rule and the scope tripwire) with a named owner. She built teams and mechanisms ground-up (public-profile inference); that is the evidence she pattern-matches.
- **Price the trade-off in her units.** She priced distance-to-customer against speed and cost for a crossdock network (public-profile inference); my versions are boarding pace vs automation maturity (S11) and partners-first vs operators' two further quarters, priced and published (S12). Say the trade and its price, not just the win.
- **Two warnings (public-profile inferences).** She tests transferability — "how would this apply to a placement / network problem?" — so practice the bridges in the twist maps before the call. And she spots borrowed LP language: never recite slogans ("I was customer-obsessed"); say the gate, the denominator, the reversal trigger in the story's own words.
- **Have the openers ready** — the 75-second "tell me about yourself" and the 60-second "why Amazon" from the master packet (newest to oldest; Sephora → Flexport → Elementum → Manhattan). Say the long version once, here.

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S1/S3 (Lan) · S2/S18 (Goforth) · S6/S8 (Hoerger) · S15/S14 (Josyula). **v2 overlap flags — confirm before the loop.** This packet deliberately tells five Flexport records that another packet also touches: **S16-A** (Snowflake data model; Josyula's primary is S16, told as Ownership and the security-review miss — here lead with grain, join and reconciliation; the S16-A failure card is that same two-week miss, so if the panel notes show Josyula tells it, give it one sentence and pivot to the data model); **S11-C** (the Crux customer transition — the loop packet's S10, "79 customers, a destination and a date each"; no other panel carries it as a primary, but Goforth's pilot bench touches the Crux shadow accounts); **S12-A** (Asia latency; Hoerger's doc bench); **S11-B** (pilot rollout schedule and testing approach; Goforth's doc bench); the audit-logs PRD and the schedules solution are Lan's bench and are not used here. If the panel notes show a clash, drop the flagged story to a one-line pointer and use its router backup. If pushed for "another example" on an LP, use the named backup in the router, the question bank or the backup bench, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers

Seven Flexport primaries: two flagships (S11, S12 — deepened, with the original pages kept) and five new (S11-A, S11-B, S11-C, S12-A, S16-A). Each: 60-second pitch → SAFE-SAY box → Situation/Task → Action in gated phases → Results → L6 evidence → five labeled facets (data analysis · solution design · math & data logic · implementation · probe questions) → four-table math → sensitivity → statements → 20-second version and NACF bridge → behavioral pack → three angles → three trap drills.

S11 — Flexport — Shipment Visibility launch: a boarding process behind published graduation gates, run from a weekly input/output review (cost-to-serve $0.66 → $0.03 across 7 → 68 customers) DEEPENED

LPs this story can serve here: Deliver Results (primary) · Ownership (accepted the critique, owned the measurement) · Customer Obsession (graduation criteria per wave, CSAT in the gate) · Earn Trust (the discarded 99.4% week) · Hire & Develop (backup — wave teams grown into owners)

Process family: Deliver — customer onboarding / service-delivery process; weekly operating cadence; launch readiness and staged rollout

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Seven to sixty-eight customers — nine point seven times — and a hundred and seventy monthly users, both landed two months early; thirty-two thousand shipments was the target I managed toward, not a result I claim.”

“What's your source for that?” “The targets and the roll-out record are in the program's own documents and my interview workbook; nine point seven is my division, sixty-eight over seven, and the cost-to-serve arc of about sixty-six cents to about three cents is the same record, rounded to about ninety-five percent.”

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Deliver Results Both public targets — 68 customers and 170 monthly active users — landed two months early with zero rollbacks (the 32,000-shipment figure was a target I managed toward, not a result), because boarding went through six published graduation criteria rather than around them.

Ownership I accepted the critique (#3 to #14 of 16) as being about evidence, not effort, and owned the measurement system myself — by hand every Monday for months — before it was handed to a pipeline.

Customer Obsession CSAT at or above the consignee average was inside the gate, every client entered on a ~50-container calibration of its own freight, and 100 real containers ran seven checks before the first paying customer.

Earn Trust A flattering 99.4% week resting on three touches was printed as a data miss, and wave 3 was held in public on an ETA-accuracy miss — trust came from numbers that survived the next Monday.

Honesty label. Program owner across nine teams / ~100 people; the WBR format was copied from an existing leadership review (the LSA review), not invented; build-week counts are illustrative; the ~$1.5M pre-production measurement saving is a separate user-anchor figure (my own estimate, not a program-document number), never merged with cost-to-serve. Nine teams built and operated; I owned the boarding process, the measurement system and the pre-prod proof.

WHAT IS SHIPMENT VISIBILITY? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The minute. Shipment Visibility — "SV" — is Flexport's client-facing visibility product in the Client App: one pane where a client watches container milestones across their whole book, Flexport-carried and non-Flexport-carried shipments alike, so track-and-trace stops being emails to an ops desk and becomes a product the client logs into. It also created a "visibility-only" client segment — companies that bought visibility first were a deliberate upsell path into

Flexport freight. The 2021 plan priced the problem it solved: roughly $3.92M a year of manual track-and-trace salary across the ops org against about $263K of vendor spend, with only 18 of 99 milestones automated at the time. RECORD-FROM-DOCS

The machinery behind the story. Product standards: coverage 90%+, accuracy 95%, timeliness 90%, a fail-safe switch, dashboards and alarms — under the operating rule "edits are the enemy": every human correction is an automation defect, not a fix. Rollout: a free pilot in graduated waves — wave 1 two named clients (the first two hand-carried customers) around Q4 2021, wave 2 four of a seven-client pool around Q1 2022, wave 3 twenty-plus around Q2 2022, full roll-out H2 2022 — each client entering on a ~50-container calibration and graduating on six published criteria. Testing: four gates before launch — a 100-container live test, biweekly bug bashes on the client job stories, shadow accounts fed real Crux data, and UAT gated on data quality (80% of shipments minor-to-no issues). And Crux — the acquired visibility product — was transitioned onto SV by end 2022: 79 customers, $185,892 printed ARR (six of nine segments — S11-C re-foots it), 294,581 containers. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the pilot schedule phrases the SLA criterion as "90% of verifications within 24–48h"; canon says "reviewed within 24h" — canon wins, say 24h. The 95% accuracy standard is the 2021 plan's bar for milestone data, not this story's ETA-accuracy metric (60% → 90%, canon) — never mix them. The $3.92M / $263K figures are the plan's ops-org business case, a different basis from canon's per-container CTS — never merged. Wave timing and the six criteria match the master packet.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's shipment-visibility product had public commitments — 68 monthly active customers, 170 MAU, 32K shipments — and was serving seven pilots at a cost-to-serve of $0.66 a container, most of it human touches. Sales wanted the entire pool boarded at once; leadership said the program was not demonstrating value and it was about to drop from #3 to #14 of 16 initiatives. The process problem was that boarding ran faster than the automation could carry, and nobody could see it. **Task.** As program owner across nine teams, I owned the boarding path, the measurement system and the pre-production proof. **Action.** I rebuilt boarding as a gated process: six graduation criteria published to Sales before wave one, a launch gate of 100 real containers across seven checks, and a weekly input/output review — seven frozen metric definitions, each signed by the owner who would be red against it, every input tagged to the output it moves — copied from a review leadership already trusted and run by hand every Monday for months. Waves boarded only through the criteria; wave three was held in public on an ETA-accuracy miss and became the cleanest wave. When a flattering 99.4% week turned out to rest on three touches, I printed a data miss instead. **Result.** Primary metric: cost-to-serve per container $0.66 → $0.03, about 95%, across customers 7 → 68 — 9.7× — with zero rollbacks and no added ops headcount; automation 40% → 90%, accuracy 60% → 90%, both public targets two months early. The ~$1.5M pre-production measurement saving is a separate user-anchor figure — my own estimate — and I keep it separate.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Senior leadership told me the visibility program was not demonstrating value at the anticipated rate — it would drop from #3 to #14 of 16 initiatives. I had been reporting activity, not evidence. The critique was about evidence, not effort, and I accepted that first.

The business process underneath: customer boarding (Deliver) ran as a sales push — board the whole pool now — against an operation where most containers were still touched by hand. Cost-to-serve was $0.66 a container (≈3.7 human interactions at the record cross-walk of $0.18 per interaction); automation covered 40% of containers; ETA accuracy was 60%. The public commitments were 68 monthly active customers, 170 MAU and 32K shipments, off seven pilots.

The cost of leaving it: a bad first impression at scale with rollbacks, ops headcount growing with the customer count, and the program losing its funding slot because value was invisible.

T — Program owner across nine teams / ~100 people: own the boarding path, the measurement system and the pre-production proof — and make value legible to the leadership that decides rollout pace and funding, with zero

rollbacks and no added ops headcount.

Baselines on the table: customers 7 (target 68) · cost-to-serve $0.66/container · automation 40% · ETA accuracy 60% · public targets 68 customers / 170 MAU / 32K shipments · no instrument leadership trusted.

KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Rollout sequencing under a free pilot.** Pilot clients pay nothing, so every wave spent engineering, data and ops capacity before revenue; the pool had to be sequenced 2 → 4-of-7 → 20+ against a ~50-container calibration per client and six graduation criteria, with GO/NO-GO decisions taken monthly. That is the doc-level version of "board no faster than the automation can carry." RECORD-FROM-DOCS
- **Testing gates before any launch.** Four gates stood between build and client: a 100-container live test watched end-to-end, biweekly bug bashes on the client job stories, shadow accounts (real Crux data in dedicated demo accounts), and UAT that did not start until 80% of shipments showed minor-to-no issues. The gates are why "zero rollbacks" was achievable. RECORD-FROM-DOCS
- **An acquired product's customers in the middle of the rollout.** Crux's 79 customers — $185,892 printed ARR (six of nine segments — see S11-C), 294,581 containers — had to be transitioned onto SV by end 2022, including cancel-vs-transition decisions by segment and per-customer calls for the largest accounts, while the pilot waves were still boarding. Migration risk ran in parallel with launch risk. RECORD-FROM-DOCS
- **Data quality as a product standard, not an ops task.** The 2021 plan set coverage 90%+ / accuracy 95% / timeliness 90% with a fail-safe switch, and named the enemy: human edits. "Edits are the enemy" is the sentence behind the story's automation 40% → 90%. RECORD-FROM-DOCS

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — wrote the honest diagnosis (activity, not evidence); inventoried what leadership actually decides on (rollout pace, funding); chose to copy the LSA review's format rather than invent one. *Exit bar:* leadership agrees the questions the review must answer weekly; ~50-container calibration entry bar per client.
- **Plan & Design** — froze seven metric definitions (coverage auto, coverage with human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% week-over-week red/green rule; the 15-day active rule as the honest denominator; six graduation criteria published to Sales before wave 1 — DQ ≥ 90%, ETA ≥ 80%, SLA 90% reviewed within 24h, ≥100 containers + 5 features, stability, CSAT ≥ consignee average; launch gate 100 real containers × 7 checks; pre-prod sample size 300 containers for ±1.6% at 95% confidence. *Exit bar:* each input owner signs the definition they will be red against; first Monday deck dry-run.
- **Build & Test** — ran the review by hand every Monday (1.5–2 hrs, later ~4); pre-decomposed every red with an owner and a dated fix; enforced the data-honesty rule — a truncated week is a data miss, not a result; pre-prod ingest with per-hop monitors and kill switches; load proof 500 containers / 20 min / 99.6%. *Exit bar:* two consecutive Mondays where every red arrives with an owner and the number verifies the following week.
- **Cutover & Go-Live (waves)** — W1 (2 clients) → W2 (4 of a 7-pool) → W3 (20+) held in public on an ETA-accuracy miss traced to one bad source → full rollout. The Monday review became the forum where the rollout decision was taken. *Exit bar:* the VP unlocks the 68-client rollout two months early off the review's own numbers.
- **Hypercare / Close** — discarded a flattering 99.4% three-touch week and printed a data miss; handed the manual pull to a pipeline with the definitions in code (S16-A in this packet tells the data model; the full S16 telling belongs to Josyula's packet); left the critic steering by the numbers. *Exit bar:* automation 40 → 90%; CTS $0.66 → $0.03; same ops team at 68 customers.

Influence and ownership. Mine: the diagnosis, the frozen definitions, the review built by hand weekly, the graduation criteria, the hold decisions and the pre-prod measurement design. Nine teams built and operated; each input owner signed the definition they would be red against; the VP took the rollout decisions in the review. The Sales push to board everything now was answered with gates published before wave 1 — so when wave 3 was held, it was the rule firing, not a judgment call; the held customers became the cleanest wave.

> **Designed the process; others own it now.** I designed the review, the definitions and the graduation gates; the input owners across the nine teams run their lines and the pipeline (S16-A) carries the definitions in code — the review steers without me. **Trade-off put to leadership.** My recommendation to leadership outlined the competing metrics explicitly — boarding pace and customer count vs automation coverage and ETA accuracy — and proposed to board no faster than the automation could carry; wave 3 was the trade made in public.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — cost-to-serve per container $0.66 → $0.03 (record; about 95%); direct CTS dollars ≈$50K/yr *modeled* on an illustrative container count — the value is scale without headcount, not the direct dollars; the ~$1.5M pre-production measurement saving is a separate user-anchor figure, never merged with CTS.
- **#** — customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; targets 68 customers / 170 MAU / 32K shipments (record).
- **%** — CTS about 95%; automation 40% → 90%; ETA accuracy 60% → 90% (record).
- **Time** — both public targets hit two months early; the rollout unlocked two months early off the review (record); pre-prod measurement path cut rollout proof from 6 months to 1.5.
- **Customer** — six graduation criteria published to Sales before wave 1 — including CSAT ≥ consignee average; wave 3 held in public on an ETA-accuracy miss and became the cleanest wave (record).
- **Quality** — a flattering truncated 99.4% week discarded and printed as a data miss; seven metric definitions frozen and later moved into code (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Leadership said the program was not demonstrating value (#3 → #14 of 16) and no measurement system existed. I wrote the diagnosis (activity, not evidence), chose the nine teams' input owners as the stakeholders, and set the goals the review had to answer weekly.
Scope & influence	Nine teams built and operated; Sales held to criteria I published; the VP took the rollout decisions inside the review I ran — influence one to two levels up, from a program seat.
Execution trade-off + mechanism others now own	Boarding speed (Sales) vs automation maturity and ETA accuracy: wave 3 held in public on the miss. Mechanism others own now: the Monday input/output review and the six graduation criteria, with the seven definitions moved into code (S16-A).
Impact (long-term)	Customers 7 → 68 (9.7×) at CTS $0.66 → $0.03 with zero rollbacks and no added ops headcount; both public targets two months early; the review became the standing rollout forum.
Process improvement (standard work institutionalized)	Seven frozen metric definitions, input→output tagging, the ±5% week-over-week red/green rule, the 15-day active denominator, and the data-honesty rule (a truncated week is a data miss, not a result).
Complexity & consensus	A critical leadership and a Sales push to board everything now — answered by gates published before wave 1, each input owner signing the definition they would be red against; the VP unlocked the 68-client rollout off the review's own numbers.
Root cause at scale	The ETA-accuracy miss traced to one bad source and fixed there; the flattering 99.4% week traced to three touches and printed as a data miss — fixed at the definition and the source, not customer by customer.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **HM — Why is this your Deliver Results story and not just a metrics story?**
>
> Because the results landed behind gates, not around them: both public targets two months early, 7 → 68 customers on the same ops team, zero rollbacks. The review and the graduation criteria are the mechanism; the about-95% cut and the 9.7× are the outcome.

HM — The direct dollars on a 95% cost reduction are small. Why lead with it?

Because it is honest and because the point was never the direct dollars: at the 32K-shipment target the direct saving is ≈$50K a year, modeled on an illustrative container count. What it bought is boarding 9.7× the customers without 9.7× the ops headcount — roughly two and a half people never hired (modeled — assumptions in the deepening's dollars table; not record headcount) — and a rollout unlocked two months early. The ~$1.5M pre-prod estimate is a separate user anchor and I never add it to this.

HM — Sales wanted everyone boarded. How did you hold the line without becoming the blocker?

By publishing the six criteria to Sales before wave one, so the rule did the refusing. Wave 3 was held in public on an ETA-accuracy miss traced to one bad source; the customers we held became the cleanest wave, and Sales had seen the gate before it fired.

HM — You discarded a 99.4% week. Walk me through that decision.

Three touches is not a week — a 95% confidence interval on N = 3 is wider than the metric. Printing it would have taught the VP a number that could not survive the next Monday. I printed a data miss. A review that flatters once is not trusted twice.

HM — What would you do differently, and how does this translate to my team?

Start the review at kickoff, not after the critique. At Amazon this is a WBR story told to the company that invented the WBR — the respectful version is that I copied a trusted format, froze definitions with their owners, tagged inputs to outputs and let the review make the rollout decision. Graduation criteria are launch gates published before the wave and held in public.

HM — What was yours and what was the team's?

Mine: the diagnosis, the seven definitions, the Monday build by hand for months, the criteria, the holds, the pre-prod measurement design. The nine teams built and operated; each input owner signed their own definition; the VP decided in the room. The format came from the LSA review — I did not invent it.

HM — Why waves at all? What did the calibration actually buy?

The pilot was free, so every boarded client cost engineering, data and ops before revenue — boarding the whole pool would have spent the program's credibility on clients we could not yet serve. The ~50-container calibration per client gave the data team a verified baseline on the client's own freight before scale, and the six published criteria converted "ready" from a feeling into a rule: wave 1 two clients, wave 2 four of a seven-client pool, wave 3 twenty-plus RECORD-FROM-DOCS. The canon version: waves boarded only through the published gates, and the held wave became the cleanest.

HM — How did you test before the first paying customer touched it?

Four gates, all from the program record RECORD-FROM-DOCS: a 100-container live test watched end-to-end; biweekly bug bashes on the client job stories; shadow client accounts — real Crux data fed into dedicated demo accounts so the clients' own data exposed the defects; and UAT that did not start until 80% of shipments showed minor-to-no issues. UAT was gated on data quality, not on the calendar — the testing version of the graduation rule.

HM — You were rolling out while an acquired product's customers sat on the old platform. How did you handle that risk?

As a parallel migration with its own decision rules, not an afterthought RECORD-FROM-DOCS: 79 Crux customers, $185,892 printed ARR (six of nine segments — see S11-C) and 294,581 containers, segmented — transition where SV fit, cancel honestly where it did not (two competitor SCM-SaaS accounts were canceled on published dates), per-customer calls for the largest accounts. Internal pushback on migration cost and feature parity was priced into the decision document, not overridden. The pilot gates kept quality honest while the migration ran beside it.

THE KEY MATH — COST-TO-SERVE, THE SCALE RESULT, AND WHY THE DOLLARS STAY SEPARATE

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve before	`$0.66 / $0.18 ≈ 3.7 touches per container`	record; $0.18 per interaction is the record cross-walk	$0.66 — ≈3.7 touches
2	Cost-to-serve after	`($0.03 − $0.66) / $0.66 = −95.45%, say "about 95%"; $0.03 / $0.18 = 0.17 touches`	record	$0.03 — ≈0.17 touches
3	Scale	`68 / 7 = 9.7×; zero rollbacks; no added ops headcount`	record	9.7× (never "10×")
4	Direct CTS dollars	`32,000 shipments × 1.5–4 containers = 48–128K; × $0.63 = $30–81K`	modeled (containers per shipment illustrative)	≈$50K/yr modeled
5	Pre-prod measurement path	`n = 300 → ±1.6% at 95% CI; rollout proof 1.5 vs 6 months`	user anchor — separate canonical	~$1.5M, never merged with CTS
6	Week-over-week rule	`WoW = (B − A)/A = (21.9 − 40.0)/40.0 = −45% → red`	±5% red/green (record design; values are the 17 Jan and 31 Jan 2022 prints)	root cause: a carrier format change — measurement, not operators

Automation 40 → 90% and the WBR hours (1.5–2 → ~4 by hand) are this packet's canon; the loop packet carries different values (§1.3 conflict) — do not mix them in the room.

Where this lands at Amazon. This is the JD's "define, create, implement and execute processes" line in its purest form: a boarding process with published graduation criteria, a weekly input/output review with frozen definitions, and the discipline to hold a wave in public. Reliability (zero rollbacks), responsiveness (two months early), cost efficiency (about 95% CTS), asset utilization (the same ops team at 9.7× the customers).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S11

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you delivered results under pressure / against a target.	Deliver Results	Open on the critique (dropping #3 → #14) and the public targets; close on both targets two months early, 7 → 68, zero rollbacks. The gates are the how.	Skip the four-system data-pull detail unless asked for the data model — then S16-A carries it.
Tell me about a time you had to make value legible to senior leadership.	Ownership / Earn Trust	Accepting that the critique was about evidence, not effort; copying a trusted format; signing each definition with its owner; the review becoming the decision forum.	Never say "I invented the WBR."
Tell me about a time you said no to a stakeholder (Sales).	Have Backbone / Highest Standards	Six criteria published before wave 1; wave 3 held in public; held customers became the cleanest wave.	Do not call Sales wrong — they were right to want speed; the gate did the refusing.
Tell me about a time you chose the harder, honest number.	Earn Trust	The 99.4% three-touch week printed as a data miss; the 15-day active rule as the honest denominator.	Skip the Wilson-interval math unless asked; say "three touches is not a week."

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
How did you decide the customer was ready to launch? (Customer Obsession)	Customer Obsession	CSAT ≥ consignee average inside the gate; ~50-container calibration entry bar; 100 real containers × 7 checks before the first paying customer.	Do not claim customer-satisfaction dollars; none exist in canon.
How would you run a WBR for my team?	Functional — operating cadence	Copy a trusted format, freeze 5–7 definitions with owners and honest denominators, tag inputs to outputs, pre-decompose reds, let the review take the decision.	Never "$1.00 → $0.03"; never "70% baseline"; never "10×".
How would your wave machine apply to my world — placement / inbound? (transferability probe — public-profile inference)	Functional — transfer	The wave machine is graduated onboarding behind published criteria — the same shape as onboarding inventory into an IXD / crossdock program: a selection bar published before the first node, a calibration volume per node (~50 containers per client ≈ a trial volume per node), and a public hold when a node misses. Scale follows the gate, never the ask.	Do not lecture on her network; bridge in one sentence and hand it back with a question.
Your CTS is per container — mine is per node. Does the logic survive? (transferability probe)	Functional — transfer	Identical logic, different denominator: cost-to-serve per node = touches per unit through the node × cost per touch; the fix is still automation coverage and source quality, and the honest denominator is still defined before the metric. $0.66 → $0.03 is the container-grain instance of it.	Never quote her profile's $50MM / $80MM as if it were mine — those are her numbers, quoted for framing only.

DEEPENING S11 — FIVE FACETS, FOUR-TABLE MATH, BEHAVIORAL PACK, ANGLES AND TRAP DRILLS NEW

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Cost-to-serve decomposed into touches (3.67 → 0.17); the 15-day active denominator; the three-touch week; wave-3 accuracy traced to one source; the pre-production evidence (500 containers in 20 minutes at 99.6%; dry run 100% processed, 86% tracked) and my own n = 300 sample design (user anchor).
2 · Solution design	Board-all vs gated waves; copy a trusted format vs invent; criteria published before wave 1; sample vs census; headcount vs automation as the scale lever.
3 · Math & data logic	Four tables: $0.66 / $0.18 = 3.67; about 95%; 9.7×; $30,240–$80,640; 80,000 × 3.5 min ≈ 2.6 FTE; the pre-production sample design n = 300 → ±1.6% (user anchor).
4 · Implementation	Waves 2 → 4-of-7 → 20+ with monthly GO / NO-GO; the Monday review; input owners across the nine teams; kill switches; the hand-off to the pipeline.
5 · Probe questions	Seven numeric probes plus three trap drills: direct dollars, "10×" and the FTE claim, and the separate ~$1.5M.

Facet 1 — Data analysis: what each analysis decided

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Cost-to-serve decomposition	dollars per container ÷ $0.18 per interaction (record cross-walk)	$0.66 is ≈3.67 human touches; $0.03 is ≈0.17	automation coverage, not headcount, is the scale lever

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Active denominator	active = cumulative inflow less containers 15 days past departed terminal, by week	cumulative counts hide a slowing program	review runs on active, not cumulative
Week-over-week instrument	(B − A) / A against ±5%	(21.9 − 40.0) / 40.0 = −45% → red; root cause a carrier format change	reds are decomposed before the meeting, measurement vs operations
Three-touch week	99.4% on N = 3 touches	a 95% interval on N = 3 is wider than the metric	printed as a data miss, not a result
Wave-3 accuracy by source	ETA accuracy per data source	the miss traced to one bad source	held in public; the held customers became the cleanest wave
Pre-production sampling	n = 300 at ≈98% expected accuracy (my sample design)	±1.6% at 95% confidence	a 1.5-month proof instead of 6 — my estimate; the separate ~$1.5M pre-production figure is a user anchor, not a program-document number

Facet 2 — Solution design: options weighed

QUESTION	OPTIONS	CHOSEN	WHY
Boarding pace	Board the whole pool now (Sales) · waves behind published criteria	Waves behind six criteria, published before wave 1	A free pilot spends engineering, data and ops before revenue; the rule does the refusing.
Review format	Invent a scorecard · copy the leadership review leadership already trusted	Copy the LSA format; freeze seven metrics with owners	Inventing would have spent credibility the program did not have.
Lever for scale	Add ops headcount · raise automation coverage	Automation 40% → 90%	Cost-to-serve is a touches problem: $0.66 / $0.18 = 3.67 touches.
Pre-production proof	Census over months · calibrated sample	Sample of 300 containers, ±1.6% at 95%	Big enough to certify the gate, small enough to afford — my estimate: 1.5 months instead of 6 (user anchor).
Bad week	Print 99.4% · print a data miss	Data miss	A number that cannot survive the next Monday teaches the VP the wrong thing.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`$0.66 per container off seven pilots`	record	$0.66
2	Residual human touches	`$0.66 / $0.18 = 3.67 touches per container`	record cross-walk ($0.18 per eliminated interaction)	≈ 3.7 touches
3	Automation and accuracy	`coverage 40%; ETA accuracy 60%`	record	most containers still touched by hand

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Customers vs target	`7 pilots against a target of 68; the program was to drop from #3 to #14 of 16`	record	value invisible to leadership

Table 2 — The fix: gates, honest denominators and the instrument, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Launch gate volume	`100 real containers × 7 checks = 700 check results before the first paying customer`	record design; product mine	700 checks
2	Load proof	`500 containers in 20 min = 25 a minute; 498 / 500 = 99.6%`	record (500 / 20 min / 99.6%); arithmetic mine	25 / min · 99.6%
3	Honest denominator	`a 99.4% week on N = 3: Wilson lower bound 3 / (3 + 1.96²) = 0.44`	record decision (99.4% on three touches, Apple A2); the Wilson interval is my arithmetic	0.44 → data miss
4	Week-over-week rule	`(21.9 − 40.0) / 40.0 = −45% → beyond ±5% → red`	record: the weekly prints were 40.0% on 17 Jan and 21.9% on 31 Jan 2022 (Apple A2)	−45% → red
5	Pre-production sample	`1.96 × √(0.98 × 0.02 / 300) = 0.0158 = ±1.6 points; rollout proof 6 − 1.5 = 4.5 months saved`	user anchor (my sample design; GFiber v8 only — the program documents show 500 containers in 20 minutes at 99.6%, and an 86%-tracked dry run)	±1.6% · 4.5 months

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`1 − $0.03 / $0.66 = 0.9545 → about 95%; $0.03 / $0.18 = 0.17 touches`	record endpoints; ratio mine	about 95% · 0.17 touches
2	Touches removed	`3.67 − 0.17 = 3.5 per container — about 95% of the touches`	record cross-walk; arithmetic mine	3.5 per container
3	Automation / accuracy	`40% → 90% = +50 points; 60% → 90% = +30 points`	record	+50 / +30 points
4	Customers	`68 / 7 = 9.71; zero rollbacks; no added ops headcount; 68 customers and 170 monthly users two months early`	record	9.7× (never rounded up)

Table 4 — The dollars: every multiplication shown (record vs modeled)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Containers at the 32,000-shipment target	`32,000 × 1.5 = 48,000; 32,000 × 2.5 = 80,000; 32,000 × 4 = 128,000`	containers per shipment illustrative (1.5–4)	48K / 80K / 128K
2	Unit saving	`$0.66 − $0.03 = $0.63`	record endpoints	$0.63
3	Direct CTS dollars	`48,000 × $0.63 = $30,240; 80,000 × $0.63 = $50,400; 128,000 × $0.63 = $80,640`	modeled	$30K · $50K · $81K
4	Headcount never hired	`80,000 × 3.5 touches = 280,000 touches; × 1 min = 4,667 h; ÷ 1,800 h per FTE = 2.6 FTE (÷ 1,920 → 2.4)`	modeled: one minute per touch and 1,800 hours per FTE-year are assumptions	≈ 2.5 people (2.4–2.6)
5	The separate user anchor	`~$1.5M pre-production measurement saving: n = 300 → 1.5 vs 6 months`	user anchor (GFiber v8 only) — a separate figure, never added to CTS or the model	~$1.5M, never merged

Facet 4 — Implementation: the wave machine and its controls

WAVE	CLIENTS (RECORD)	ENTRY	GRADUATION / DECISION	TIMING (RECORD)
1	2 named pilots	~50-container calibration	six criteria; monthly GO / NO-GO	around Q4 2021
2	4 of a 7-client pool	~50-container calibration	six criteria; monthly GO / NO-GO	around Q1 2022
3	20+	~50-container calibration	held in public on an ETA-accuracy miss; held customers became the cleanest wave	around Q2 2022
Full	the pool to 68	graduated clients	rollout unlocked two months early off the review	H2 2022

Controls. Nine input owners each sign the definition they will be red against; reds are pre-decomposed with an owner and a dated fix; per-hop monitors and kill switches on the pre-production ingest; the manual pull is the review's source until the pipeline matches within 1% (S16-A). **Lean / Kaizen mapping.** The six criteria are standard work for onboarding; the hold on wave 3 is jidoka — stop the line when a quality criterion fails; the Monday review is the PDCA Check; "edits are the enemy" is the 2021 plan's own waste statement — every human correction is a defect, not a fix.

Facet 5 — Probe questions (numeric)

> **Why is the direct dollar figure so small?**
>
> Because the unit economics is the point: $0.63 saved on 48,000–128,000 containers is $30,240–$80,640, modeled on an illustrative containers-per-shipment. What it bought was 68 customers on the same ops team — about 2.5 people never hired, modeled — and a rollout unlocked two months early.

> **Prove the 2.5 people.**
>
> It is a model with two assumptions I name: 80,000 containers × 3.5 touches removed × one minute = 4,667 hours; at 1,800 hours a year that is 2.6 FTE, at 1,920 it is 2.4. I would replace the minute with the ticket-grain automation data.

> **Why ±1.6%?**
>
> That is my sample design, a user anchor — the program's own evidence was 500 containers in 20 minutes at 99.6% and a dry run at 100% processed, 86% tracked. n = 300 at about 98% accuracy: 1.96 × √(0.98 × 0.02 / 300) = 0.0158. Big enough to certify the gate, small enough to afford — it cut the rollout proof from 6 months to 1.5, which is the separate ~$1.5M pre-production estimate (user anchor), never merged with cost-to-serve.

> **Is it 10× or 9.7×?**
>
> 68 / 7 = 9.71. Say 7 to 68, or 9.7 times; no other multiple belongs in the room.

> **What does −45% week-over-week mean?**
>
> (21.9 − 40.0) / 40.0 = −45%, red against ±5%. The weekly prints were 40.0% on 17 January and 21.9% on 31 January 2022 (record); the root cause was a carrier format change — measurement, not operators.

> **How do you know a wave is ready?**
>
> Six criteria all pass — data quality at least 90%, ETA accuracy at least 80%, SLA 90% reviewed within 24 hours, at least 100 containers and 5 features, stability, CSAT at or above the consignee average — after a ~50-container calibration.

> **What would you change?**
>
> Start the review at kickoff instead of after the critique, and price the pipeline hand-off earlier — the pull stayed with me for months.

SENSITIVITY — CONTAINERS PER SHIPMENT (THE SOFT INPUT)

CONTAINERS PER SHIPMENT	CONTAINERS A YEAR	DIRECT CTS SAVING (× $0.63)	FTE-EQUIVALENT (ILLUSTRATIVE)
1.5 (low)	48,000	$30,240	1.6
2.5 (base)	80,000	$50,400	2.6
4.0 (high)	128,000	$80,640	4.1

Both columns are modeled on an illustrative containers-per-shipment; the record gives shipments, not containers. The ~$1.5M pre-production figure is excluded from every row.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> Honestly, the cost-to-serve number is a unit-economics result, not a big-dollar number: ninety-five percent off per container is measured, and at the thirty-two-thousand-shipment target it is only about fifty thousand dollars a year of direct cost — my model, on an illustrative container count. What it bought is scale: sixty-eight customers on the same ops team with zero rollbacks, about two and a half people we never had to hire — modeled — and a rollout unlocked two months early. The million and a half on the pre-production measurement path is a separate figure — my own estimate — and I keep it separate. So: the ninety-five percent and the seven-to-sixty-eight are record; every dollar except that one-and-a-half is my model.

IF CHALLENGED — THE FULL ANSWERS

"Your containers-per-shipment is a guess."

It is, and I said so: 32,000 shipments is a target, not a count, and it is shipments, not containers. At 1.5 to 4 containers per shipment the direct saving runs $30,240 to $80,640. I would pull the active-container count from the 15-day active rule the review already uses.

"Fifty thousand a year is small — why lead with it?"

Because it is honest, and the point was never the direct dollars: it was boarding 9.7 times the customers without 9.7 times the ops headcount.

"Where does the million and a half come from, then?"

The pre-production measurement path: 300 containers at ±1.6% proved the product in 1.5 months instead of 6. It is my estimate (a user anchor, not a program-document number), it is not cost-to-serve, and I never add it to the fifty thousand.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I owned the boarding process for Shipment Visibility: six published graduation criteria and a Monday input-output review took cost-to-serve from $0.66 to $0.03 per container while customers went from 7 to 68, with zero rollbacks and 68 customers and 170 monthly users landed two months early.

NACF BRIDGE — ONE SENTENCE

It is the shape of a network launch: onboard each node behind criteria published before the first one, calibrate on a trial volume, and hold a node in public when it misses — scale follows the gate, never the ask.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you not only met a goal but considerably exceeded expectations. How were you able to do it? What challenges did you have to overcome?" AIQB: Deliver Results Q3 Deliver Results FACET: IMPLEMENTATION

Open (first two sentences): "Both public targets landed two months early with zero rollbacks — and the way they landed is the point: boarding went through six published criteria, not around them."

90-second STAR. **S** Seven pilots against public targets of 68 customers and 170 monthly active users (plus a 32,000-shipment target); Sales wanted the whole pool boarded at once. **T** Hit the targets without a rollback or added ops headcount. **A** Waves — 2, then 4 of 7, then 20-plus — each entering on a ~50-container calibration and graduating on six criteria published to Sales before wave 1, with a launch gate of 100 real containers across seven checks and a Monday review taking the rollout decision. **R** Customers 7 → 68, cost-to-serve $0.66 → $0.03, automation 40% → 90%, accuracy 60% → 90%; zero rollbacks.

Land: 7 → 68 (9.7×); 68 customers and 170 users two months early · **Pivot to NACF:** launching nodes in a network: published gates, calibrated volumes, a public hold when a node misses.

Do NOT say: "10×" or any other multiple; the $1.5M added to cost-to-serve.

B2 · "Give me an example of a tough or critical piece of feedback you received. What was it and what did you do about it?" AIQB: Earn Trust Q2 Ownership FACET: DATA ANALYSIS

Open (first two sentences): "Senior leadership told me the program was not demonstrating value and would drop from third to fourteenth of sixteen. I had been reporting activity, not evidence — and I accepted that first."

90-second STAR. **S** A program about to lose its funding slot because value was invisible. **T** Make value legible without arguing the critique. **A** I inventoried what leadership actually decides on, copied a review format they already trusted, froze seven metrics with owners who signed the definitions they would be red against, and ran the pull by hand every Monday for months. **R** The same critic steered the rollout by the review's numbers, and unlocked the 68-client rollout two months early.

Land: #3 → #14 of 16 → the VP unlocked the rollout · **Pivot to NACF:** owning a metric nobody owned: build it by hand first, automate it once the definitions are signed.

Do NOT say: that I invented the weekly review — I copied the LSA format.

B3 · "Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?" AIQB: Customer Obsession Q8 Customer Obsession FACET: SOLUTION DESIGN

Open (first two sentences): "Wave three was held in public because ETA accuracy missed on one bad source. Sales wanted speed; the customers we held became the cleanest wave."

90-second STAR. **S** A free pilot where every boarded client cost engineering, data and ops before revenue. **T** Protect each client's first impression at scale. **A** Six criteria published before wave 1 — including CSAT at or above the consignee average — a ~50-container calibration on each client's own freight, and a hold on wave 3 traced to one bad source. **R** Zero rollbacks; the held wave became the cleanest.

Land: zero rollbacks; wave 3 held · **Pivot to NACF:** inbound readiness: do not take the first container until the node's criteria pass.

Do NOT say: that Sales was wrong — they were right to want speed; the gate did the refusing.

B4 · "Tell me about a time when you uncovered a significant problem in your team. What was it and how did you communicate it to your manager and other stakeholders? What did you do to address the problem?"

AIQB: Earn Trust Q6 | Earn Trust | FACET: MATH AND DATA LOGIC

Open (first two sentences): "One week showed ninety-nine point four percent — on three touches. A 95% confidence interval on three is wider than the metric, so I printed a data miss instead."

90-second STAR. **S** A flattering week landing just before a funding decision. **T** Keep the review credible. **A** I computed the interval — Wilson lower bound 0.44 on N = 3 — and held the review to the 15-day active denominator; the week was printed as a data miss. **R** A number that survived the next Monday; the review was trusted twice.

Land: 99.4% on N = 3 → data miss · **Pivot to NACF:** any capacity report: a rate on a tiny denominator is not a result.

Do NOT say: the Wilson math unless asked — say "three touches is not a week".

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Cost-to-serve was sixty-six cents a container — which is three and two-thirds human touches. The program was a touches problem, not a headcount problem."

Headline metric: $0.66 / $0.18 = 3.67 → 0.17 touches

- Decompose dollars into touches with the record cross-walk.
- Active denominator: cumulative inflow less containers 15 days past departed terminal.
- WoW rule: −45% is red; root cause a carrier format change.
- Say "modeled" on every dollar.

Conflict / influence / backbone angle — Have Backbone; Disagree and Commit

Opener: "Sales wanted the entire pool boarded at once; I published the gates to Sales before wave one, so when wave three was held, it was the rule firing — not me."

Headline metric: six criteria; wave 3 held

- The rule did the refusing; no one was called wrong.
- The held customers became the cleanest wave.
- The trade: boarding speed for a rollout that never rolled back.
- Leadership took the decision in the Monday review.

Honest failure-and-learning angle — Learn and Be Curious

Opener: "I had been reporting activity, not evidence — leadership said so, and they were right. The review should have started at kickoff, not after the critique."

Headline metric: #3 → #14 of 16

- What I got wrong: measuring effort instead of outcomes.
- What I changed: seven frozen definitions, input → output tagging.
- What I would do differently: kickoff review; earlier pipeline hand-off.
- What held: zero rollbacks.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Customers 7 to 68 — that's a 10× result?"

Numeric answer. 68 / 7 = 9.71. The only multiple to say is 9.7× — or just 7 to 68. Rounding up to 10× overstates a record number by 3%.

Say exactly: "Seven to sixty-eight — nine point seven times, and I won't round that up."

"You said two and a half people never hired. Show me."

Numeric answer. 80,000 containers × 3.5 touches removed × 1 minute = 4,667 hours; ÷ 1,800 h = 2.6 FTE (÷ 1,920 h = 2.4). Modeled: containers per shipment, one minute per touch and the hours per FTE are my assumptions; the record is zero added ops headcount at 68 customers.

Say exactly: “It's modeled: about two and a half people on an illustrative container count and a minute a touch — the record is that headcount didn't grow.”

"So the saving is $1.5M plus $50K?"

Numeric answer. No. The ~$1.5M is the pre-production measurement saving (n = 300 → ±1.6%, 1.5 vs 6 months) — my estimate and a user anchor, kept separate. The $50,400 is modeled direct cost-to-serve at 80,000 containers. They never add.

Say exactly: “They're separate. The million and a half is my estimate of the pre-production proof path, kept separate; the fifty thousand is my model of direct cost-to-serve.”

S11-A — Flexport — The Monday review as an analysis: taking a 22× cost-to-serve drop apart (6.0× coverage × 3.67× touch efficiency), with honest denominators, seven frozen metrics and one discarded week NEW

LPs this story can serve here: Dive Deep (primary) · Insist on the Highest Standards · Earn Trust · Are Right, A Lot (backup)
Process family: Plan / Deliver — operating cadence, measurement design and cost-to-serve analysis; weekly business review (WBR)

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep I took $0.66 → $0.03 apart: 22.0× = 6.0× fewer containers needing a human (60% → 10%) × 3.67× fewer touches on each one that did — and showed the split depends on the order you attribute it.

Insist on the Highest Standards Seven metric definitions frozen and signed by the owners who would be red against them; a 15-day active denominator; a truncated 99.4% week (three touches) thrown out.

Earn Trust I printed a data miss instead of a flattering number, because three touches is not a week.

Are Right, A Lot Every red was decomposed before the meeting; a −45% week (40.0% on 17 Jan to 21.9% on 31 Jan 2022, record) traced to a carrier format change — measurement, not operators — and the same number was checked the next Monday.

Honesty label. Same program family as S11, a different cut: S11 is the boarding process and its gates; this story is the measurement system and the analysis behind it. The review format was copied from the LSA weekly review, not invented, and I built it by hand every Monday for months. The 22× decomposition and the 56% / 44% attribution are *my model* on record endpoints — they assume touches fall only on non-automated containers, and I say the split is a convention, not a measurement (the order-dependence is shown in Table 3). The week-over-week values (40.0% on 17 Jan → 21.9% on 31 Jan 2022), the three grains (19.4% · 54.1% · 40.2% → 69.2%) and the discarded 99.4% week are **record** from my own review workbook (Apple v11 A2, extract SV_28.xlsx); the hand-built review ran for months. The real LSA weekly-review workbook (Table 5) is the format I copied — its FFA tab is another team's slide, used here to show the mechanics on real data, not as my program's numbers. An early draft note in the source folder carries much larger growth multipliers for customers and volume — they conflict with canon (7 → 68 = 9.7×) and are **never said**.

WHAT THE RECORD SAYS ABOUT THE REVIEW (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The format. The LSA weekly-review instructions separate *output* from *input* metrics; a week runs Monday–Sunday; the view carries five consecutive weeks plus months; week-over-week is the relative change (B − A)/A; green is favorable beyond 5%, red unfavorable beyond 5%. RECORD-FROM-DOCS

What the program asked of it. The Feb 1, 2022 E-Team takeaways name Goal 1 as lowering cost-to-serve, ask for cost-to-serve measured for 3PL + 4PL versus 4PL only, have the visibility team publish its own weekly review, and set the improvement priority rule: rank carriers, terminals and milestones on the two priority lanes by cost-to-serve reduction first, ease of getting the data second. RECORD-FROM-DOCS

What the review ran on. The BRD names the seven metrics and the as-is: a manual Zendesk pull matched to a manual container pull in Google Sheets, 1.5–2 hours per week-over-week snapshot (a BRD measure; my own account is "about four hours a week at its worst" — two different measures, never stacked). My review workbook (Apple A2) is the container-status-history extract SV_28.xlsx — 10,289 events, week ending 28 Feb 2022. RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's visibility program was being judged by leadership that said it was not demonstrating value, and the only instrument was a spreadsheet I stitched from four systems every Monday. Cost-to-serve was $0.66 a container; nobody could say whether it would fall because automation covered more containers or because each human touch got cheaper, and one flattering week could steer a rollout. **Task.** As program owner, make the numbers decision-grade. **Action.** I copied the LSA weekly review's mechanics — Monday-to-Sunday weeks, five weeks in view, week-over-week as (B − A)/A, red or green beyond ±5% — froze seven metrics with owners who signed the definitions they would be red against, tagged every input to the output it moves, and fixed the denominator: active is the cumulative inflow less containers 15 days past departed terminal. Then I took the headline apart: $0.66 to $0.03 is 22×, which is 6× fewer containers needing a human — 60% down to 10% — times about 3.7× fewer touches on each one that still did. When a week printed 99.4% on three touches I discarded it, because a 95% interval on three is 44% to 100%. **Result.** Primary metric: cost-to-serve $0.66 → $0.03 per container — about 95% — across customers 7 → 68, with every red arriving with an owner and a dated fix — and the same seven definitions now live in code.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. "Cost-to-serve went from sixty-six cents to three cents a container — about ninety-five percent. The split into six-times coverage and three-and-two-thirds touches is my model on those two record endpoints, and I call it a convention, not a measurement."

"What's your source for that?" "The endpoints and the weekly prints come from the review workbook I built — the container-status extract — and from the program's own documents; the split and the sample-size arithmetic are mine, and I label them modeled."

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The program had public targets — 68 customers and 170 monthly active users (both landed two months early) and a 32,000-shipment target I managed toward — and leadership that said value was not legible; it was about to drop from #3 to #14 of 16 initiatives. The instrument was my Monday spreadsheet: four systems (tickets, container pulls, platform data, Jira), about four hours a week at its worst, one person who could break it. The analysis problem underneath had three parts: a headline that moved ($0.66 → $0.03) with no explanation of *what* moved it; denominators that could flatter or flatten a week; and small-N weeks that could mislead a VP.

T — Own the measurement design and the analysis: define what is measured, at which grain, against which denominator, and what the number must be able to explain — so leadership could steer rollout pace and funding off it.

Baselines on the table: CTS $0.66/container (≈3.7 touches at the record $0.18 per interaction) · automation 40% · ETA accuracy 60% · customers 7 · review built by hand — about four hours a week at its worst · seven metrics · five Mon–Sun weeks in view · ±5% week-over-week rule.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is (four systems, Sheets, a person-dependent manual build); asked what leadership decides on (rollout pace, funding) and wrote the questions the review must answer weekly; chose the LSA review as the pattern to copy. *Exit bar:* leadership agrees the weekly questions; each input has a named owner.
- **Plan & Design** — froze seven metric definitions (coverage auto, coverage with human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% week-over-week rule; the 15-day active denominator; a parking lot for new metrics. *Exit bar:* every owner signs the definition they will be red against; first Monday deck dry-run.
- **Build & Test** — built the deck by hand every Monday; pre-decomposed every red into source, definition or operator before the meeting; enforced the data-honesty rule (a truncated week is a data miss, not a result); tied Jira

blocker/critical trends to the input metrics. *Exit bar:* two consecutive Mondays where every red arrives with an owner and a dated fix, and the number verifies the following Monday.

- **Cutover & Go-Live** — the Monday review became the forum where rollout decisions were taken; waves boarded only through the published criteria; the decomposition went into the deck so leadership could see which lever moved cost-to-serve. *Exit bar:* the VP takes the rollout decision in the room off the numbers.
- **Hypercare / Close** — discarded the flattering 99.4% three-touch week and printed a data miss; moved the definitions into code with the Snowflake pipeline (S16-A) so the review no longer depended on one person. *Exit bar:* same seven numbers within 1% for consecutive weeks; the critic steers by them.

Influence and ownership. Mine: the diagnosis, the seven definitions, the denominator rule, the decomposition, the data-honesty rule and the Monday build. The input owners — Data Services, Support, GTM and the engineering teams — signed their own definitions and carried their own reds; the VP decided in the room. The format came from the LSA review.

> **Designed the process; others own it now.** I designed the definitions, the denominator and the reading rules; the owners run their lines, and the pipeline (S16-A) carries the definitions in code. **Trade-off put to leadership.** Coverage of the page (every metric anyone wanted) versus stability of the page (seven that never change definition) — I recommended stability and gave the rest a parking lot.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — cost-to-serve $0.66 → $0.03 per container (record; about 95%); direct dollars ≈$50K/yr *modeled* at the 32,000-shipment target on an illustrative 2.5 containers per shipment; ≈$21K/yr of by-hand review labor *modeled* (≈4 h/wk at its worst, at $100/h — an upper bound); the ~$1.5M pre-prod figure (S11, user anchor) stays separate.
- **#** — seven frozen metrics, each with a named owner; customers 7 → 68 (9.7×) with zero rollbacks; 68 customers and 170 monthly users landed two months early (record); the 32,000-shipment figure is a target, not a result.
- **%** — automation 40% → 90%; ETA accuracy 60% → 90% (record); coverage-versus-touch-efficiency split ≈56% / 44% (modeled).
- **Time** — by hand for months — about four hours a week at its worst — until the pipeline matched within 1% (record, Apple A1/A2); rollout unlocked two months early off the review (record).
- **Customer** — the review carried the graduation criteria and CSAT ≥ consignee average into the rollout decision — the customer lens was on the Monday page (record).
- **Quality** — a truncated 99.4% week (three touches) discarded and printed as a data miss; a −45% week (40.0% → 21.9%, 17 → 31 Jan 2022) traced to a carrier format change, not operators; definitions frozen and later moved into code (record).

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No product-analytics function existed. I defined the questions, the metrics and the denominators, and chose the owners — then explained the headline, not just reported it.
Scope & influence	Seven owners across Data Services, Support, GTM and engineering signed definitions; a VP decided in the room — influence one to two levels up from a program seat.
Execution trade-off + mechanism others now own	Stability over coverage: seven metrics, a parking lot, no mid-flight redefinition. Mechanism others own: the frozen definitions, the 15-day rule and the ±5% rule, now in code (the Snowflake pipeline, S16-A).
Impact (long-term)	A cost-to-serve story leadership could act on: which lever moved it, which to fund next, and a rollout paced by evidence.
Process improvement (standard work institutionalized)	Standard work for the weekly review: frozen definitions, honest denominators, input→output tags, every red decomposed with an owner and a dated fix, verified next Monday (PDCA).
Complexity & consensus	Owners wanted their metric on the page; I held seven and gave them a parking lot. Sales wanted to board everything; the review carried the gates.
Root cause at scale	Reds traced to source, definition or operator before escalation — the −45% week was a carrier format change; the 99.4% week was three touches.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Three grains from SV_28.xlsx (event 19.4% · ticket queue 54.1% · weekly stock 40.2% → 69.2%); ticket ↔ container join; cuts by coverage, SLA and 4PL vs 3PL; what it revealed: a two-lever decomposition, a measurement-caused red, a small-N week.
2 · Solution design	Copy the LSA format vs invent; freeze seven vs expand; 15-day active vs cumulative denominator; ±5% week-over-week vs raw levels.
3 · Math & data logic	3.67 touches → 22.0× = 6.0× × 3.67×; order-dependent attribution ($0.55/$0.08 vs $0.48/$0.15) → fair split 56/44; Wilson interval on N = 3 → [0.44, 1.0]; $50.4K + $20.8K = $71.2K modeled.
4 · Implementation	Monday standard work: build, decompose, owner + dated fix, verify next Monday; owners sign; parking lot; hand-build for months, then pipeline parity within 1%.
5 · Probe questions	Six numeric probes plus three trap drills; Table 5 recomputes the real LSA workbook (relative vs absolute WoW, a constant "weekly" metric, a dubious-data flag).

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

Grain and joins. The instrument was the container-status-history extract (SV_28.xlsx, week ending 28 Feb 2022): the grain was the container-status *event*, not the shipment, rolled up to a weekly stock. Three grains, one formula family, never blended: **event** 1,991 of 10,289 human-touched = 19.4%; **ticket queue** 538 auto vs 456 human = 54.1% auto; **weekly stock** (the headline) 40.2% on 13 Dec 2021 → 69.2% on 21 Feb 2022 RECORD (APPLE V11 A2). Sources: Zendesk tickets (a ticket is created when Visibility fails to auto-initiate tracking), the container pull from Crux, platform tracking data and Jira blocker/critical counts RECORD-FROM-DOCS. The join is ticket ID ↔ container number; tickets carried no container ID, so the first versions matched by hand — the gap S16-A closes with a cross-reference table.

Cuts. Coverage automated vs with human input; tickets and tickets per *active* container; SLA for 4PL vs 3PL and the delta between them; 3PL + 4PL vs 4PL only (the E-Team ask). Every cut is an output metric tagged to the inputs that move it.

What the analysis revealed. (1) The headline was two multiplied levers: 22.0× = 6.0× (containers needing a human 60% → 10%) × 3.67× (touches per touched container 6.11 → 1.67) — a *modeled* split on record endpoints. (2) The attribution is order-dependent, so I report the fair split (average of both orders): ≈56%

coverage, ≈44% touch efficiency. (3) A −45% week-over-week red — 40.0% on 17 Jan to 21.9% on 31 Jan 2022 — traced to a carrier format change, measurement not operators RECORD. (4) The 21 Feb print of 69.2% moved with the denominator (active book 287 → 546 the week a big client's 283 containers hit), and the 28 Feb print of 99.4% rested on three touches — a truncated week, discarded RECORD.

CUT	QUESTION IT ANSWERS	INPUT OWNER TYPE
Coverage auto vs with human	Is the machine doing the work, or the desk?	Data Services / engineering
Tickets per *active* container	Is cost-to-serve falling per unit of live freight?	Support / Data Services
4PL vs 3PL SLA and delta	Do we serve non-Flexport freight as well as our own?	Visibility / Data Services
Jira blocker/critical WoW	Which defects explain the reds?	Engineering managers

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Review format	Invent a new format · copy the LSA weekly review	Copy	It already had leadership's trust; inventing one would have spent credibility I did not have (canon).
Metric set	Everything anyone asks for · freeze seven + parking lot	Freeze seven	A number that changes definition cannot steer a rollout — stability over coverage (canon).
Denominator	Cumulative containers · touched this week · active = cumulative inflow less containers 15 days past departed terminal	15-day active	The honest denominator for per-container rates; cumulative flatters. Defined in the workbook as cumulative inflow less containers 15 days past departed terminal (Apple A2) VERIFY-WITH-RAJESH · V2
Reading rule	Raw levels · week-over-week (B − A)/A with ±5% red/green	WoW ±5%	Same rule as the review leadership already read; a red needs an owner and a dated fix (canon).
Small weeks	Print every week · mark insufficient data	Mark a data miss	The interval on three touches is wider than the band the number is read against.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Touches per container at the baseline	`$0.66 / $0.18 = 3.67`	record; $0.18 per interaction is the record cross-walk	≈3.7 touches per container
2	Containers needing a human	`1 − 0.40 = 60%`	record (automation coverage 40%)	60%
3	Touches per human-needed container	`3.67 / 0.60 = 6.11`	modeled — assumes every touch lands on a non-automated container	≈6.11 touches
4	By-hand build time	`about four hours a week at its worst (my own account); the BRD as-is is 1.5–2 h per snapshot — a different measure, not stacked`	record (Apple A1/A2); BRD figure labeled record-from-docs	≈4 h/wk at its worst

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
5	Three grains, three honest numbers	`event: 1,991 / 10,289 = 19.4% human-touched (80.6% auto); ticket queue: 538 / (538 + 456) = 54.1% auto; weekly stock: 40.2% (13 Dec) → 69.2% (21 Feb)`	record — Apple A2, extract SV_28.xlsx; the ratios are my arithmetic	one formula family, three grains — never blended
6	The denominator moved under the headline	`active book 287 → 546 (21 Feb, a big client's 283 containers hit): (546 − 287)/287 = +90.2%`	record inputs (Apple A2); the ratio is my arithmetic	a rate can jump because the base did

Table 2 — The fix: the instrument and its rules, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Seven frozen metrics	`2 coverage + 2 cost-to-serve + 3 timeliness = 7`	record (BRD metric list)	7 definitions, 7 owners
2	Week-over-week rule	`(B − A)/A = (21.9 − 40.0)/40.0 = −45% → |−45%| > 5% → red`	record: the ±5% rule is the LSA review (Instructions row 9); 40.0% on 17 Jan and 21.9% on 31 Jan 2022 are the printed weekly values (Apple A2)	red; root cause a carrier format change
3	Small-N rule	`Wilson lower bound for 3 of 3 = 3 / (3 + 1.96²) = 0.44 → interval [0.44, 1.00]`	record: 99.4% on three touches (Apple A2); the Wilson interval is my arithmetic	width 0.56 vs a 0.10-wide band
4	Sample floor I would write down now	`n = 1.96² × 0.90 × 0.10 / 0.05² = 138.3 → 139 active containers for ±5 points at 90%`	illustrative threshold — not in the record (what I would add)	≈139 containers
5	Honest denominator	`Active_d = cumulative inflow − containers 15 days past departed_terminal; rate = human-touched containers ÷ Active_week; automation = 1 − rate`	record rule (Apple A2)	every per-container metric

Table 3 — After: the metric, before → after, and the decomposition

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cost-to-serve	`1 − $0.03/$0.66 = 1 − 0.0455 = 0.9545 → I say "about ninety-five percent"`	record endpoints (Apple A1/A2); ratio mine	$0.66 → $0.03, about 95%
2	Coverage lever	`human-needed share 60% / 10% = 6.0×`	record endpoints (40% → 90%)	6.0×
3	Touch-efficiency lever	`6.11 / 1.67 = 3.67× (1.67 = $0.03 / $0.18 / 0.10)`	modeled	3.67×
4	Identity check	`6.0 × 3.67 = 22.0 = $0.66 / $0.03`	arithmetic	22.0×
5	Attribution, coverage first	`$0.66 − 0.10 × 6.11 × $0.18 = $0.66 − $0.11 = $0.55 from coverage; $0.11 − $0.03 = $0.08 from touches`	modeled	$0.55 / $0.08
6	Attribution, touches first	`$0.66 − 0.60 × 1.67 × $0.18 = $0.66 − $0.18 = $0.48 from touches; $0.18 − $0.03 = $0.15 from coverage`	modeled	$0.48 / $0.15
7	Fair split (average of both orders)	`coverage ($0.55 + $0.15)/2 = $0.35; touches ($0.08 + $0.48)/2 = $0.28; $0.35 / $0.63 = 56%, $0.28 / $0.63 = 44%`	modeled	56% / 44%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
8	Scale	`68 / 7 = 9.7×; automation 40% → 90%; accuracy 60% → 90%; 170 monthly users is the public target, landed`	record	9.7× — the only multiple to say

Table 4 — The dollars, every multiplication shown

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Unit saving	`$0.66 − $0.03 = $0.63 per container`	record endpoints	$0.63
2	Containers per year at the 32,000-shipment target	`32,000 × 2.5 = 80,000 (range 48,000–128,000 at 1.5–4.0)`	the shipment count is a target, not an outcome; containers per shipment illustrative	80,000
3	Direct cost-to-serve saving	`80,000 × $0.63 = $50,400`	modeled	≈$50K/yr
4	Split by the fair attribution	`80,000 × $0.35 = $28,000 (coverage); 80,000 × $0.28 = $22,400 (touch efficiency)`	modeled	$28,000 / $22,400
5	Review labor by hand	`4 h/wk × 52 = 208 h; 208 × $100 = $20,800 ($16,640 at $80; $24,960 at $120)`	modeled; 4 h/wk is the at-its-worst figure, so an upper bound; rate is a benchmark range	≈$21K/yr
6	Total modeled run-rate	`$50,400 + $20,800 = $71,200`	modeled — master sensitivity base ≈$71K	≈$71K/yr
7	One-time revenue pull-forward (excluded from the run-rate)	`61 × $25,000 × 2/12 = $254,167`	modeled; ARR per customer is an illustrative benchmark; the pilot itself is free	≈$254K one-time

Table 5 — The review format run on a real workbook: LSA WBR 2022, FFA tab (my pandas / openpyxl recompute; the FFA slide is another team's, used to show the mechanics)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Exceptions per shipment, Wk 35 → Wk 1	`(2.900821 − 4.068702) / 4.068702 = −28.7% (lower is better, so favorable)`	record: sheet FFA row 11, cols Wk 35–Wk 1; arithmetic mine	−28.7% in 18 weeks
2	Same metric, monthly view	`Apr 6.0 → Sep 4.0: (4.0 − 6.0) / 6.0 = −33.3%`	record: FFA row 11, monthly columns	−33.3%
3	Red / green weeks under the ±5% rule	`over the 18 week-over-week steps: unfavorable (> +5%) = 1, favorable (< −5%) = 4, inside the band = 13`	my recompute of FFA row 11 under Instructions row 9	1 red · 4 green · 13 quiet
4	The holiday week	`import-partner on-time share Wk 51 → Wk 52: (0.290 − 0.459) / 0.459 = −36.8% → red; Wk 52 → Wk 1: (0.506 − 0.290) / 0.290 = +74.5%`	record: FFA row 5; the owner's callout attributes Wk 52 to the holiday season	a red that arrives with a cause
5	Mean with and without the holiday week	`nine weeks: 4.2785 / 9 = 47.5%; without Wk 52: (4.2785 − 0.290) / 8 = 49.9%`	record inputs FFA row 5 (Wk 45–Wk 1); mean mine	47.5% vs 49.9%
6	Written rule vs printed column	`the Instructions say WoW = (B − A)/A; the WoW cell for Wk 52 → Wk 1 prints 0.216 = 0.506 − 0.290 — an absolute difference; under the rule it is +74.5%`	record: Instructions row 8 and FFA row 5 WoW cell; my observation	definition drift to flag, not to fix silently
7	A "weekly" metric that never moves	`import tasks structured into work items = 0.1346 in all 19 weeks; 7 / 52 = 0.1346`	record: FFA row 12; the 7 and 52 are my inference, the sheet does not say	static, not weekly — mark it

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
8	Data the owner flagged themselves	`notification read-rate Wk 43 → Wk 1: (0.050 − 0.0725) / 0.0725 = −31.0%, beside the sheet's own note: "data looks dubious and will double check"`	record: FFA row 16 and its note; arithmetic mine	a data miss until checked — the same rule as the 99.4% week

The ~$1.5M pre-production measurement saving (S11) is a separate user-anchor figure and is excluded from every row above. Automation is 40% → 90%; the loop packet's "60 → 97" is wrong (§1.3) — do not mix them in the room.

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

CONTROL	STANDARD WORK	OWNER	LEAN / KAIZEN WORD
Definitions	Seven written definitions, signed, frozen; new metrics enter a parking lot, never the page	Each input owner	Standard work
Weekly build	Pull four systems → join ticket ↔ container → seven metrics on the 15-day active base → WoW ±5%	Me, by hand, then the pipeline	Takt: the Monday cadence
Red handling	Decompose to source / definition / operator before the meeting; owner + dated fix; verify the same number next Monday	Metric owner	Root cause (5 Whys), PDCA
Data honesty	A truncated or tiny-N week is a data miss, never a result	Me	Jidoka — stop the line on bad data
Hand-off	Pipeline parity within 1% for consecutive weeks, then retire the manual build (S16-A)	Infra + me	Remove waste (motion) and the single point of failure

Risks and mitigations. Measurement-caused reds (decompose first); a flattering wrong number (data-honesty rule); a single point of failure (the manual review); an expanding page (parking lot). **Rollout of the review itself:** dry-run deck → two consecutive clean Mondays → the VP takes a decision in the room → the review becomes the standing forum. A review that skips a week teaches leadership it can be skipped, so it ran every Monday.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Is the 6.0× / 3.67× split real, or an artifact of your assumption?**
>
> It is a model, and I say so. The endpoints are record: $0.66 → $0.03 and coverage 40% → 90%. The split assumes every touch lands on a non-automated container; if exception handling touches automated containers too, the touch-efficiency lever is understated. The honest convention is the average of both attribution orders — 56% coverage, 44% touches — and the order-dependence itself (coverage's share runs 24%–87%) is why I call it a convention, not a measurement.

> **How do you know the −45% week was not operators?**
>
> Because I decomposed it before the meeting: source, definition, operator, in that order. The drop traced to a carrier format change — measurement. The owner took a dated fix and I looked at the same number the next Monday. The weekly prints were 40.0% on 17 Jan and 21.9% on 31 Jan 2022; the arithmetic is (21.9 − 40.0)/40.0 = −45%.

> **Why a 15-day active window, and what would change it?**
>
> It is the honest denominator for per-container rates — cumulative containers would let the base swamp the signal — in the real extract the active book went 287 → 546 in the week one big client's 283 containers hit, and the headline moved from the low 40s to 69.2% with it. Fifteen days past departed terminal was frozen with the owners. What would change it: showing the rate at 10, 15 and 30 days and freezing the window where the rate stops moving VERIFY-WITH-RAJESH · V2.

Three touches — why discard instead of caveat?

Because the interval was wider than the band. At 3 of 3 the 95% Wilson interval is 0.44 to 1.00 — 0.56 wide — against a ±5% (0.10-wide) red/green band. A number you cannot read against its own threshold is not a reading. To read a rate to ±5 points at 90% you need about 139 active containers.

You froze seven metrics. What did that cost?

Coverage of the page — engineering wanted more end-to-end input metrics, and the review parked them. The trade was stability: a number that changes definition cannot steer a rollout. The parking lot was real, and the pipeline (S16-A) is where the extra inputs could be added without touching the seven.

What decisions did the review actually change?

Three, all record: waves boarded only through the published criteria; wave 3 held in public on an ETA-accuracy miss traced to one bad source; the VP unlocked the 68-client rollout two months early off the review's numbers. The E-Team's improvement rule — rank by cost-to-serve reduction first, ease second — is the same Pareto the reds were sorted by.

BUSINESS IMPACT MODEL — KPI TREE

```
Leadership steers rollout pace and funding off a number that survives the next Monday
+- OUTPUT  cost-to-serve per container   $0.66 -> $0.03   (record, about 95%)   22.0x
|    +- INPUT  containers needing a human   60% -> 10%    (coverage 40% -> 90%, record)   6.0x
|    +- INPUT  touches per touched container 6.11 -> 1.67 (modeled)                        3.67x
|    +- fair split of the $0.63 saving: coverage ~56% / touch efficiency ~44% (modeled)
+- OUTPUT  ETA accuracy 60% -> 90%; customers 7 -> 68 (9.7x)   (record)
+- TAGS    Jira blocker/critical trend  <->  the input metric it explains
+- DENOM   active = inflow less containers 15 days past departed terminal; tiny-N weeks = data miss
(loop)   red -> owner -> dated fix -> same number verified next Monday -> trust -> decision taken in the
room
```

SOLUTION FLOW

```
Sun night   pull 4 systems (tickets, containers, platform, Jira)
   -> join ticket <-> container (xref until the real key exists)
   -> 7 metrics on the 15-day active base, container-week grain
   -> WoW = (B - A)/A ; flag beyond +/-5%
   -> decompose each red: source? definition? operator?   [5 Whys]
   -> owner + dated fix   -> Monday review: decision taken in the room
   -> next Monday: same number verified   [PDCA: Check / Act]
   -> pipeline parity within 1% for consecutive weeks -> manual build retired
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Containers per shipment (illustrative)	1.5	2.5	4.0
Containers per year at 32,000 shipments	48,000	80,000	128,000
Direct saving ($0.63 × containers)	$30,240	$50,400	$80,640
Review labor by hand ($80 / $100 / $120 per hour × 208 h)	$16,640	$20,800	$24,960
Total modeled $/yr	≈$47K	≈$71K	≈$106K

Revenue pull-forward ($254K) is one-time and the ~$1.5M pre-prod estimate is a separate user anchor; both are excluded from the run-rate.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

Cost-to-serve fell from sixty-six cents to three cents a container — twenty-two times. When I took it apart it was two levers multiplied: six times fewer containers needing a human, sixty percent down to ten, and about three and two-thirds times fewer touches on each one that still did. That split is my model — it assumes the touches fall on the non-automated containers — and the attribution depends on the order you take it, so I report the average, roughly fifty-six percent coverage and forty-four percent touch efficiency, and I call it modeled. At the thirty-two-thousand-shipment target the direct saving is about fifty thousand dollars a year on an illustrative container count, plus about twenty-one thousand of review labor I stopped doing by hand — both modeled. What is record is the sixty-six cents, the three cents and seven to sixty-eight; what the review added is that every red arrived with an owner and a dated fix, and that when a week printed ninety-nine point four percent on three touches I printed a data miss instead.

IF CHALLENGED — THE FULL ANSWERS

"Your 22× decomposition is just algebra."

The product identity is algebra; the analysis is knowing it needs an attribution convention. If touches per touched container had stayed flat, the whole result would have been coverage; because both moved, the order matters, so I show both orders and report the average. The use is practical: it tells the review which lever to fund next.

"Why throw out a number that looked good?"

Three touches gives an interval of 44% to 100% — wider than the ±5% band the review reads against. Printing it would have taught the VP a number that could not survive the next Monday, and a review that flatters once is not trusted twice.

"An early note claims a far bigger growth multiple than 7 → 68. Which is it?"

The record is 7 → 68, 9.7×, against public targets of 68 customers and 170 monthly active users — both landed two months early — and a 32,000-shipment tracked target. The larger multiples are from an early draft note and are not canon; I do not use them.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I ran the weekly review that steered a visibility rollout: seven frozen metrics, a 15-day active denominator, ±5% red/green. I took the 22× cost-to-serve drop apart — 6× fewer containers needing a human, 3.7× fewer touches on each — and threw out a 99.4% week built on three touches.

NACF BRIDGE — ONE SENTENCE

The same review shape runs an inbound network — doors × hours × unload rate on one side, UPH and cubic fill on the other, each with a frozen weekly definition — so when door utilization crosses the ~85% knee or weeks-of-cover leaves its 2–4 week band, the red arrives with an owner, a dated fix and an honest denominator (doors in use, not doors installed).

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a problem you had to solve that required in-depth thought and analysis. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?" AIQB: Dive Deep Q3 **Dive Deep** FACET: DATA ANALYSIS

Open (first two sentences): "A cost-to-serve number fell ninety-five percent and nobody could tell me why, so I took it apart. It was two multiplied levers, not one."

90-second STAR. S Leadership doubted the program, and the headline was $0.66 → $0.03 per container. **T** Explain what moved it, so the review knew what to fund next. **A** I converted dollars to touches at the record $0.18 per interaction — 3.67 down to 0.17 — then split coverage (containers needing a human, 60% → 10%, 6.0×) from touches per touched container (6.11 → 1.67, 3.67×). I noticed the attribution changes with order — $0.55/$0.08 coverage-first, $0.48/$0.15 touches-first — so I reported the average, 56/44. **R** The split was modeled and I said so; the review had a shared way to talk about where the next fix should go, and the record endpoints stayed untouched.

Land: 22.0× = 6.0× × 3.67× · **Pivot to NACF:** the same move on dock-to-stock cost: split a throughput gain into door-hours opened and unload rate before deciding which to fund.

Do NOT say: that the 56/44 split was measured; any customer-growth multiple other than 9.7×; any dollar from the split as record.

B2 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 **Insist on the Highest Standards** FACET: DEFINITIONS AND DENOMINATORS

Open (first two sentences): "Every owner wanted their metric on the page, and I froze seven and parked the rest. A number that changes definition cannot steer a rollout."

90-second STAR. S A weekly review drifts: metrics get added, definitions bend, the page stops meaning anything. **T** Keep the review decision-grade for a VP who would steer funding off it. **A** I wrote seven definitions, got each input owner to sign the one they would be red against, fixed the denominator at 15-day active, offered a parking lot for new metrics and held the line after sign-off. **R** Every red arrived with an owner and a dated fix; the same seven definitions later moved into code with no drift.

Land: seven metrics, 15-day active · **Pivot to NACF:** freeze door-utilization and weeks-of-cover definitions before the quarter, signed by the node owner, so a red means the same thing in every building.

Do NOT say: "I invented the WBR" — it was copied from the LSA review, which was the point.

B3 · "Tell me about a time when you didn't have enough data to make the right decision. What did you do? What path did you take? Did the decision turn out to be the correct one?" AIQB: Are Right, A Lot Q1 **Earn Trust** FACET: DATA HONESTY

Open (first two sentences): "I had a week that printed ninety-nine point four percent and I threw it out. It rested on three touches, so I printed a data miss in its place."

90-second STAR. S The program was being doubted, and a flattering week landed. **T** Protect leadership's trust in the number over the next month, not the next meeting. **A** I computed the interval: 3 of 3 gives a 95% Wilson interval of 0.44 to 1.00 — wider than the ±5% band. I told the VP it was a data miss, not a result. **R** The number never had to be walked back, and the critic who started this later steered by the review.

Land: N = 3 → interval 44%–100% · **Pivot to NACF:** a unit-fill or WOC reading built on a handful of trailers or SKUs gets marked insufficient volume, not headlined.

Do NOT say: what the "real" number was — I do not have one and will not invent it.

B4 · "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?"

AIQB: Dive Deep Q2 | Are Right, A Lot | FACET: ROOT CAUSE ON MEASUREMENT

Open (first two sentences): "The weekly automation rate fell from forty percent on the seventeenth of January to twenty-one point nine on the thirty-first — minus forty-five, deep red. Before the meeting I decomposed it, and it was a carrier format change, not the operators."

90-second STAR. **S** A red week landed on a desk that could have been blamed. **T** Find the cause before anyone defended or accused. **A** The standard work: source, definition, operator, in that order. The source had changed format; I assigned the owner a dated fix and checked the same number the next Monday. **R** The red arrived with an owner, the operators were not blamed, and the order of questions became standard work.

Land: (21.9 − 40.0)/40.0 = −45% — printed weekly values 17 Jan → 31 Jan 2022 (record) · **Pivot to NACF:** a sudden UPH or weeks-of-cover swing gets the same three questions — source, definition, operator — before an ops root cause is declared.

Do NOT say: the carrier name; that operators had slipped.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Cost-to-serve fell 22× and I can tell you which lever did it — and why the answer depends on the order you ask."

Headline metric: 22.0× = 6.0× coverage × 3.67× touch efficiency; fair split 56% / 44% (modeled)

- Grain: container-week; join ticket ↔ container; 15-day active base.
- Identity: 6.0 × 3.67 = 22.0.
- Order test: $0.55/$0.08 coverage-first vs $0.48/$0.15 touches-first.
- Say "modeled" once, early; the record is the endpoints.

Conflict / influence / backbone angle — Insist on the Highest Standards

Opener: "The owners wanted their metric on the page. I held seven and gave them a parking lot — and a signature line."

Headline metric: seven frozen metrics, each signed by the owner who would be red against it

- The objection: more end-to-end input metrics, now.
- The answer: stability over coverage; a changing definition cannot steer a rollout.
- The mechanism: sign-off, freeze, parking lot — no argument by seniority.
- The proof: the same definitions moved into code with no drift.

Honest failure-and-learning angle — Ownership

Opener: "I started the review after the critique, not at kickoff — and for months I was its single point of failure."

Headline metric: by hand for months, about four hours a week at its worst; no longer a single point of failure once the pipeline matched within 1%

- What I would do differently: instrument at kickoff, not after the program fell from #3 to #14 of 16.
- What I would add now: a written sample floor (~139 active containers for ±5 points at 90%) instead of a judgment call on each small week.
- What fixed it: definitions in code (S16-A), not heroics.
- No dollar claimed for this angle.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"How do you know the drop was your review and not the automation teams — or seasonality?"

Numeric answer. I do not claim the review built the automation; nine teams did (coverage 40% → 90%). The review's claim is narrower: it made boarding follow coverage — 9.7× the customers, zero rollbacks, both targets two months early. Cost-to-serve is a per-container ratio, so seasonal volume does not move it; a mix shift could, and I would check that by lane.

Say exactly: "I don't claim the review made the automation — the teams built that. The review's claim is narrower and checkable: it made boarding follow coverage, and the record shows zero rollbacks at 9.7 times the customers."

"Your decomposition rests on an assumption. What number would change your mind?"

Numeric answer. The endpoints do not depend on it; only the split does. Depending on attribution order coverage's share runs $0.15 to $0.55 of the $0.63 saving — 24% to 87% — so no single split is a measurement. Ticket-grain data showing touches on automated containers would move me to a two-factor model with an interaction term.

Say exactly: "The endpoints are record; only the split is modeled — and the split runs from twenty-four to eighty-seven percent depending on order, so I report the average and call it a convention."

"Three touches — why not just report it with a caveat?"

Numeric answer. The 95% interval was 0.44 to 1.00 — 0.56 wide — against a 0.10-wide red/green band, 5.6 times wider than the threshold. A caveat does not fix a reading the threshold cannot interpret. To read a rate to ±5 points at 90% I need about 139 active containers.

Say exactly: "A number whose interval is wider than the band you read it against isn't a reading. At three touches the interval was forty-four to a hundred; to read ninety percent within five points you need about a hundred and forty."

S11-B — Flexport — Launch readiness as standard work: four test gates, a 100-container hop-by-hop ledger and six graduation criteria — and the pilot that launched in December 2021, later than the first testing plan NEW

LPs this story can serve here: Insist on the Highest Standards (primary) · Customer Obsession · Dive Deep · Learn and Be Curious (the honest miss)
Process family: Deliver — launch readiness, test strategy and staged rollout; entry / exit criteria and go / no-go governance

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Insist on the Highest Standards Business UAT did not open until 80% of shipments showed minor-to-no issues, and no wave graduated until six published criteria passed — the gate was data quality, not the calendar.

Customer Obsession Shadow accounts ran two real Crux clients' live data beside the old product, and every new client supplied at least 50 containers of its own freight to calibrate on before onboarding.

Dive Deep The 100-container ledger traced real containers hop by hop; of 59 failing rows, 41 (69.5%) first broke at route creation or the milestone on screen — two hops, not seven.

Learn and Be Curious My test-effort estimates were optimistic and monitoring arrived late, so the pilot launched later than the first testing plan and bugs took hours to localize across six-plus layers — the lesson is to size testing on hop count and instrument first.

Honesty label. One of two technical program managers on the program (the record also names a co-TPM and a DRI). I owned the gate definitions, the readiness read-out and the go/no-go inputs; the 100-container log was run day to day by the QA team named in the testing approach, the hop-by-hop design is credited in the secondary loop packet to a test lead, and business users ran UAT. My own account (Apple v11 A5) is that I ran decision week as a two-gate go/no-go with criteria published beforehand, made the 100-container run the UAT spine with a Jira ID on every missed row, and set the UAT entrance criteria as exposure gates; I say "I defined the gates and ran the read-out", not "I ran the test". The ledger numbers below are *my recount of the 100-container working log* — a partial log where blank means not recorded.
Timing, doc against doc: the testing-approach document planned an early-November pilot; the program's year-end summary shows launch in December 2021 with delays among its misses; the pilot roll-out schedule has Wave 1 in Q4 2021, and my own workbook calls go-live "on schedule". I therefore say "later than the first testing plan", never "on time" and never a specific number of weeks late.

WHAT THE RECORD SAYS ABOUT READINESS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

Four test gates. A "100 container test" (random container and MBL numbers traced and logged, bugs in Jira, daily stand-ups); bug bashes every other week with job-story ranges assigned per person; shadow client accounts (client data uploaded to the legacy product fed into dedicated demo accounts, starting with two named Crux clients); and UAT with semi-weekly syncs, dependent on data quality first — 80% of shipments with minor-to-no issues. RECORD-FROM-DOCS

Go / no-go and waves. Two go/no-go meetings (business and tech) read: stable production, first-client data quality 80%+, no show-stoppers. Waves then graduated on six criteria — verification SLA 90% (max 48 h in "Verification in progress"), data quality 90%+ at least as accurate as the legacy product, milestone estimates 80%+, stable production, at least 100 containers plus key features used, and CSAT on par with the consignee average — with at least 50 containers of a new client's own data to calibrate on first, a free pilot and light-weight onboarding. RECORD-FROM-DOCS

Where it landed. The Nov 2021 status read Yellow with both checkpoints passed, a performance test of 500 containers in 20 minutes at 99.6% connectivity accuracy, end-to-end 100% processed and 86% trackable on 11/18, job stories 98.4% complete. The year-end summary: launched in December 2021, nine teams and close to 100 people, three environments built — and the misses: delays, optimistic testing estimates, and monitoring implemented late (bugs took hours to localize across six-plus system layers). RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport's shipment-visibility pilot was first planned for early November 2021 and launched in December. It ran on a new stack across nine teams and close to a hundred people, with data crossing six or more system layers before it reached a client's screen. **Task.** As one of the program's two TPMs, I owned the readiness gates: what had to be true before a client touched the product, and before each wave graduated. **Action.** We tested in four layers — a 100-container live test, biweekly bug bashes, shadow accounts that ran two real clients' data beside the old product, and business UAT that opened only when 80% of shipments showed minor-to-no issues. The 100-container ledger traced real containers hop by hop — created, subscribed, data back, route, milestone, exception, webhook — and its Pareto was blunt: of 59 failing rows, 41 first failed at route creation or the milestone on screen. Graduation was six published criteria, with at least 50 containers of the client's own freight to calibrate on first. **Result.** Primary metric: the gate held — a 500-container load test in 20 minutes at 99.6% connectivity, both go/no-go checkpoints passed, no wave rolled back. The honest part: my test-effort estimates were optimistic and monitoring arrived late, so the pilot launched later than first planned and bugs took hours to localize; I now size testing on hop count and put the monitors in first.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Of fifty-nine failing rows in the hundred-container log, forty-one — about seventy percent — first broke at two hops, route creation and the milestone on screen; that is my recount of a working log, so it tells you where to look, not a defect rate.”

“What's your source for that?” “The pre-launch evidence is in the program's own status documents — five hundred containers in twenty minutes at ninety-nine point six, the dry run a hundred percent processed and eighty-six tracked — and the hop recount is mine from the log file.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The visibility pilot had to go live to Crux-grade clients on a brand-new stack: nine teams, close to 100 people, three environments, data traveling through six-plus system layers before it reached the Client App. The testing plan targeted an early-November pilot; the status record shows end of November and the year-end summary shows December 2021 (the roll-out schedule keeps Wave 1 inside Q4, and my own workbook records go-live on schedule — the documents disagree on whether December was late). The program's own retrospective calls the testing estimates "overly optimistic for the complexity of the product" and names monitoring as implemented too late.

T — Define "ready": the gates before a client touches the product, the criteria to graduate each wave, and the evidence the go/no-go checkpoints would read — so rollout followed data quality, not the calendar.

Baselines on the table: First testing plan early Nov 2021 → launched Dec 2021 (docs disagree whether late) · 9 teams / ~100 people · 6+ system layers · 3 environments · 4 test gates · 6 graduation criteria · ≥50 containers per new client to calibrate · UAT entry gate 80% minor-to-no issues · ledger: 474 containers listed, 142 marked tested, 105 with hop results.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — defined "working" per container as seven checks (created, subscribed, data back, route, milestone, exception, webhook); read the product standards (coverage 90%+, accuracy 95%, timeliness 90%) and what each go/no-go would read. *Exit bar:* seven checks and four gates named; the questions the go/no-go answers written down.
- **Plan & Design** — four test gates; the UAT entry gate at 80% minor-to-no issues; six graduation criteria published before wave 1; at least 50 containers of each new client's own freight before onboarding; free pilot, light-weight onboarding, monthly go/no-go. *Exit bar:* criteria published to Sales and the business before wave 1.
- **Build & Test** — the 100-container log in daily stand-ups; bug bashes by job-story range; shadow accounts on real Crux data; production connectivity monitors for the hops; a trace ID and per-hop ratios proposed after the slow bug

hunts — the logs showed no unique ID tying events from downstream to the screen, so finding the source system took 2–4 hours per blocker; the recommendation cut blocker-bug time-to-resolve by about half (3PL ingestion one-pager). *Exit bar:* 500 containers / 20 minutes / 99.6%; end-to-end 100% processed, 86% trackable on 11/18.

- **Cutover & Go-Live** — two go/no-go checkpoints (business, tech) on stable production, first-client data quality 80%+ and no show-stoppers; wave 1 started in December with two hand-carried customers (the schedule and my workbook agree on those two; two other accounts ran as shadow UAT accounts, and the Nov 19 status note's "first pilot client" is the earlier plan — I say "the first two hand-carried customers", no names, unless asked). *Exit bar:* both checkpoints passed; wave 1 live.
- **Hypercare / Close** — feedback session about a week after kickoff; a 50-container calibration sample from each wave-2 client before onboarding; monthly go/no-go on the six criteria; production data-quality testing on the same per-hop ratios. *Exit bar:* wave 2 opened on wave 1's graduation read; zero rollbacks.

Influence and ownership. Mine: the definition of ready, the gate and criteria design, the go/no-go inputs and the pre-production measurement design (S11). The QA team ran the log, a test lead designed the hop columns, the engineering teams instrumented their hops and fixed what the log found, business users ran UAT, and product and engineering leadership took the go/no-go.

> **Designed the process; others own it now.** The gates and criteria are published artifacts, the ledger columns are the standard checklist, and the per-hop ratios became standing monitors — the launch reads off documents, not off me.
> **Trade-off put to leadership.** Launch date versus data quality: I recommended the data-quality gate and accepted the later date it implied.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — no record dollar for the testing layer. The ~$1.5M pre-production estimate is S11's separate user anchor and is never merged. An illustrative escape cost at the record $0.18 per touch is under $1K a year — the real cost of an escaped defect is the rollback and the first impression, which is what the gates bought.
- **#** — four gates; six graduation criteria; 500 containers in 20 minutes at 99.6%; end-to-end 11/18 100% processed, 86% trackable; job stories 98.4% complete; both go/no-go checkpoints passed (record-from-docs).
- **%** — UAT entry gate 80% minor-to-no issues; graduation bars 90% data quality, 80% milestone estimates, 90% verification SLA; ledger first-failure Pareto 69.5% in two hops (my read of the log).
- **Time** — pilot launched December 2021, later than the early-November testing plan (record-from-docs; the docs disagree whether it was late); bug localization took hours across six-plus layers (record); about half that after the trace ID (3PL ingestion one-pager: blocker-bug time-to-resolve −50%).
- **Customer** — shadow accounts on two real clients' data; ≥50 containers of each new client's freight to calibrate on; CSAT par with the consignee average inside the graduation gate (record-from-docs).
- **Quality** — zero rollbacks through the waves (record, S11 canon); every ledger miss ticketed to the owning team.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	"Ready" was undefined for a new-stack product. I defined the seven checks, the four gates and the graduation criteria, and what each go/no-go would read.
Scope & influence	Nine teams and ~100 people across three environments; I held the gates without owning any team — the criteria were published, the checkpoints were business and tech leadership's.
Execution trade-off + mechanism others now own	Date versus data quality: the UAT gate cost schedule and saved the first impression. Mechanism others own: the published criteria, the per-hop ratios as monitors.
Impact (long-term)	A launch with no rollbacks and a repeatable wave machine from 2 → 4 → 20+ clients; the monitors outlived the launch.
Process improvement (standard work institutionalized)	Standard work for launch: entry gate, calibration volume per client, six graduation criteria, monthly go/no-go, per-hop ratios.
Complexity & consensus	Nine teams with separate backlogs, an acquired product's clients in the middle of the rollout, and a date everyone wanted; the gates made the trade explicit.
Root cause at scale	A Pareto of first-failing hop located the defects (route, milestone) instead of counting symptoms; the missing seam — a trace across hops — was the root cause of slow debugging.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	The 100-container ledger as a hop-by-hop Pareto: 142 tested, 105 with results, 59 failing, first failure at route (23) or milestone (18); partial-log caveat.
2 · Solution design	Hop ledger at the base of a four-gate pyramid; UAT gated on data quality; waves behind six criteria; calibration volume per client; light-weight onboarding.
3 · Math & data logic	Detection power: 1 − 0.95^50 = 92.3%; 1 − 0.97^100 = 95.2%; 1 − 0.99^300 = 95.1%; rule of three 3/100; load 500 / 20 = 25 per minute.
4 · Implementation	Four gates with owners and cadence; two go/no-go checkpoints; waves; monitors; three environments; risks and the late-monitoring lesson.
5 · Probe questions	Seven numeric probes plus three trap drills: zero-rollback luck, the partial log, the later launch and its leading indicator.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What the ledger is. One row per container or MBL; one column per hop — created, subscribed in SV, data back, route created, milestone shown, exception shown, webhook created — plus an overall Success flag, a date and the bug IDs RECORD-FROM-DOCS. Population as I read it: 474 containers listed, 142 marked tested, 105 with at least one hop result, 53 with an overall verdict (19 Yes, 34 No) — a partial working log; blank means not recorded, and twelve numbered issues sit in the bug column.

The right Pareto is first-failing hop, not No-counts per column. Downstream hops cascade — a failed route makes the milestone fail — so counting every "No" double-counts one fault. Of the 105 rows with results, 59 had at least one "No"; each was assigned to the first hop in order that failed:

FIRST FAILING HOP	ROWS	SHARE OF 59	CUMULATIVE
Route created?	23	39.0%	39.0%
Milestone shown in UI?	18	30.5%	69.5%
Data back from the legacy source	7	11.9%	81.4%
Exception shown in UI?	6	10.2%	91.5%

FIRST FAILING HOP	ROWS	SHARE OF 59	CUMULATIVE
Shipment created?	5	8.5%	100.0%

What it revealed. Two hops — route creation and the milestone on screen — hold 41 of 59 first failures (69.5%); the ingestion hops (created, data back) hold 12 (20.3%). Only 3 of the 105 rows passed all seven hops. So the defects were downstream of ingestion, which is where the bug bashes and the owning teams went first. Caveats I would say out loud: this is a working log, not a random sample, and I would re-run a fresh random 100 and log all seven hops to turn the Pareto into a defect-rate estimate. The Pareto is my recount of the 100-Container Debugging Session log, not a number printed in any document.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY
Test structure	End-to-end pass counts only · a pyramid with a hop-by-hop ledger at the base	Pyramid + hop ledger	End-to-end says that it broke, not where; the hop column routes the ticket to the right team the same day (the seven checks and the Jira ID on every missed row are mine per my workbook; credit the QA team for running the log)
UAT entry	Open on the calendar (mid-October) · open on a data-quality gate (80% minor-to-no issues)	Quality gate	Business testers should validate the product, not rediscover known bugs (record-from-docs).
Pilot entry	Board the pool at once · waves 2 → 4 of 7 → 20+ behind six criteria	Waves	A free pilot spends engineering, data and ops before revenue; the gate must fire before the next wave (record-from-docs; S11).
Calibration	Onboard, then measure · at least 50 containers of the client's own freight first	Calibrate first	The product holds data on the client's lanes by the time they onboard; 50 containers detect a 5% defect 92.3% of the time.
Onboarding weight	API integration and LOAs from day one · light-weight until about wave 2	Light-weight	Limits the client's investment if the product misses expectations (record-from-docs).
Monitoring	Debug as you go · instrument the hops first (trace ID, per-hop ratios, connectivity monitors)	Instrument first — I did it late	The program's own retrospective: bugs took hours to localize at six-plus layers; the lesson is the story's honest half.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Ledger population	`142 marked tested / 474 listed = 30.0%; 105 with hop results / 142 = 73.9%`	record-from-docs; my count of the working log (blank = not recorded)	142 of 474; 105 with results
2	Overall verdicts	`19 Yes / 53 verdicts = 35.8%`	record-from-docs; partial log	35.8% passing at the time logged
3	Rows with a failing hop	`59 / 105 = 56.2%; all seven hops Yes: 3 / 105 = 2.9%`	record-from-docs; my count	59 failing; 3 clean
4	Localization cost	`hours per blocker across 6+ system layers; 2–4 h in the loop-packet telling`	record-from-docs (hours); secondary source (2–4 h)	2–4 h per blocker

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
5	Plan versus actual	`testing-approach plan: early November 2021; launch: December 2021 (year-end summary)`	record-from-docs, doc against doc (month-level resolution)	later than the first testing plan

Table 2 — The fix: gates and detection power, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Calibration sample	`P(detect ≥ 1 defect) = 1 − 0.95^50 = 92.3% at a 5% defect rate`	record sample size (≥50 containers); defect rate illustrative	92.3%
2	Launch gate sample	`1 − 0.97^100 = 95.2% at a 3% defect rate`	record sample size (100 containers); defect rate illustrative	95.2%
3	If the 100 all pass	`upper bound = 1 − 0.05^(1/100) = 2.95% (rule of three: 3/100 = 3.0%)`	arithmetic	defect rate below ≈3% at 95%
4	Pre-production production test	`1 − 0.99^300 = 95.1% at 1% defects; 1 − 0.98^300 = 99.8% at 2%`	record sample size (300 containers, S11); defect rates illustrative	95.1% / 99.8%
5	Entry and graduation bars	`UAT opens at ≥80% minor-to-no issues; pilot entry at first-client data quality ≥80%; graduation at ≥90% data quality, ≥80% milestone estimates, 90% verification SLA`	record-from-docs thresholds	staged bar: enter safe, graduate proven

Table 3 — After: the metric, before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Load proof	`500 containers / 20 min = 25 per minute; 99.6% × 500 = 498 connected`	record-from-docs (Nov 2021 status)	25 per minute; 498 of 500
2	End-to-end on 11/18	`100% processed; 86% trackable → 100 − 86 = 14% not trackable`	record-from-docs	14% untrackable that day
3	Job stories	`98.4% complete → 100 − 98.4 = 1.6% outstanding`	record-from-docs	1.6% outstanding
4	Go / no-go	`2 of 2 checkpoints passed (business, tech); criteria: stable production, first-client data quality 80%+, no show-stoppers`	record-from-docs	launch cleared
5	Scale outcome	`68 / 7 = 9.7×; zero rollbacks`	record (S11 canon)	9.7×

Table 4 — The dollars, every multiplication shown

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Containers per year	`32,000 shipments × 2.5 = 80,000`	illustrative containers per shipment (S11)	80,000
2	Escaped defects at a 5% rate	`80,000 × 0.05 = 4,000 containers`	illustrative defect rate	4,000
3	Direct touch cost of the escapes	`4,000 × $0.18 = $720 per year`	record $0.18 per interaction; one touch each (illustrative)	$720 — tiny: the cost is the rollback, not the touch

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Localization labor (illustrative)	`5 blockers/wk × (3 h − 1.5 h) = 7.5 h/wk; 7.5 × 52 × $145 = $56,550`	illustrative: 5 blockers/wk is my example, not a source figure; 2–4 h per blocker and the ~50% cut are record-from-docs; $145/h from the S16-A model	≈$57K/yr — priced, not claimed; say the hours, not this dollar
5	Pre-production measurement path	`n = 300 → ±1.6% at 95%; rollout proof 1.5 vs 6 months`	user anchor (GFiber v8 only) — separate (S11), prefer the documents' 500 containers / 20 min / 99.6%	~$1.5M, never merged

No dollar in this story is a claimed saving. The two priced rows are illustrations built on the record $0.18 per touch and on a labeled localization model; the ~$1.5M stays with S11.

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

GATE	ENTRY → EXIT	CADENCE AND OWNER	CONTROL
100-container test	random container / MBL numbers → every hop logged; bugs to Jira	ongoing; QA team; daily stand-up	every miss ticketed to the owning team; re-test the same container
Bug bash	job-story ranges assigned per person → issues logged	every other week; whole team	UI, data and performance in one pass
Shadow accounts	legacy-client uploads fed into dedicated demo accounts → side-by-side read	ongoing; two named clients first	real client data exposes defects synthetic data hides
UAT	opens at ≥80% minor-to-no issues → semi-weekly syncs (daily in November)	business users	quality gate, not calendar
Go / no-go ×2	stable production · first-client DQ 80%+ · no show-stoppers	business and tech leadership	evidence pack prepared in advance
Waves	6 criteria → next wave; ≥50 containers calibration; monthly go/no-go	program + business	hold in public if a criterion misses

Risks and mitigations. Environment outages (three environments; a "path to green" process to protect UAT availability); late monitoring (per-hop ratios and connectivity monitors — built later than they should have been); optimistic estimates (the lesson). **Lean / Kaizen mapping.** The ledger is genchi genbutsu — trace the real unit; the UAT gate is jidoka — stop the line on data quality; the criteria are standard work; the monthly go/no-go is PDCA at wave grain; defects found at the bottom cost a ticket, found by a client they cost trust.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Was the pilot late, and whose miss was it?**
>
> The first testing plan said early November and it launched in December; the program's retrospective lists delays, optimistic testing estimates and late monitoring, and I was one of its two TPMs, so I own my share of the estimate. I won't say how much was mine against the co-TPM's — I wasn't the only estimator. I now size test effort on hop count and put the monitors in before the debugging starts.

> **What does your ledger actually prove?**
>
> Where to look first, not a defect rate. It is a partial working log: 142 marked tested, 105 with results, 59 failing. By first-failing hop, route (23) and milestone (18) hold 41 of 59 — 69.5% — and only 3 of 105 passed all seven hops. To estimate a rate I would re-run a fresh random 100 and log every hop.

Why is entry at 80% but graduation at 90%?

The bars answer different questions. Entry (first-client data quality 80%+, no show-stoppers) says "safe enough for a calibrated client"; graduation (90%+ at least as accurate as the legacy product, 80%+ milestone estimates, 90% SLA) says "proven enough for the next wave" (record-from-docs).

Why at least 50 containers to calibrate?

Because 50 detects a 5% defect 92.3% of the time (1 − 0.95^50), and 100 detects it 99.4% (1 − 0.95^100). Fifty is the cheapest sample that gives the data team a verified baseline on the client's own lanes before onboarding.

How did you decide UAT could open?

On the 80% minor-to-no-issues gate: business testers are further from product development, so opening earlier would have had them rediscovering known bugs. The gate was data quality, not the calendar — which is also why it moved the date.

Is 500 containers in 20 minutes a lot?

25 containers a minute; 99.6% connected is 498 of 500. At the S11 illustrative 80,000 containers a year (≈219 a day) one 20-minute run is ≈2.3 days of average volume — headroom, not proof of peak behavior, and I said so.

What would you change about the testing plan?

Monitors before debugging, effort sized on hop count, the trace ID agreed in design rather than after the first slow bug hunts, and the three environments treated as the critical path — the record's own list of what went wrong.

BUSINESS IMPACT MODEL — KPI TREE

```
Launch readiness: no client touches the product until the evidence says it is ready
+- GATE 1  100-container live test: 7 hop checks per container, every miss a ticket
+- GATE 2  bug bashes every other week: job-story ranges per person
+- GATE 3  shadow accounts: real legacy-client data beside the old product
+- GATE 4  UAT opens at >=80% of shipments minor-to-no issues
+- GO / NO-GO x2 (business + tech): stable prod, first-client DQ >=80%, no show-stoppers
+- WAVES   2 -> 4 of 7 -> 20+: six criteria; >=50 containers of calibration per new client
+- RESULT  500 containers / 20 min / 99.6%; zero rollbacks; customers 7 -> 68 (9.7x)
(loop)  hop ledger -> first failing hop -> right team same day -> fewer escapes -> shorter next cycle
(miss)  optimistic estimate + late monitors -> pilot: first plan early Nov, launched Dec; bugs took hours
to localize
```

SOLUTION FLOW

```
random container/MBL -> trace 7 hops -> log row -> first failing hop?
   -> Jira ticket to the owning team -> daily stand-up -> fix -> re-test the same container
shadow account (real client data) -> side-by-side with the old product -> defects
UAT opens at >=80% minor-to-no issues -> semi-weekly syncs
GO / NO-GO (business + tech) -> wave 1 -> six criteria -> monthly go/no-go
   -> wave 2 (>=50 containers calibration first) -> wave 3 -> full roll-out
```

SENSITIVITY — LOW / BASE / HIGH

TRUE DEFECT RATE (ILLUSTRATIVE)	FORMULA	N = 50	N = 100	N = 300
1%	1 − 0.99^50	39.5%	63.4%	95.1%
3%	1 − 0.97^n	78.2%	95.2%	99.99%
5%	1 − 0.95^n	92.3%	99.4%	≈100%

Probability that a sample of n containers contains at least one defective container. The record sample sizes are 50 (calibration), 100 (launch gate) and 300 (pre-production); defect rates are illustrative.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

Honestly, the testing layer has no record dollar and I don't invent one. What it has is a gate that held: a hundred-container live test traced hop by hop, shadow accounts on real client data, UAT that stayed closed until eighty percent of shipments showed minor-to-no issues, and six published criteria for each wave — with a fifty-container calibration first, which detects a five-percent defect more than nine times in ten. The load proof was five hundred containers in twenty minutes at ninety-nine point six percent, both go/no-go checkpoints passed, and no wave rolled back. The ledger told us where to look — forty-one of fifty-nine failing containers broke first at route or milestone — and it is a working log, so I use it to aim the fix, not to estimate a rate. The honest cost is the schedule: my testing estimate was optimistic and monitoring came late, so the pilot launched in December, later than the first testing plan. I would take that trade again, and I would instrument first.

IF CHALLENGED — THE FULL ANSWERS

"You had zero rollbacks — isn't that just luck with only three waves?"

Zero events in three waves proves little statistically — the 95% upper bound on a per-wave rollback rate is 1 − 0.05^(1/3) = 63%. The evidence is the gates' detection power (92.3% at 50 containers for a 5% defect, 95.2% at 100 for 3%) and that exposure was bounded: wave 1 was two clients, not the whole seven-client pool.

"Why trust a log with blanks?"

I don't use it as a sample. I use it to rank where to look: 59 failing rows, first failure at route or milestone in 41. To estimate a rate I would re-run a random 100 with every hop recorded.

"What did the later launch cost?"

The record does not price it and the pilot was free, so there was no revenue delay to claim. The cost was engineering capacity, client expectation and credibility — I do not put a number of weeks on it, because the documents disagree — which is why the lesson is about estimates and monitoring, not about the gate.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I defined "ready" for a visibility launch: four test gates — a hop-by-hop 100-container ledger, bug bashes, shadow accounts on real client data and UAT that opened only at 80% data quality — and six graduation criteria per wave. The gates held and nothing rolled back; the honest miss was optimistic test estimates and late monitoring, which put the launch in December, later than the first testing plan.

NACF BRIDGE — ONE SENTENCE

A new inbound node launches the same way: trace real units hop by hop (ASN → appointment → door → receive → putaway → available), publish the entry gate and a calibration volume per node before the first trailer, and put the monitors in before the first week of debugging — doors × hours × unload rate only means something once each hop is measured.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you had to make a decision between standards and delivery. What tradeoffs did you have make? What was the outcome? Knowing what you know now, would you have done anything differently?" AIQB: Highest Standards Q4 **Insist on the Highest Standards** FACET: THE UAT GATE

Open (first two sentences): "We kept business UAT closed until eighty percent of shipments showed minor-to-no issues. Opening it on the calendar would have had business testers rediscovering bugs we already knew about."

90-second STAR. S A new-stack launch with a calendar everyone wanted. **T** Make "ready" a rule, not a feeling. **A** I helped define four gates and set UAT's entry at 80% minor-to-no issues, with graduation criteria for each wave published before wave 1; the first-client go/no-go read stable production, 80%+ data quality and no show-stoppers. **R** Both checkpoints passed and nothing rolled back — and the price was real: the pilot launched in December, later than the first testing plan.

Land: UAT entry 80%; six graduation criteria · **Pivot to NACF:** a new node does not take inbound until its entry gate is met — doors staffed, WMS receiving verified, first trailers traced — whatever the launch calendar says.

Do NOT say: that the later date was "worth it" without naming the cost; any dollar for the gate.

B2 · "Tell me about a time when you evaluated the customer experience of your product or service. What did you do? What was the result?" AIQB: Customer Obsession Q5 **Customer Obsession** FACET: SHADOW ACCOUNTS AND CALIBRATION

Open (first two sentences): "We ran two real clients' live data from the old product through the new one in demo accounts, side by side. And every new client gave us at least fifty containers of their own freight before they onboarded."

90-second STAR. S Synthetic tests were not going to show what a client's lanes and carriers actually do. **T** Find the defects on the client's own freight, before the client sees them. **A** Shadow accounts fed the legacy-product uploads of two named clients into dedicated demo accounts; wave-2 clients supplied a 50-container sample for calibration; the six criteria included CSAT on par with the consignee average. **R** Defects surfaced in our accounts, not theirs; the onboarding was light-weight because the product had calibrated on their freight.

Land: ≥50 containers → 92.3% chance to detect a 5% defect · **Pivot to NACF:** a node's first inbound sample — a trial volume per node — calibrates receiving and putaway before the node takes real peak volume.

Do NOT say: client names beyond what is confirmed; "customers loved it" — the record has no such claim.

B3 · "Tell me about a time when you were trying to understand a complex problem on your team and you had to dig into the details to figure it out. Who did you talk with or where did you have to look to find the most valuable information?" AIQB: Dive Deep Q1 **Dive Deep** FACET: THE HOP LEDGER

Open (first two sentences): "We traced real containers through seven hops and logged every one. When I counted by first failing hop instead of by column, forty-one of fifty-nine failures were at just two hops."

90-second STAR. S Six-plus system layers and a pile of defects, with every team seeing only its slice. **T** Locate the defects instead of counting symptoms. **A** I read the log by first-failing hop because downstream hops cascade: route (23), milestone (18), data back (7), exception (6), created (5). I took the top two to the owning teams first and kept the caveat — it is a working log, not a sample. **R** The fixes were aimed at route and milestone first; the log became the standing checklist.

Land: 41 of 59 first failures at two hops (69.5%) · **Pivot to NACF:** a receiving-defect Pareto by first failing step, not by total flags — so one dock-scheduling fault is not counted four times downstream.

Do NOT say: that the log estimates a defect rate; the share as precise.

B4 · "We all have things about ourselves we'd like to improve on at work. Give me an example of something that you've worked on to improve your overall work effectiveness. What resources did you identify to help you develop? What was the impact?" AIQB: Learn and Be Curious Q4 **Learn and Be Curious** FACET: THE HONEST MISS

Open (first two sentences): "The first testing plan said early November; the pilot launched in December. The program's own retrospective says our testing estimates were overly optimistic and monitoring came too late — and I was one of its two TPMs."

90-second STAR. **S** A launch across nine teams on a new stack. **T** Land a safe pilot; the estimate was part of the job. **A** I held the data-quality gate and accepted the later date rather than launch on the plan. The slow part was debugging: data crossed six-plus layers with little monitoring, and bugs took hours to localize. We added a trace across hops and per-hop ratios. **R** The pilot launched clean; the lesson is to size testing on hop count and instrument before debugging starts.

Land: first plan early November → launched December 2021 · **Pivot to NACF:** size a new-node launch plan on the number of hand-offs, and put the receiving monitors in before the first inbound wave.

Do NOT say: blaming other teams; "about a month" as a precise figure — the record is month-level.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "Of fifty-nine failing containers, forty-one first broke at one of two hops — and only three of a hundred and five passed all seven."

Headline metric: 41 of 59 (69.5%) at route + milestone; 3 of 105 all-seven-clean

- Read by first-failing hop, not by column.
- Order: route 23, milestone 18, data back 7, exception 6, created 5.
- Caveat first: partial log, blank = not recorded.
- Next step: a fresh random 100 with every hop logged.

Conflict / influence / backbone angle **Ownership**

Opener: "Nine teams, none reporting to me, each with its own backlog — I had to get every one to instrument its own hop in the middle of a launch crunch."

Headline metric: nine teams aligned on one seam; time to localize cut about 50% (secondary source)

- The seam nobody owned: no unique ID from downstream events to the screen.
- The ask was framed in each team's own bug queue, not the program's need.
- The sell was arithmetic: a few hours per blocker, every team paying it.
- Source: the 3PL ingestion one-pager (TPM across 6 systems and 9 teams; no unique ID; 2–4 hours to find the source system per blocker; blocker-bug time-to-resolve cut about 50%).

Honest failure-and-learning angle **Learn and Be Curious**

Opener: "The first testing plan said early November and we launched in December, and the program's own retrospective says why: testing estimates were optimistic and monitoring was late."

Headline metric: first plan early November → launched December 2021; hours to localize a bug at six-plus layers

- What I would do differently: size testing on hop count; monitors before debugging.
- What the record calls a win, I read as a warning: the environment was built "through long hours" — optimistic estimates are paid for by people.
- What held: the gates — no wave rolled back.
- No dollar claimed.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"With only three waves, how do you know your gates caused zero rollbacks — and not luck?"

Numeric answer. I don't, statistically: zero events in three waves gives a 95% upper bound of 1 − 0.05^(1/3) = 63% on a per-wave rollback rate. The case rests on detection power — 92.3% at 50 containers for a 5% defect, 95.2% at 100 for 3% — and on bounded exposure: wave 1 was two clients, not seven.

Say exactly: “Zero in three waves proves almost nothing on its own — the upper bound is sixty-three percent. What I can show is the gates' detection power and that wave one was two clients, not the whole pool.”

"Your ledger is a partial log. Why should I trust the Pareto?"

Numeric answer. Because I only claim what it supports: 105 rows had results, 59 had a failing hop, 41 of those first failed at route or milestone. Blank is not "pass," and a working log over-records failures, so the shares are indicative; the robust finding is that failures sat downstream of ingestion. I would re-run a random 100 with all seven hops logged.

Say exactly: “It's a working log, not a sample — blank means not recorded. I use it to decide where to look first, not to estimate a defect rate.”

"The pilot launched later than planned. What number would have told you earlier?"

Numeric answer. Hours to localize a blocker: the record says bugs took hours across six-plus layers (2–4 h in the secondary telling). A weekly chart of that figure against a ≤1 h target, from week one, would have shown the monitoring gap before the date did.

Say exactly: “The number I should have charted from week one was hours to localize a blocker — it was the leading indicator, and the date was the lagging one.”

S11-C — Flexport — Retiring an acquired product's book: a segment-by-segment RAPID decision for 79 Crux customers — 55 onto Shipment Visibility, 24 on rule-based exits, and three customers carrying 75% of the containers NEW

LPs this story can serve here: Customer Obsession (primary) · Success and Scale Bring Broad Responsibility · Earn Trust · Have Backbone, Disagree and Commit
Process family: Deliver — customer transition and product retirement; segmentation, decision rights (RAPID) and cut-over planning

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Customer Obsession Every one of the 79 customers got a destination by segment — transition, partner route, or API-or-cancel with dates — instead of an end-of-life email.

Success and Scale Bring Broad Responsibility Three truck brokers were 3.8% of customers and 75.0% of containers, so they got a per-customer partner route; 39 small accounts at ≈$770 a year each got the standard path.

Earn Trust The decision document kept the dissent on the page — cost of two systems, an unwritten parity list, "ejecting customers is never a good experience" — and cancellations ran on published dates.

Have Backbone; Disagree and Commit The decider committed the organization with the objections recorded, and each impacted party committed to perform — nobody smoothed the disagreement away.

Honesty label. This is a decision-document story, and the seat matters: the RAPID table names the Recommend, Agree, Perform, Input and Decide people, and the decider is an executive — *the decision was not mine*. My seat, from my own workbook (Apple v11 A4): I owned the mitigation of the decommission risk — I put it on the risk register as the top risk, rebuilt the comparison apples to apples, priced the 79-customer base per segment, and wrote the one-canonical-source rule; the transition document's owners built the per-customer paths with me, and the decision itself was the executive decider's. The segment table is the doc's; the footing, concentration arithmetic, break-even rule and parity matrix are my analysis. **Footing flag:** the doc's printed total row ($15,491 MRR / $185,892 ARR) equals the sum of only the six non-transition segments; all nine rows sum to $21,246 MRR / $254,952 ARR, and the appendix lists 77 names against 79 counted (37 SMB names vs 39) The number I say in the room is the printed one — 79 customers, $185,892 ARR, 294,581 containers (Apple A4 prints the same) — and I add that the printed total equals six of the nine rows. Overlap: the loop packet tells this as S10 for another interviewer.

WHAT THE RECORD SAYS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The opportunity. Migrate Crux clients to Shipment Visibility by the end of 2022. GKA, Enterprise and SMB customers transition and are upsell opportunities; for Motor Carriers, Truck Brokers, Freight Forwarders, Customs Brokers, 3PL and SCM SaaS the choice was cancel now, or keep serving them on Crux while working out missing features, support and Data Services cost-to-serve. RECORD-FROM-DOCS

The decisions. Truck brokers and motor carriers: transition AV Logistics and Loadsmart to partners in Transmission (like ITS, already there); cancel the remaining motor carriers at wind-down. Freight forwarders, customs brokers, 3PL: no new investment — recommend the SV API (no extra dev), else cancel at wind-down. SCM SaaS: cancel both — dated August 1 and October 24, 2022; the customs broker was cancelled September 1, 2022. A named executive decides; two people build the per-segment transition plans and review them with the exec team. RECORD-FROM-DOCS

The dissent. One reviewer wanted the cost of maintaining two systems and the stalled "Crux Absorption" project brought into the decision; another asked what Crux costs to keep running, whether the missing-feature list is written down, and warned that ejecting customers is never a good experience and that cancellations should be timed to win the competitors' consignees; others argued to cut direct competitors and to keep only partners. The 2021 plan adds that about 75% of Crux revenue came from clients who are not trade clients, so Crux's front end could not be sunset until the VaaS portal was stood up. RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Flexport had acquired a visibility product, Crux, and the goal was to move its clients to Shipment Visibility by the end of 2022 — but 79 paying customers and about 295 thousand containers a year were on it, and not all of them fit. The easy versions were an end-of-life email or running two systems forever. **Task.** Produce a decision rule per customer type that a decider could commit the company to, with the dissent visible. **Action.** I segmented the 79 into nine types and read each on three questions: does Shipment Visibility serve them, are they a partner or a competitor, and does serving them need extra build. Then I checked concentration: three truck brokers were 4% of customers and 75% of containers, so they got a per-customer partner route; the 39 small accounts, about $770 a year each, got the standard path. The decision: 55 customers to Shipment Visibility, three brokers to partners in Transmission, forwarders and 3PLs to the SV API or cancel at wind-down, nine motor carriers cancel at wind-down, and three dated cancellations. The dissent — cost of two systems, an unwritten parity list — stayed on the page. **Result.** Primary metric: 79 of 79 customers with a destination and a date — 55 + 3 + 9 + 9 + 3 — and a table footed by container share, because the printed ARR total summed only six of nine rows.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “The document prints seventy-nine customers, $185,892 of annual revenue and 294,581 containers; I add that the printed revenue total equals six of its nine segment rows, so I lead with containers.”

“What's your source for that?” “The base is the transition document's own table, which my workbook also quotes; the footing, the three-brokers concentration and the break-even are my arithmetic on that table, and I label them that way.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Flexport's acquired visibility product, Crux, still billed 79 customers — 294,581 containers in 2021 — while Shipment Visibility was in pilot. The aim was to migrate Crux clients by the end of 2022, but the decision document itself says not all customers fit the new service and a plan was needed. Keeping both meant two systems, double maintenance and a prospect set comparing Flexport against itself; closing Crux carelessly meant stranding customers who used it as infrastructure — and the plan's own dependency was that non-trade clients could not use the Client App at all until the VaaS portal existed.

T — Own the segmentation analysis and the per-segment recommendation a decider could commit to — transition, partner route, API-or-cancel, or cancel, each with a date — with the cost of two systems and the parity gaps on the page, not behind it.

Baselines on the table: 79 customers · 294,581 containers (2021) · nine segments · doc-printed ARR total $185,892 (six non-transition rows) vs nine-row sum $254,952 · 12 customers flagged API · 6 flagged churned in the appendix · goal: migrate by end of 2022 · ~75% of Crux revenue from non-trade clients (2021 plan).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — footed the customer table (count, containers, ARR, MRR) and read each segment's use, revenue and containers; separated partners from competitors; listed what the plan itself said Crux customers could not do on SV. *Exit bar:* segment table reconciled to the appendix names; the footing differences written down.
- **Plan & Design** — a three-axis rule — does SV serve them, partner or competitor, extra build needed — mapped to five decisions; concentration analysis (brokers vs the long tail); the parity matrix drafted; the open questions listed (keep-alive cost, written parity list). *Exit bar:* RAPID roles named: Recommend, Agree, Perform, Input, Decide.
- **Build & Test** — collected stakeholder input verbatim, including the dissent; ran per-customer conversations for the large accounts with the trucking group; drafted the per-segment transition plans. *Exit bar:* the decider can read every objection beside the recommendation.
- **Cutover & Go-Live** — the decider committed the organization; each impacted party committed to perform; the SV-bound segments entered through the pilot waves; competing SCM SaaS and the customs broker were cancelled on dates. *Exit bar:* dated actions on the page: August 1, September 1 and October 24, 2022.

- **Hypercare / Close** — API-or-cancel conversations for forwarders and 3PLs; partner conversations for the brokers; wind-down of the Crux front end as VaaS / API paths landed; churn flagged in the appendix tracked. *Exit bar:* every customer on a path with a date; Crux wound down by the end-2022 goal.

Influence and ownership. Mine (as I would tell it): the segmentation, the concentration read, the plan inputs and the parity questions. Not mine: the recommendation owners, the decider and the per-segment transition plans' authors named in the RAPID table; the trucking group owned the partner conversations. I influenced by putting the numbers and the dissent on one page.

> **Designed the process; others own it now.** The segment rules, the dated cancellations and the parity questions live in the decision document, not in my head; the transition plans have named performers. **Trade-off put to leadership.** Revenue surrendered (the six non-SV segments, $185,892 of printed ARR) against two systems, double maintenance and competing with ourselves — and a keep-alive cost the record never priced.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — the doc's printed $185,892 ARR is the revenue on the six non-SV segments (modeled footing — mine); the three SV-bound segments add $69,060, for $254,952 across all nine rows. No Crux keep-alive cost exists in the record, and I do not invent one — the break-even is $15,491 a month (modeled).
- **#** — 79 of 79 customers with a decision: 55 to SV, 3 to Transmission partners, 9 SV-API-or-cancel, 9 cancel at wind-down, 3 cancelled now on dates (record-from-docs; tally mine).
- **%** — brokers 3.8% of customers and 75.0% of containers; SV-bound segments 69.6% of customers, 13.3% of containers, 27.1% of ARR (my footing of the doc table).
- **Time** — target: migrate Crux clients by end of 2022; cancellations dated August 1, September 1 and October 24, 2022 (record-from-docs).
- **Customer** — each customer type has a stated destination; the largest accounts got partner conversations, not a template; six customers were already flagged churned in the appendix.
- **Quality** — two open questions flagged rather than hidden — keep-alive cost and a written feature-parity list; the doc's own footing differences caught and reconciled.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	An acquired product, nine customer types and no stated rule for who fits. I built the rule (serve? partner or competitor? extra build?) and the concentration read from the data.
Scope & influence	Executives decided; product, trucking, data services and finance gave input; I influenced by putting numbers and dissent on one page — a recommendation to leaders, one to two levels up.
Execution trade-off + mechanism others now own	Revenue for simplicity, with the cost of keeping both unpriced in the record. Mechanism others own: the segment rules, the dated cancellations and the decision document.
Impact (long-term)	One visibility product to run, the acquired book moved or exited deliberately, the partner route for the largest accounts preserved.
Process improvement (standard work institutionalized)	A reusable product-retirement template: segment by use, find concentration, rule per segment, dated actions, dissent recorded, commitments confirmed.
Complexity & consensus	Real dissent (cost of two systems, parity, customer experience, competitor timing) resolved by a named decider with commitments from each impacted party.
Root cause at scale	The root question was "who is this product for" — the plan's non-trade clients could not use SV, which is why the rule is by segment, not by customer.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Nine-segment table; footing (printed total vs nine rows); concentration (3 brokers = 75.0% of containers); ARR per customer 60× spread; API and churn flags.
2 · Solution design	Cancel all vs keep all vs segmented rules; broker route options; parity matrix; decision rights via RAPID.
3 · Math & data logic	Tally 55 + 3 + 9 + 9 + 3 = 79; shares 69.6 / 13.3 / 27.1% vs 30.4 / 86.7 / 72.9%; break-even $185,892 / 12 = $15,491 a month.
4 · Implementation	Per-segment plan, dated cancellations, commitments confirmed, SV pilot waves for the 55, partner conversations for the three.
5 · Probe questions	Seven numeric probes plus three trap drills: which total, will the brokers accept, what number would flip the cancel decision.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. The doc's segment table (customers, MRR, ARR, 2021 containers) and its appendix of names, with API and churn flags RECORD-FROM-DOCS. I added three columns the doc does not carry — ARR per customer, share of containers and the decision — and footed everything from the rows.

SEGMENT	CUSTOMERS	ARR (DOC)	CONTAINERS 2021	ARR / CUSTOMER	% OF CONTAINERS	DECISION (DOC)
GKA	2	$16,500	17,061	$8,250	5.8%	transition to SV
Enterprise	14	$22,560	17,983	$1,611	6.1%	transition to SV
SMB	39	$30,000	4,153	$769	1.4%	transition to SV
Motor Carrier	9	$17,760	12,359	$1,973	4.2%	cancel at wind-down
Truck Broker	3	$139,512	221,044	$46,504	75.0%	Transmission partner route (else cancel)
Freight Forwarder	5	$14,820	11,871	$2,964	4.0%	SV API, else cancel at wind-down
Customs Broker	1	$600	432	$600	0.1%	cancel now (dated)
3PL	4	$2,400	4,218	$600	1.4%	SV API, else cancel at wind-down
SCM SaaS	2	$10,800	5,460	$5,400	1.9%	cancel now (dated)

What it revealed. (1) *Footing:* customers foot to 79 and containers to 294,581, but the printed $185,892 ARR is only the six non-transition rows — the nine rows sum to $254,952, so I lead with container shares, which foot. (2) *Concentration:* three truck brokers are 3.8% of customers, 75.0% of containers and 54.7% of nine-row ARR. (3) *Spread:* ARR per customer runs from $769 (SMB) to $46,504 (truck broker) — 60.5×. (4) *The SV-bound 55 are a long tail:* 69.6% of customers but 13.3% of containers and 27.1% of ARR — light-weight onboarding fits; the brokers need per-customer work. (5) The 2021 plan's "~75% of Crux revenue is non-trade" sits within two points of my 72.9% for the six non-SV segments. (6) Flags: 12 customers carry an API flag (5 of them SV-bound) and 6 are flagged churned. I quote 79 customers as printed (Apple A4 prints 79 too); I do not quote a per-customer average beyond the SMB tail.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Whole book	Cancel everything not on SV now · keep every customer on Crux until parity · segmented rules	Segmented rules	Cancel-all strands the large partner-like accounts; keep-all leaves two systems running with no end date and misses the end-2022 goal.
Truck brokers and motor carriers	Standalone service · white-label / partner via Transmission · cancel	Partner route for brokers; cancel the nine motor carriers at wind-down	The 2021 strategy doc gives no standalone service for these types; ITS already uses Transmission.
Forwarders, customs, 3PL	Build something extra · SV API (no extra dev) · cancel	SV API, else cancel at wind-down	No new investment; they are not competitors in the doc's reading and some give intel on their customers.
SCM SaaS	Explore data partnerships · cancel now	Cancel now (dated)	They could undermine selling our own branded services; one went through legal review.
Decision rights	Consensus · RAPID with a named decider	RAPID	Recommend, Agree, Perform, Input, Decide named; each impacted party commits.

Parity matrix (drafted from the documents; gaps flagged, not guessed)

DIMENSION	CRUX TODAY	SHIPMENT VISIBILITY	GAP AND HANDLING
Who can use it	Any customer type — shippers, carriers, brokers, forwarders, 3PL, SaaS	Client App for trade clients; non-trade clients cannot use it until a VaaS portal exists (2021 plan)	Rule by segment; non-trade types go API, partner or cancel
How tracking is requested	Front-end manual search and API	Single input, CSV or API in the Client App	Manual search is not offered on the VaaS-portal plan — a documented gap
Data-quality bar	The incumbent benchmark	Graduation: at least as accurate as Crux ≥90% of the time	Built into the six graduation criteria
Human in the loop	not in the documents — say "I don't know how Crux handled exceptions" VERIFY-WITH-RAJESH · V3	Tickets to Data Services when auto-tracking fails	Unknown on the Crux side
Webhooks / API	12 customers flagged API	Public API and client webhooks	Decision: "SV API, no extra dev effort"
Price	MRR by segment, $50–$11,626	Free pilot ≈6 months, then the SV charge model	The price step is the thing to explain to the SV-bound 55
Feature-parity list	A reviewer asked: "written down somewhere?"	—	Not in the record — the document records the question, not an answer; I do not claim the list existed in writing

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers (footing the doc's own table)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Customers	`2 + 14 + 39 + 9 + 3 + 5 + 1 + 4 + 2 = 79`	record-from-docs rows	79
2	Containers, 2021	`17,061 + 17,983 + 4,153 + 12,359 + 221,044 + 11,871 + 432 + 4,218 + 5,460 = 294,581`	record-from-docs rows	294,581
3	ARR, all nine rows	`16,500 + 22,560 + 30,000 + 17,760 + 139,512 + 14,820 + 600 + 2,400 + 10,800 = $254,952`	my footing of the doc rows	$254,952
4	The printed total row	`17,760 + 139,512 + 14,820 + 600 + 2,400 + 10,800 = $185,892; $254,952 − $185,892 = $69,060 = the three SV-bound rows`	my footing: the printed ARR is the six non-transition rows	$185,892 (six rows)
5	MRR cross-check	`$21,246 (nine rows) − $15,491 (printed) = $5,755; $5,755 × 12 = $69,060`	my footing	$5,755 MRR missing from the printed total

Table 2 — The fix: the rules, the tally and the concentration, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Decision tally	`55 (SV) + 3 (Transmission partners) + 9 (API-or-cancel) + 9 (cancel at wind-down) + 3 (cancel now) = 79`	record-from-docs decisions; tally mine	79 of 79 with a destination
2	Broker concentration	`3 / 79 = 3.8% of customers; 221,044 / 294,581 = 75.0% of containers; 139,512 / 254,952 = 54.7% of ARR`	my footing	3.8% → 75.0% / 54.7%
3	ARR per customer spread	`$30,000 / 39 = $769 (SMB); $139,512 / 3 = $46,504 (brokers); 46,504 / 769 = 60.5×`	my footing	60.5×
4	SV-bound segments	`55 / 79 = 69.6% of customers; 39,197 / 294,581 = 13.3% of containers; 69,060 / 254,952 = 27.1% of ARR`	my footing (39,197 = 17,061 + 17,983 + 4,153)	69.6% / 13.3% / 27.1%
5	Exit and re-route segments	`24 / 79 = 30.4%; 255,384 / 294,581 = 86.7% of containers; 185,892 / 254,952 = 72.9% of ARR`	my footing (255,384 = 294,581 − 39,197)	30.4% / 86.7% / 72.9%

Table 3 — After: where each customer goes (customers · containers)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	To Shipment Visibility	`GKA 2 + Enterprise 14 + SMB 39 = 55 customers; 39,197 containers`	record-from-docs decision	55 · 39,197
2	To partners in Transmission	`truck brokers: 3 customers; 221,044 containers (else cancel at wind-down)`	record-from-docs decision	3 · 221,044
3	SV API, else cancel at wind-down	`forwarders 5 + 3PL 4 = 9 customers; 11,871 + 4,218 = 16,089 containers`	record-from-docs decision	9 · 16,089
4	Cancel at wind-down	`motor carriers: 9 customers; 12,359 containers`	record-from-docs decision	9 · 12,359
5	Cancel now (dated)	`customs 1 + SCM SaaS 2 = 3 customers; 432 + 5,460 = 5,892 containers`	record-from-docs decision	3 · 5,892
6	Footing check	`39,197 + 221,044 + 16,089 + 12,359 + 5,892 = 294,581`	arithmetic	every container accounted for

Table 4 — The dollars, every addition shown (ARR; pilot is free, so none of it is "saved")

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	ARR on the SV-bound segments	`$16,500 + $22,560 + $30,000 = $69,060`	my footing of the doc rows; flows into the charge model after the pilot	$69,060
2	ARR on the exit and re-route segments	`$17,760 + $139,512 + $14,820 + $600 + $2,400 + $10,800 = $185,892`	the doc's printed total, footed	$185,892
3	Of which hard-cancel	`$17,760 (motor carriers) + $600 (customs) + $10,800 (SCM SaaS) = $29,160`	my footing	$29,160
4	Of which conditional on SV-API uptake	`$14,820 (forwarders) + $2,400 (3PL) = $17,220`	my footing	$17,220
5	Of which the partner route	`$139,512 / $185,892 = 75.1% of the exit ARR; 29,160 + 17,220 + 139,512 = $185,892`	my footing	$139,512
6	Break-even keep-alive cost	`$185,892 / 12 = $15,491 per month; if running Crux for these segments costs more, cancelling dominates on direct economics`	arithmetic on the printed total; the keep-alive cost is not in the record, and I do not state one	$15,491 / month

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

SEGMENT	ACTION	TIMING	OWNER (RECORD) / CONTROL
GKA · Enterprise · SMB (55)	Transition through the pilot waves; upsell opportunity; light-weight onboarding; API customers (5) need an integration path	pilot waves through H2 2022	transition-plan authors; SV program
Truck brokers (3)	Partner in Transmission (white-label, like ITS); cannot track non-Flexport containers there; else cancel at wind-down	with the Crux wind-down	trucking group; per-customer conversations
Forwarders · 3PL (9)	SV API (no extra dev) or cancel at wind-down	at wind-down	transition-plan authors
Motor carriers (9)	Cancel at wind-down	at wind-down	transition-plan authors
Customs (1) · SCM SaaS (2)	Cancel now	Sept 1 · Aug 1 · Oct 24, 2022	dated in the document

Controls and risks. Commitment step: each impacted party commits to perform, verbally in the decision meeting or by email. Risks named in the record: cost of two systems and the stalled Crux Absorption project; an unwritten parity list; legal concerns over web scraping; cancellations that hand a competitor's consignees to someone else unless timed. **Lean / Kaizen mapping.** Segmenting by what flows through the process is value-stream thinking; the dated cancellations are standard work with a due date; the open questions are the PDCA "Check" I would run first.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why not cancel everyone who is not on SV and be done?**
>
> Because the three brokers are 75.0% of containers and 54.7% of nine-row ARR — ITS already uses Transmission and the decision routes AV Logistics and Loadsmart there too. Cancelling them would strand the largest flows to save a template. The other 21 customers are exactly the ones the decision exits or re-routes.

What does it cost to keep Crux alive?

The record has no figure — a reviewer asked the question and it is open. I will not invent one; I give the break-even: the printed total of $185,892 a year is $15,491 a month, so if keeping Crux alive for those six segments costs more than that, cancelling dominates on direct economics. The decision also carried strategic inputs the break-even ignores.

Your totals do not foot. Which ARR is right?

Both, once you say what they are: the printed $185,892 is the six non-SV rows; all nine rows sum to $254,952, and the $69,060 difference is the three SV-bound segments. I lead with container shares because they foot to 294,581, and the appendix lists 77 names against 79 counted — I would confirm that before quoting a per-customer figure.

How many of the 55 will actually move?

The record does not say how many did. What I can show is the shape of the problem: 55 customers, $1,256 average ARR (69,060 / 55), 5 of them API integrations (9.1%) — a long tail suited to light-weight onboarding behind the pilot waves, with six already flagged churned across the whole book (7.6%).

Who disagreed, and what did they get right?

Four inputs: the cost of two systems and the stalled absorption project; a missing, unwritten parity list; "ejecting customers is never a good experience" and the timing of competitor cancellations; and cut-the-competitors. All four were right about something the decision then had to carry — the parity list is the one the record never closes.

How do you know the SMB path is cheap enough?

ARR per SMB customer is $769 against $46,504 per broker, a 60.5× spread — a bespoke migration for 39 accounts at that revenue does not pay, so the SMB path is the standard one. I would price the onboarding hours per account to confirm; there is no record figure.

What would you do on day one of a similar retirement?

Foot the table, find the concentration, write the rule per segment, put the dissent and the open questions on the same page, and date every action. Then price the keep-alive cost before the memo, not after.

BUSINESS IMPACT MODEL — KPI TREE

```
One visibility product to run; every acquired customer has a destination and a date
+- 55 -> Shipment Visibility (GKA 2, Enterprise 14, SMB 39): 69.6% of customers, 13.3% of containers,
27.1% of ARR
+- 3  -> Transmission partners (truck brokers): 3.8% of customers, 75.0% of containers, 54.7% of ARR
+- 9  -> SV API else cancel at wind-down (forwarders 5, 3PL 4)
+- 9  -> cancel at wind-down (motor carriers)
+- 3  -> cancel now, dated (customs 1, SCM SaaS 2)
+- Printed ARR $185,892 = the six non-SV rows; nine rows = $254,952 (footed)
+- Open: keep-alive cost (break-even $15,491 / month); written parity list
(loop)  segment by use -> find concentration -> rule per segment -> dissent on the page -> decider
commits -> dated actions
```

SOLUTION FLOW

```
foot the table -> customers 79, containers 294,581, ARR rows vs printed total
   -> axes per segment: SV serves them? partner or competitor? extra build?
   -> concentration: who carries the containers?
   -> RAPID: Recommend / Agree / Perform / Input / Decide
   -> dissent + open questions on the page -> Decide commits -> each party commits
   -> dated actions: SV waves | partner route | API-or-cancel | cancel now
   -> wind-down of the old front end by the end-2022 goal
```

SENSITIVITY — LOW / BASE / HIGH

KEEP-ALIVE COST (ILLUSTRATIVE)	PRINTED MRR − COST	DIRECT-ECONOMICS READ
$5,000 / month	$15,491 − $5,000 = $10,491	keep Crux for the six segments
$15,491 / month	$15,491 − $15,491 = $0	indifferent
$30,000 / month	$15,491 − $30,000 = −$14,509	cancel dominates on direct economics

The keep-alive cost is not in the record; the three values are illustrative bookends around the break-even. The strategic inputs — competitor timing, partner routes — are not in this arithmetic.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> Honestly, retiring Crux was not a money-making decision and I don't dress it as one. The doc's printed total, about one hundred eighty-six thousand a year, is the revenue on the six segments that don't move to Shipment Visibility — I footed the table myself, and all nine rows sum to about two hundred fifty-five thousand. What the analysis changed was the shape of the work: three truck brokers were four percent of customers and seventy-five percent of the containers, so they got a partner route; fifty-five customers — the long tail — move through the pilot waves; nine are API-or-cancel, nine motor carriers cancel at wind-down, three were cancelled on dates. Every customer has a destination and a date. The record doesn't price keeping Crux alive, so I give the break-even — about fifteen and a half thousand a month — and I say the strategic inputs sit outside it. Every dollar here is the document's, footed, or labeled; none of it is claimed as savings.

IF CHALLENGED — THE FULL ANSWERS

"You are telling me a $186K book was a good decision to exit?"

I'm telling you what the record shows: the printed total is the six non-SV segments, and the rule exits or re-routes them by segment, with the brokers — 75% of containers — taking a partner route. Whether the economics beat the keep-alive cost is exactly the number the record never priced; I give the break-even, $15,491 a month, and flag it.

"Why trust your footing over the doc?"

Because the rows foot and the printed total does not: customers foot to 79, containers to 294,581, but the printed ARR equals the six non-transition rows only — the other three rows add $69,060. That is arithmetic, and it is why I use container shares first.

"What did you actually decide?"

The decider decided. My part, as I tell it, was the segmentation, the concentration read and getting the dissent and the open questions on one page beside the recommendation.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

When Flexport retired an acquired visibility product, 79 customers needed a destination. I segmented them into nine types, found three truck brokers were 4% of customers and 75% of containers, and wrote a rule per segment: 55 to the new product, three to a partner route, the rest API-or-cancel with dates — with the dissent on the page.

NACF BRIDGE — ONE SENTENCE

Retiring or re-pointing a node or lane in a network is the same problem: segment what flows through it, find the concentration (a few lanes carry most of the volume), give each segment a destination and a date, and price keeping both running — a placement trade-off of cost-to-serve against distance-to-customer.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?"

AIQB: Customer Obsession Q8 | **Customer Obsession** | FACET: SEGMENTATION AND DESTINATIONS

Open (first two sentences): "Every one of seventy-nine customers got a destination, not an end-of-life email. I segmented them by what they actually were and wrote a rule per segment."

90-second STAR. **S** An acquired product with 79 customers and a goal to move them by the end of 2022 — and not every customer fit. **T** Give each customer a path, not a notice. **A** Nine segments, three questions each — does SV serve them, partner or competitor, extra build — giving 55 to Shipment Visibility, three brokers to a partner route, nine API-or-cancel, nine cancel at wind-down and three cancelled on dates. **R** 79 of 79 with a destination and a date; the largest accounts got conversations instead of a template.

Land: 79 of 79: 55 + 3 + 9 + 9 + 3 · **Pivot to NACF:** retiring a lane or node: every flow through it gets a destination and a date before the old path closes.

Do NOT say: "nobody was stranded" or "zero escalations" — the record does not say; dollars as savings.

B2 · "Tell me about a time when you considered everyone who was affected by your work, not just your direct customers or people within your organization."

AIQB: Broad Responsibility Q5 | **Success and Scale Bring Broad Responsibility** | FACET: CONCENTRATION

Open (first two sentences): "Three customers were four percent of the book and seventy-five percent of the containers. A template would have been irresponsible for them and overkill for thirty-nine small accounts."

90-second STAR. **S** A single end-of-life policy would have treated 79 customers alike. **T** Match the effort to what each customer needs and what is at stake. **A** I computed concentration: brokers 3.8% of customers, 75.0% of containers; ARR per customer from $769 (SMB) to $46,504 (broker), 60.5×. The brokers got a partner route and per-customer conversations; the tail got the standard path. **R** Effort followed stakes: partner route for the largest flows, standard path for the tail.

Land: 3.8% of customers = 75.0% of containers · **Pivot to NACF:** a Pareto of lanes or nodes before a network change — the few that carry the volume get bespoke handling.

Do NOT say: that the partner route succeeded — the record shows the decision, not the outcome.

B3 · "Describe a time when you felt really strongly about something on a project but the team decided to go in a different direction. How hard did you press the issue? How did you approach that project afterward?"

AIQB: Have a Backbone Q4 | **Have Backbone; Disagree and Commit** | FACET: THE DISSENT

Open (first two sentences): "The document carried real dissent: the cost of running two systems, a parity list nobody had written down, and a warning that ejecting customers is never a good experience. We put all of it on the page."

90-second STAR. **S** A recommendation to retire a product that others had reasons to keep. **T** Get a decision the organization would commit to, with the objections visible. **A** The RAPID named who recommends, who agrees, who performs, who gives input and who decides; the input section kept each reviewer's words, including the parity and keep-alive questions; the decider committed with those on the page, and each impacted party confirmed they would perform. **R** A decision with named performers and dated actions — and two open questions flagged instead of hidden.

Land: four dissenting inputs; two open questions · **Pivot to NACF:** a network decision memo — putting the cost of the status quo and the unresolved questions beside the recommendation, then committing.

Do NOT say: that I overruled anyone — the decider decided; or what my exact seat was without checking.

B4 · "Tell me about a time when you uncovered a significant problem in your team. What was it and how did you communicate it to your manager and other stakeholders? What did you do to address the problem?"

AIQB: Earn Trust Q6 | Dive Deep | FACET: FOOTING

Open (first two sentences): "When I re-footed the decision document's table, the customer and container counts added up but the dollar total did not — the printed ARR was only six of the nine rows."

90-second STAR. **S** A table everyone quoted: 79 customers, $185,892. **T** Make sure the numbers under a decision were right. **A** I summed each column from its rows: customers 79, containers 294,581, ARR $254,952 against the printed $185,892 — the difference, $69,060, is exactly the three SV-bound segments. I switched the analysis to container shares and flagged the 77-versus-79 name count. **R** The decision rested on numbers that foot; the dollar total was relabeled as the exit segments' ARR.

Land: $254,952 vs $185,892; difference $69,060 · **Pivot to NACF:** every network metric gets footed from its rows before a decision — totals that silently drop a segment are the commonest error.

Do NOT say: that I caught it live in the room — say "when I re-footed it preparing this".

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle — Dive Deep

Opener: "Three customers were under four percent of the book and three-quarters of the containers — so the plan could not be one policy."

Headline metric: 3.8% of customers = 75.0% of containers = 54.7% of ARR

- Foot first: containers foot, the printed ARR does not.
- Concentration: 3 brokers vs 39 SMBs — $46,504 vs $769 per customer.
- The SV-bound 55 are the long tail: 69.6% of customers, 13.3% of containers.
- Say "my footing" once, early.

Conflict / influence / backbone angle — Have Backbone; Disagree and Commit

Opener: "The document carried real dissent, and the decision was only as strong as the objections left standing on the page."

Headline metric: four dissenting inputs; one decider; each impacted party commits

- Cost of two systems; a missing parity list; ejecting customers; cut the competitors.
- RAPID: Recommend, Agree, Perform, Input, Decide named.
- The decider committed with the objections recorded.
- My part: the comparison, the per-segment pricing and the dissent on one page; the decider was an executive.

Honest failure-and-learning angle — Learn and Be Curious

Opener: "The decision memo had two questions it could not answer — what Crux costs to keep running, and whether the feature-parity list was written down — and the record never closes either."

Headline metric: 2 open questions; break-even $15,491 / month

- What I would do differently: price the keep-alive cost before the memo, not after.
- What I would add: a written parity matrix as an exit criterion for each segment.
- What held: dated actions and named performers.
- No dollar claimed as savings.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"You quote 79 customers and $185,892. Your own table says $254,952. Which is it?"

Numeric answer. Both, and I say what each is: $185,892 is the six non-transition rows (24 customers); the three SV-bound rows add $69,060 for $254,952. Customers (79) and containers (294,581) foot either way, so container shares lead. The appendix lists 77 names against 79 counted, so I would confirm before quoting a per-customer average.

Say exactly: "The printed total is the six segments that don't move to Shipment Visibility; all nine rows sum to about two hundred fifty-five thousand. Customers and containers foot either way, so I lead with container shares."

"What if the three brokers refuse the partner route?"

Numeric answer. Then they cancel at wind-down per the decision, and the exposure is the largest in the book: 221,044 containers (75.0%) and $139,512 (54.7% of nine-row ARR). One of three already uses Transmission, which is why it is the route; the other two are per-customer conversations, not a template.

Say exactly: "If they decline, they cancel at wind-down — and that is seventy-five percent of the containers, which is exactly why they got per-customer conversations instead of a template."

"What number would have flipped your decision to exit the forwarders and 3PLs?"

Numeric answer. The keep-alive cost. The nine customers pay $17,220 a year; the six exit segments together pay $15,491 a month. If keeping Crux alive costs less than that monthly figure, deferral wins on direct economics; the record never prices it, which is why I would price it before the memo.

Say exactly: "The keep-alive bill. The six exit segments pay fifteen thousand four hundred ninety-one a month; if running Crux costs less than that, deferral wins — and the record never prices it."

S12 — Flexport — FF App export launch: re-sequencing a committed 18-month roadmap on cost-to-serve evidence (value in month 6, 66% less effort, ~$5M cost avoidance)

DEEPENED

LPs this story can serve here: Think Big (primary — the flip) · Frugality (primary — ~$5M avoided by sequencing) · Have Backbone, Disagree & Commit (changing the manager's plan) · Earn Trust (privately first) · Bias for Action (the reversal trigger written before the swing)
Process family: Deliver — export documentation / operator and partner workflow; launch sequencing; unit-economics-driven roadmap governance

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “About five million dollars of avoided cost — engineering effort never spent plus value pulled forward — said once, an avoidance and not a saving, and never added to the unit-cost arithmetic.”

“What's your source for that?” “It is the figure in my own interview workbook for the export-app re-plan; my split into engineering effort and value pulled forward comes to about four point four million, a model I label as mine, so I say about, and I never call it annual.”

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Think Big The destination changed — first value in month 6 instead of month 18, and half again the unit saving — by re-opening a committed, socialized roadmap on evidence at the work-item grain.

Frugality About $5M of cost avoidance — engineering never spent plus value pulled forward — said once; 66% less effort to first value; a reuse rule for any module over 8 dev-weeks.

Have Backbone; Disagree and Commit I changed my manager's plan privately first, gave engineering two days to break the model, and the onsite ratified it with his name on it.

Earn Trust The cost of the trade — operators on the old screens two more quarters — was priced and published, not hidden.

Honesty label. The roadmap was my manager's; I owned the sequencing recommendation. Engineering produced the re-estimates; the ~$5M is the record's cost-avoidance figure, said once; the run-rate decomposition and the engineer-count split are my model. Loop-packet conflict (§1.3): that document tells this as a Sephora roadmap — the record is Flexport FF App; say Flexport.

WHAT IS THE FF EXPORT APP? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

The minute. The Freight Forwarding App — "FFA" — is Flexport's operator-facing execution platform: where import and export shipments actually get run. Its core tenet is work-item-driven execution — every unit of work is a work item assigned to exactly one operator ("one operator, one work item"), so nothing ships on tribal knowledge and every action is attributable. For export, the heart of it is schedule management: a "Generate Schedule" work item builds the shipment's milestone schedule automatically, and a "Replan Schedule" work item fires on defined exceptions — an LCL CFS-cutoff miss, a vessel late against truck pickup, a vessel late past the quoted delivery date, a change of vessel — each with a defined rework path instead of ad-hoc heroics. It shipped FCL first (2022), with Air and LCL behind it (2023). RECORD-FROM-DOCS

The machinery behind the story. The schedule work sat inside a five-work-stream FFA program (Schedule Management; Cascading Schedules; Auto-Build Initial Schedule; End-to-End Operator Experience; Operator MiMeC), with schedule automation completing end Q3/early Q4 2022 and export-partner UAT aimed at a March 2023 launch. Because the app serves external partner organizations, it also carries a compliance bar: an immutable audit log of who changed what and when — nine event categories at a 100%-captured standard across ~100 partner orgs and thousands of changes a day, phased Alpha → Beta → MVP with CISO sign-off. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the module documents carry forward-plan dates (export-partner UAT March 2023; schedule automation completing end Q3/early Q4 2022) — the build-out clock after MVP. Canon's "value in month 6" is the sequencing decision's clock: different milestones, no conflict — never merge them. The audit-logs "~100 partner orgs" is platform-wide; canon's 53 is the import-partner migration set — never merge those either.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Freight Forwarding App export roadmap was operator-first — 18 months to first value, already socialized, with my manager's name on it. The process baselines were 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment. The question nobody had asked was time-to-value and unit cost — the plan had been built around persona preference. **Task.** With no authority over the roadmap, re-sequence it on cost-to-serve evidence without abandoning the operators. **Action.** I wrote the baseline memo, had the business cut a bare-minimum scope per persona, and got engineering to re-estimate with confidence levels: operator-first reached −25% cost-to-serve in 9 months with operators still spending 20% of their time in the old app; partner-first reached −50% in 6 months and built the chassis operators would later ride. I added a rebuild-vs-reuse rule per module — over eight dev-weeks, reuse — a scope tripwire in the burn-up, and a written reversal trigger, then read it to my manager privately first; engineering had two days to break it, and the Bellevue onsite ratified the flip with his name on it. Operators' two further quarters on old screens were priced and published, not hidden. **Result.** Primary metric: engineering effort to first value — the 18-month plan → 66% less effort, value in month 6 instead of 18. ~$5M of cost avoidance — the record figure, said once; cost-to-serve −50% versus −25% at the work-item grain; 48 of 53 import partners migrated within a month.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The inherited launch plan for the freight-forwarding app was operator-persona first, 18 months, already socialized, and my manager owned it. Process baselines: 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment — the two numbers everything prices from.

The business problem was time-to-value and unit cost, not persona preference. Left as inherited, operators would reach first value in month 18 while still spending 20% of their time in the old app — half the unit saving (−25% vs −50% at the work-item grain), reached a year later. The risks of re-opening it: a committed, socialized plan; optimistic estimates; and operators feeling abandoned while partners went first.

T — Own the sequencing recommendation — and change a plan with my manager's name on it, without authority over the roadmap, in a way he could own.

Baselines on the table: 339.5 min/shipment · 2,200 operator interactions per departed shipment · first value in month 18 on the inherited path · cost-to-serve −25% (operator-first) vs −50% (partner-first) at the work-item grain.

KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Asia latency surfaced right after the April 2022 import-side launch.** The monthly business review read P95 page load at 21.19s in Asia against 3.3s in the US; static assets ran ~5s in China against ~500ms in the US; the app's domain had no geo-based DNS or reverse proxy, so Asia traffic rode the public internet. The fixes were sequenced — WAF enablement, a double CDN removed, a regional CDN proposal with smoke tests per change, then China and Asia tuning — with 48 of 53 import partners (about 90%) already migrated and expecting the app to hold. RECORD-FROM-DOCS
- **Compliance was a launch feature, not an afterthought.** The Audit Logs PRD demanded an immutable who/what/when trail across ~100 partner orgs and thousands of changes a day — nine categories at a 100%-captured bar, phased Alpha → Beta → MVP — and the CISO's definition of "important events" was deliberately vague, so PM buy-in and CISO sign-off were written into the plan as the mitigation. RECORD-FROM-DOCS
- **Exceptions had to become work items.** Replanning ran on a defined taxonomy — LCL CFS-cutoff miss, vessel late vs truck pickup, vessel late past quoted delivery date, change of vessel — each with a rework path, under the

"one operator, one work item" tenet. The exception list, not the happy path, is where operator trust is won. RECORD-FROM-DOCS

- **A hard external deadline.** Export-partner launch UAT was set for March 2023, with schedule automation completing end Q3/early Q4 2022 and FCL ahead of Air/LCL — a date-driven constraint on top of the unit-economics sequencing. RECORD-FROM-DOCS

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — baseline memo (339.5 min; 2,200 interactions); bare-minimum scope per persona cut with the business; the question restated as time-to-value and unit cost. *Exit bar:* business signs the bare minimums; engineering agrees to re-estimate with confidence levels.
- **Plan & Design** — re-estimates 9 months (operator) vs 6 months (partner) with confidence buffers (×1.0 / ×1.4 by confidence, +15% testing); the 20%-still-in-old-app finding; work-item-grain cost-to-serve translation (−25% vs −50%); rebuild-vs-reuse rule per module (>8 dev-weeks → reuse; rank by benefit × impact ÷ dev-weeks); reversal trigger written; read privately to my manager first; engineering given two days to break it. *Exit bar:* my manager puts his name on the flip at the Bellevue onsite; sequencing signed.
- **Build & Test** — 15 domains × 6 modules; Jira burn-up with the scope tripwire — new work estimated and swapped, never absorbed; operator-interactions-eliminated as the success metric; the tenet "no go-live without every work item assigned to one operator." *Exit bar:* burn-up 55.8% → 83.2% across two quarters; every work item owned.
- **Cutover & Go-Live (waves)** — the import-side app live April 2022 with 11 partners in 10 countries; partner migration plan published; operators' two further quarters on old screens priced and published. *Exit bar:* 48 of 53 import partners migrated within a month.
- **Hypercare / Close** — tracked interactions eliminated against the 2,200 baseline by work item; sequenced FCL 2022 → Air/LCL 2023 off that read; close-out at ~$5M cost avoidance and 66% less effort. *Exit bar:* value in month 6 instead of 18; operators moved onto the shared chassis on the published schedule.

Influence and ownership. Mine: the baseline memo, the comparison model, the rebuild/reuse rule, the scope tripwire and the reversal trigger. The business signed the bare minimums; engineering produced the re-estimates; my manager signed the sequencing and owned the flip. He argued two points, I adjusted one, and the onsite ratified it — I have never won a disagreement by seniority; I rank, privately first, and let the room ratify.

> **Designed the process; others own it now.** I designed the comparison model, the rebuild-vs-reuse rule and the scope tripwire; the delivery teams run the burn-up and the swap rule, and the owner of the original plan owned the flip. **Trade-off put to leadership.** The recommendation to leadership put the competing metrics side by side — time-to-first-value and partner cost-to-serve (6 months, −50%) vs operator effort (9 months, −25%) — with the operators' two further quarters on old screens priced, not hidden.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~$5M cost avoidance (record, said once — never "avoidance" and "annual reduction" as two results); underneath it a ≈$1.5M/yr run-rate *modeled* at −50% CTS on an illustrative 20K export shipments; the decomposition ≈$2.9M engineering avoided + ≈$1.5M pulled forward ≈ $4.4M is my model, within ~12% of the record.
- **#** — 48 of 53 import partners migrated within a month; the import-side app live April 2022 with 11 partners in 10 countries; baselines 339.5 min/shipment and 2,200 interactions (record).
- **%** — engineering effort to first value −66%; CTS −50% partner-first vs −25% operator-first at the work-item grain; burn-up 55.8% → 83.2% across two quarters (record).
- **Time** — first value in month 6 instead of month 18 (record); FCL 2022 → Air/LCL 2023 sequenced off the interactions read.
- **Customer** — partners completed end-to-end in the new app; operators' two further quarters on old screens priced and published, not hidden (record).
- **Quality** — new scope estimated and swapped, never absorbed; no go-live without every work item assigned to one operator (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	An 18-month committed roadmap with no cost-to-serve read behind it. I restated the question as time-to-value and unit cost, built the baseline memo (339.5 min; 2,200 interactions) and set the success metric — operator interactions eliminated.
Scope & influence	15 domains × 6 modules; the business signed the bare minimums, engineering produced the re-estimates, my manager signed the sequencing and owned the flip at the onsite — influence on the owner of the original plan, one level up.
Execution trade-off + mechanism others now own	Partner-first (6 months, −50% CTS) vs operator-first (9 months, −25%), with the operators' two further quarters on old screens priced and published. Mechanism others own now: the rebuild-vs-reuse rule per module and the scope tripwire on the burn-up.
Impact (long-term)	~$5M cost avoidance (record); first value in month 6 instead of 18; FCL 2022 → Air/LCL 2023 sequenced off the interactions read; 48 of 53 import partners migrated within a month.
Process improvement (standard work institutionalized)	Confidence-buffered estimates (×1.0 / ×1.4, +15% testing); the tenet "no go-live without every work item assigned to one operator"; new scope estimated and swapped, never absorbed.
Complexity & consensus	A disagreement with my manager, privately first and with numbers; two days for engineering to break the model; his name on the flip — the onsite ratified it.
Root cause at scale	The 20%-still-in-old-app finding and the work-item-grain cost-to-serve translation located where value actually accrued, so sequencing ran on unit economics rather than feature count.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

HM — You changed your manager's plan. How, and would you do it to me?

Privately first, with the numbers — 9 versus 6 months, 20% still in the old app, −25% versus −50%. He argued two points, I adjusted one, and the onsite ratified it with his name on it. Yes — the same way: the model, the owner named, the reversal trigger written, and two days for your engineers to break it before anyone else sees it.

HM — Is the $5M a saving or an avoidance?

Avoidance — engineering effort never spent plus a year of unit savings pulled forward — and I say it once. The recurring run-rate underneath is a separate, smaller modeled number (≈$1.5M/yr on an illustrative 20K shipments); I never present the avoidance as an annual reduction.

HM — What stopped the operators from being abandoned?

A priced, published two-quarter schedule — the trade was visible and scored, not hidden — and the partner backend was the chassis operators later rode, so their move got cheaper, not later.

HM — Your engineer count and rate are yours, not the program's.

Correct — twelve engineers at $20K a month are my numbers; the record does not itemize the $5M. At 10–15 engineers and $15–25K the effort piece runs $1.8–4.5M; adding the pulled-forward run-rate (≈$1.5M at the base) puts the whole decomposition at $2.9–6.7M, and the ~$5M record figure sits inside that range — the effort piece alone does not reach it. I would firm it with the two scoping envelopes (66.2 and 99.8 engineer-months) and Finance's loaded rate.

HM — What made the big swing safe?

Three mechanisms: confidence buffers on low-confidence estimates, a written reversal trigger, and the scope tripwire in the burn-up — new work estimated and swapped, never absorbed — which is why the burn-up read 55.8 → 83.2% honestly.

HM — Lesson, and how it shows up in your first quarter here?

Bring the model, name the owner, write the reversal trigger. In the first quarter: re-baseline any committed plan in time-to-value and unit cost at the work-item grain before defending or changing it.

> **HM — What broke after the April 2022 import-side launch that the plan had not seen?**
>
> Asia latency RECORD-FROM-DOCS: the business review showed P95 page load at 21.19 seconds in Asia against 3.3 in the US — static assets ~5s in China against ~500ms in the US — because the domain had no geo-based DNS or reverse proxy and Asia traffic rode the public internet. The fix was infrastructure sequencing, not re-architecture: WAF enablement, the double CDN removed, a regional CDN proposal with smoke tests per change, China tuning then Asia tuning. The lesson imported into every later plan: the pilot's geography is a performance requirement, not a rollout detail.

> **HM — What compliance bar does a partner-facing app carry, and what did it cost?**
>
> An immutable audit trail RECORD-FROM-DOCS: who/what/when for logins, work-item changes, route updates, messages, documents, schedule updates, exceptions, assignment rules and permission changes — nine categories at a 100%-captured bar, across ~100 partner orgs and thousands of changes a day. The honest cost answer: the PRD phased it Alpha → Beta → MVP precisely because capturing everything on day one was not feasible; the residual risk — the CISO's vague "important events" — was mitigated with sign-off, not silence.

> **HM — What did the March 2023 export-partner deadline change about the sequencing?**
>
> It made the module order do double duty RECORD-FROM-DOCS: schedule automation had to complete by end Q3/early Q4 2022 so UAT could start for the March 2023 export-partner launch, and FCL went ahead of Air/LCL because that is where the volume and the interactions baseline (2,200 per departed shipment) lived. The unit-economics sequencing and the deadline sequencing pointed the same way — which is exactly what made the flip ratifiable at the onsite.

THE KEY MATH — THE TWO PATHS AT THE WORK-ITEM GRAIN, AND THE RECORD'S ~$5M SAID ONCE

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	CTS per shipment	`339.5 min × $0.45/min ≈ $153`	record inputs; blended loaded rate derived, labeled	≈$153/shipment
2	Operator-first path	`9 months → −25%; operators still 20% in old app`	engineering re-estimates (record)	≈$38/shipment saved, month 18 to full value
3	Partner-first path	`6 months → −50%; partner backend = the chassis`	engineering re-estimates (record)	≈$76/shipment saved, 12 months earlier
4	Effort to first value	`100 − 66 = 34% of the inherited path`	record	−66%
5	Run-rate	`$76 × 20,000 export shipments = $1.52M`	volume illustrative (modeled)	≈$1.5M/yr modeled
6	Decomposition of the avoidance	`12 eng × 18 mo × 66% × $20K ≈ $2.9M + 12 mo × $1.5M = $1.5M → $4.4M`	counts and rate mine; within ~12% of the record	~$5M record headline, said once
7	Confidence rule	`(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 wks vs 20 naive`	two illustrative 10-week modules	buffers inflate only low-confidence items

Burn-up trajectory 55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% (record). Rebuild-vs-reuse threshold of 8 dev-weeks was the engineering median (record rule).

Where this lands at Amazon. Think Big with Frugality attached — re-open a committed plan when the goal matters more than the plan, price it in unit economics, and have the owner whose name was on the original ratify the change. For a NACF process roadmap: sequence by time-to-value and cost-to-serve at the work-item grain, with a reversal trigger and a scope tripwire written before the swing.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S12

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you thought big / changed the direction of a program.	Think Big	The destination changed — value in month 6 not 18 — on evidence at the work-item grain; the mechanisms that made the swing safe.	Do not tell it as a Sephora story (loop-packet conflict); the record is Flexport.
Tell me about a time you saved money / did more with less.	Frugality	~$5M avoided by sequencing, said once; 66% less effort; rebuild-vs-reuse over 8 dev-weeks.	Never "$5M annual cost reduction" in the same breath as the avoidance.
Tell me about a time you disagreed with your manager.	Have Backbone	Private first, with numbers; two days for engineering to break it; his name on the flip at the onsite.	Skip S14 (the quiet-fix refusal) — held for the bar raiser.
How do you prioritize when everything is committed?	Functional — roadmap governance	Benefit × impact ÷ dev-weeks per module; new work estimated and swapped, never absorbed; the burn-up as the honest read.	Do not claim the roadmap was mine — it was my manager's.
Tell me about a trade-off you made that hurt someone.	Earn Trust / Customer Obsession (internal)	Operators on old screens two more quarters — priced, published and scored, not hidden.	Do not present the operator delay as costless.
What's a decision you'd reverse if the data changed?	Are Right, A Lot	The reversal trigger was written before the flip; the interactions-eliminated read sequenced Air/LCL 2023, not a date on a slide.	No YoY framing; the avoidance is one-time.
How does re-sequencing a roadmap relate to launch readiness? (transferability probe — public-profile inference)	Functional — transfer	Her world is 50+ FC launches: readiness discipline is the same shape as the flip — a baseline memo (the readiness scorecard), gates that fire on evidence (the reversal trigger ≈ a launch-readiness gate), and the owner of the original plan ratifying the change. Sequence by time-to-value at the work-item grain, never by the date on the slide.	Do not claim FC-launch experience; bridge from the mechanism, not the domain.
How would you apply cost-to-serve thinking to a placement network? (transferability probe)	Functional — transfer	CTS per container is cost-to-serve at one node grain; placement is the same arithmetic at the node's position — distance-to-customer priced against speed and cost per unit moved. The work-item grain (339.5 minutes; 2,200 interactions) is how I keep it honest.	Never quote her $50MM / $80MM savings as shared ground — those are her numbers, framing only.

DEEPENING S12 — FIVE FACETS, FOUR-TABLE MATH, BEHAVIORAL PACK, ANGLES AND TRAP DRILLS NEW

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Baseline memo: 339.5 minutes per shipment and 2,200 interactions per departed shipment; per-persona bare-minimum scope; re-estimates by path at the work-item grain; the 20% still in the old app.
2 · Solution design	Operator-first vs partner-first; rebuild vs reuse by module; confidence buffers; swap-never-absorb scope tripwire; a written reversal trigger; the manager's name on the flip.
3 · Math & data logic	Four tables: $152.78 ≈ $153; −25% vs −50%; 18 → 6 months; −66% ⇒ 34%; 27.6 vs 20 weeks; $2.85M + $1.52M ≈ $4.4M vs the ~$5M record.
4 · Implementation	15 domains × 6 modules; burn-up 55.8% → 83.2%; import-side app live April 2022 with 11 partners in 10 countries; 48 of 53 import partners (about 90%) migrated; FCL 2022, Air and LCL 2023; the export launch is the later Q1 2023 milestone.
5 · Probe questions	Seven numeric probes plus three trap drills: avoidance vs saving, invented engineers, and the 27.6-week rule.

Facet 1 — Data analysis: what each analysis decided

ANALYSIS	DATA AND GRAIN	WHAT IT SHOWED	DECISION IT DROVE
Baseline memo	minutes and interactions per departed shipment	339.5 min and 2,200 interactions — the two numbers everything prices from	the debate became arithmetic, not persona preference
Cost-to-serve translation	339.5 min × ≈ $0.45 per min (derived blend)	$152.78 ≈ $153 per shipment	each path priced as a percent of $153
Re-estimate by path	engineering re-estimates with confidence levels, work-item grain	operator-first 9 months, −25%; partner-first 6 months, −50%	partner-first, with operators on the shared chassis later
Legacy residue	share of operator time still in the old app	20% under operator-first	caps operator-first at half the unit saving
Module cost vs benefit	dev-weeks per module; benefit × impact	median 8 dev-weeks (record rule)	over 8 dev-weeks → reuse, ranked by benefit × impact ÷ dev-weeks
Burn-up	completed ÷ pointed work	55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% over two quarters	new work estimated and swapped, never absorbed

Facet 2 — Solution design: options weighed

QUESTION	OPTIONS	CHOSEN	WHY
Who goes first	Operators (inherited) · partners · both in parallel	Partners first; operators ride the same chassis later	6 months and −50% against 9 months and −25%; the partner backend is reused, so operator work gets cheaper, not later.
Build or reuse	Rebuild each module · reuse legacy · rule by size	Over 8 dev-weeks → reuse	Engineering's own median; module by module, not ideology.
Estimate honesty	Single-point estimates · confidence buffers	× 1.0 / × 1.4 by confidence, plus 15% testing buffer	Buffers only inflate low-confidence items: 27.6 weeks against 20 naive.

QUESTION	OPTIONS	CHOSEN	WHY
Change control	Absorb new scope · swap	Estimate and swap, never absorb	The burn-up reads honestly: 55.8 → 83.2%.
How to change the plan	Escalate over the manager · private first with numbers	Private first, two days for engineering to break it, then the room ratifies	The fastest path to a better decision was making it his decision.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Inherited plan	`operator-persona first: 18 months to first value, socialized, my manager's plan`	record	month 18
2	Unit baselines	`339.5 minutes of handling per shipment; 2,200 operator interactions per departed shipment`	record	the two numbers everything prices from
3	Cost-to-serve per shipment	`339.5 min × $0.45 per min = $152.78 ≈ $153`	record inputs (FCL $0.56, Network $0.61, Data $0.24 per min; the blend is derived, labeled)	≈ $153 per shipment
4	Operator-first path	`9 months to −25%; operators still spend 20% of their time in the old app`	engineering re-estimates (record)	−25%, capped by legacy residue

Table 2 — The fix: paths, rules and tripwire, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Partner-first path	`6 months to −50%; the partner backend is the chassis operators later ride`	engineering re-estimates with confidence levels (record)	−50%, 12 months earlier
2	Unit saving per path	`$153 × 50% = $76.50 ≈ $76; $153 × 25% = $38.25 ≈ $38`	from Table 1	double the unit saving
3	Effort to first value	`100 − 66 = 34% of the inherited path`	record	−66% ⇒ 34%
4	Confidence rule	`(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 weeks against 20 naive; 27.6 / 20 = +38%`	two illustrative 10-week modules; the multipliers and 15% buffer are the record rule	27.6 vs 20 weeks
5	Rebuild-vs-reuse	`dev-weeks > 8 → reuse; rank by benefit × impact ÷ dev-weeks`	the 8-week rule was engineering's median (record rule)	module by module
6	Scope tripwire	`burn-up 55.8 → 83.2% = +27.4 points over two quarters; new work swapped, never absorbed`	record	+27.4 points

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Time to first value	`month 18 → month 6`	record	12 months earlier
2	Engineering effort to first value	`100% → 34%`	record	−66%

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
3	Cost-to-serve per shipment	≈ $153 → ≈ $76 (−50%)	record inputs; derived	≈ $76
4	Partners migrated	48 of 53 import partners within a month: 48 / 53 = 0.9057 → about 90% (the documents print 90.5%)	record (resume, Asia-latency doc); ratio mine	48 of 53 — about 90%
5	Pilot reach	import-side app live April 2022 with 11 partners in 10 countries; 15 domains × 6 modules = 90 modules	record; product mine	11 partners · 90 modules

Table 4 — The dollars: every multiplication shown; the ~$5M said once

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Unit saving	$153 × 50% = $76 per shipment	from Table 2	≈ $76
2	Run-rate	$76 × 20,000 export shipments = $1,520,000	modeled (volume illustrative)	≈ $1.5M a year
3	Engineering avoided	12 eng × 18 mo = 216 eng-months; × 66% = 142.6; × $20,000 = $2,851,200	modeled (counts and rate mine; benchmark $15–25K)	≈ $2.9M one-time
4	Value pulled forward	12 months × ($1.52M ÷ 12) = $1.52M	modeled, from the run-rate	≈ $1.5M one-time
5	Decomposition check	$2.9M + $1.5M = $4.4M; ($5.0M − $4.4M) / $5.0M = 12% under the record	modeled on rounded inputs; the ~$5M is the record, said once	$4.4M vs ~$5M

Facet 4 — Implementation: the flip, the build and the cutover

PHASE	WHAT HAPPENED (RECORD UNLESS MARKED)	CONTROL
Decide	private read to my manager first; engineering two days to break the model; the onsite ratifies with his name on the flip	reversal trigger written before the flip
Build	15 domains × 6 modules; Jira burn-up 55.8% → 83.2% across two quarters; no go-live without every work item assigned to one operator	scope swapped, never absorbed
Cutover	import-side app live April 2022 with 11 partners in 10 countries; 48 of 53 import partners (about 90%) migrated within a month; operators' two further quarters on old screens priced and published	partner migration plan published
Sequence	FCL 2022, then Air and LCL 2023, read off interactions eliminated against the 2,200 baseline	by work item

Lean / Kaizen mapping. Time-to-value is the lead-time waste; the 20% still in the old app is inventory of unfinished transition; the rebuild-vs-reuse rule is standard work for the make-or-reuse decision; the swap-never-absorb tripwire is a pull system for scope; the written reversal trigger is the PDCA "Check" fixed in advance.
Honest note. The record's single engineering-effort figure is −66%; the effort-envelope numbers sometimes quoted alongside it are not used here.

Facet 5 — Probe questions (numeric)

> **Saving or avoidance? Annual?**
>
> Avoidance, one-time, said once: about $5M, record. My decomposition — $2.85M of engineering never spent plus $1.52M of unit saving pulled forward — is $4.4M, within 12% of it. The recurring run-rate underneath, $1.52M a year, is a separate modeled number; I never present the avoidance as annual.

Your engineers and rate are invented.

They are my model, and I say so: ten to fifteen engineers at $15–25K a month moves the effort piece from $1.8M to $4.5M; with the pulled-forward piece, the decomposition ranges $2.9M to $6.7M, and the record's ~$5M sits inside that range — not inside the effort piece alone.

Why 27.6 weeks and not 20?

(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6: the 1.4 applies only to the low-confidence module and the 15% is testing buffer, so the plan is 38% longer than naive and honest about which parts are risky.

Why is partner-first twice the saving?

Because at the work-item grain operator-first reaches −25% in 9 months with 20% of operator time still in the old app, and partner-first reaches −50% in 6 months: $76 against $38 per shipment, 12 months earlier.

Why 8 dev-weeks?

It was engineering's median module cost — the record rule — so the threshold is theirs, not mine: over 8 dev-weeks, reuse; otherwise rebuild, ranked by benefit × impact ÷ dev-weeks.

What did operators pay?

Two more quarters on the old screens — priced, published and scored, not hidden. The partner backend was the chassis they later rode, so their move got cheaper, not later.

What broke after the April 2022 import-side launch?

Asia latency — the monthly review showed 21.19 seconds at P95 against 3.3 in the US; see S12-A for the full investigation. The lesson: the pilot's geography is a performance requirement, not a rollout detail.

SENSITIVITY — LOW / BASE / HIGH

CASE	EFFORT INPUTS	ENGINEERING AVOIDED (ONE-TIME)	EXPORT SHIPMENTS	RUN-RATE / PULLED FORWARD	DECOMPOSITION TOTAL
low	10 engineers × $15,000	$1.8M	15,000 shipments	$1.1M	$2.9M
base	12 engineers × $20,000	$2.9M	20,000 shipments	$1.5M	$4.4M
high	15 engineers × $25,000	$4.5M	30,000 shipments	$2.3M	$6.7M

Rows are modeled, not record; the program's ~$5M is the single signed figure and the base decomposition ($4.4M) is within 12% of it. Note: the high case here reads $6.7M on unrounded inputs (the master rounds to $6.8M).

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> About five million of cost avoidance, and I say that once. Roughly two point nine of it is engineering effort we never spent — sixty-six percent less on a plan that was eighteen months long — and about one and a half is a year of unit savings pulled forward because value landed in month six instead of month eighteen. Behind that sits a run-rate of about a million and a half a year at half the cost-to-serve on twenty thousand export shipments. The sixty-six percent, the six versus eighteen months and the minus fifty versus minus twenty-five are the record; the five million is the program's signed figure; the split and the shipment volume are my model.

IF CHALLENGED — THE FULL ANSWERS

"Twelve engineers and twenty thousand a month are yours, not the program's."

Correct, and the record does not itemize the five million, so I modeled the split. At ten to fifteen engineers and $15–25K a month the effort piece is $1.8M–$4.5M and the whole decomposition $2.9M–$6.7M; the record's ~$5M sits inside the decomposition range.

"Where do twenty thousand shipments come from?"

An illustrative FCL export volume, and the softest input: at 15,000 to 30,000 the run-rate runs $1.14M to $2.28M. I would replace it with the departed-shipment count from the system that produced the 2,200-interaction baseline.

"Are you double counting the pulled-forward value and the five million?"

No. The five million is the total; the pulled-forward year is one slice of how it decomposes, and the run-rate underneath is not a second result.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport I re-sequenced a committed 18-month roadmap on cost-to-serve evidence — partners first, operators on the same chassis later — so value landed in month 6, effort to first value fell 66%, and the program recorded about $5M of cost avoidance.

NACF BRIDGE — ONE SENTENCE

It is the discipline of sequencing launches or placement moves by time-to-value at the unit grain, with a written reversal trigger and the owner of the original plan ratifying the change.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Give me an example of how you have changed the direction or view of a specific function/department and helped them embrace a new way of thinking. Why was a change needed? What was the outcome?"

AIQB: Think Big Q2 | Think Big | FACET: SOLUTION DESIGN

Open (first two sentences): "The roadmap said eighteen months to first value, operators first. I re-opened it, and the new plan put value in month six and delivered twice the unit saving."

90-second STAR. **S** A committed, socialized 18-month plan owned by my manager. **T** Change the plan without authority over the roadmap, in a way he could own. **A** I priced both paths at the work-item grain — partner-first 6 months and −50%, operator-first 9 months and −25% with 20% of operator time still in the old app — added confidence buffers, a rebuild-vs-reuse rule, a scope tripwire and a written reversal trigger. **R** First value in month 6 instead of 18; effort to first value −66%; ~$5M cost avoidance, said once.

Land: 18 → 6 months; −66% effort · **Pivot to NACF:** sequencing launches by time-to-value at the unit grain.

Do NOT say: "$5M annual savings" or the roadmap as mine.

B2 · "Tell me about a time when you strongly disagreed with your manager or peer on something you considered very important to the business. What was it and how did you handle it? Knowing what you know now, would you do anything differently?" AIQB: Have a Backbone Q1 **Have Backbone; Disagree and Commit** FACET: IMPLEMENTATION

Open (first two sentences): "The plan had my manager's name on it. I read it to him privately first, with the numbers, and gave engineering two days to break my model before anyone else saw it."

90-second STAR. S A plan my manager owned, with socialized commitments. **T** Change it without becoming a political fight. **A** Private read; he argued two points and I adjusted one; engineering got two days to break the model; the onsite ratified with his name on the flip. **R** He owned the flip; the plan changed; the model survived.

Land: two points argued, one adjusted · **Pivot to NACF:** changing a launch sequence: model first, private first, owner of the old plan on the new one.

Do NOT say: that I won on seniority or escalated — I never have.

B3 · "Give me an example of how you have helped save costs or eliminate waste within your role or organization. What was the situation? What was the impact?" AIQB: Frugality Q1 **Frugality** FACET: MATH AND DATA LOGIC

Open (first two sentences): "About five million dollars of cost avoidance — engineering we never spent plus a year of unit savings pulled forward — and I say it once, because it is avoidance, not an annual reduction."

90-second STAR. S An 18-month plan that would have consumed the full engineering effort. **T** Reach first value with less spend. **A** Reuse over 8 dev-weeks, partner backend as the shared chassis, confidence buffers, scope swaps instead of absorption. **R** Effort −66%; decomposition $2.85M + $1.52M = $4.4M against the record ~$5M.

Land: ~$5M cost avoidance (record), said once · **Pivot to NACF:** reuse over rebuild in a network program: the 8-week rule is an engineering-median threshold.

Do NOT say: avoidance and annual reduction as two results; the engineers and rate as the program's.

B4 · "Tell me about a time when you had to communicate a change in direction that you anticipated people would have concerns with. What did you do to understand the concerns and mitigate them?"

AIQB: Earn Trust Q1 **Earn Trust** FACET: DATA ANALYSIS

Open (first two sentences): "Operators would stay on the old screens two more quarters. I priced that and published it instead of hiding it."

90-second STAR. S A sequencing choice that delayed one group for the benefit of the whole. **T** Be transparent about who pays. **A** The baseline memo showed 20% of operator time still in the old app under the inherited path; I priced two more quarters for operators, published the schedule, and made the partner backend the chassis they later ride. **R** No hidden trade; operators' move got cheaper, not later.

Land: two quarters, priced and published · **Pivot to NACF:** a network change that burdens one node temporarily: price it, publish it, give the date it ends.

Do NOT say: that the delay was costless.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "Three hundred thirty-nine and a half minutes and twenty-two hundred interactions per shipment — with those two numbers the roadmap debate became arithmetic."

Headline metric: 339.5 min × $0.45 ≈ $153; −25% vs −50%

- Price each path at the work-item grain.
- The 20% residue caps operator-first at half.
- Confidence buffers: 27.6 vs 20 weeks.
- Say "derived" on the $153.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: “It was my manager's plan. I read it to him alone, with numbers, and gave engineering two days to break it.”

Headline metric: two points argued; one adjusted

- Private first, then the room.
- His name on the flip.
- Reversal trigger written in advance.
- Never won on seniority.

Honest failure-and-learning angle **Are Right, A Lot**

Opener: “What I did not see was Asia: right after the April 2022 import-side launch, the review showed 21 seconds at P95 against 3.3 in the US.”

Headline metric: 21.19 s vs 3.3 s

- The plan sequenced by value and effort, not by the pilot's geography.
- The fix was infrastructure sequencing (see S12-A).
- The lesson imported into later plans: geography is a performance requirement.
- What held: the sequencing decision itself.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Five million — is that saved every year?"

Numeric answer. No. It is one-time avoidance, the record's signed figure. My decomposition is $2.85M engineering never spent + $1.52M pulled forward = $4.4M; the recurring run-rate is $1.52M a year, a separate modeled number on illustrative volume. They are never both claimed.

Say exactly: “It's avoidance — one time, said once. The run-rate underneath is a separate, smaller model; I never present the avoidance as annual.”

"Your twelve engineers — where did they come from?"

Numeric answer. My model: 12 × 18 months × 66% × $20,000 = $2,851,200. At 10–15 engineers and $15–25K the effort piece is $1.8M–$4.5M; with the pulled-forward piece the decomposition is $2.9M–$6.7M, and the record's ~$5M sits inside that range.

Say exactly: “Those are my modeled inputs — the record doesn't itemize the five million. Across ten to fifteen engineers it moves from about two to four and a half million; I'd firm it with Finance's rate.”

"Why is your plan 27.6 weeks when two 10-week modules is 20?"

Numeric answer. (10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 — the 1.4 applies only to the low-confidence module, plus a 15% testing buffer; 38% over naive, and the buffer is not spread evenly.

Say exactly: “The buffer only goes where confidence is low — twenty-seven point six against twenty naive — so the extra eight weeks are the risky module, not padding.”

S12-A — Flexport — Why Asia waited 21 seconds: a P95 latency investigation that split one complaint into a static-content problem and a routing problem, then ran it as a dated, smoke-tested change plan NEW

LPs this story can serve here: Dive Deep (primary) · Customer Obsession · Bias for Action · Insist on the Highest Standards
Process family: Improve — root-cause investigation of a regional performance gap; change control for network and security changes

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep The review said "Asia is slow"; the work was to find where the 21.19 seconds went — region split, static versus dynamic, and a China static-JS P95 of 5 s against 0.5 s in the US — before anyone bought a CDN.

Customer Obsession Asia was 45.3% of the 13.92 thousand page views in the window, so the slowest region was the largest region — the plan was ordered by where the partners actually were.

Bias for Action Reversible fixes went first — double CDN removed in China, WAF on globally — and the expensive one (CDN for the dynamic host) was deferred until the metrics said it was needed.

Insist on the Highest Standards Every WAF or CDN change ran a written pre-check, a post-check and a rollback-able DNS override — 32 smoke checks a change — and I will not claim a single-cause before-and-after that the record does not support.

Honesty label. This is an investigation story with a split of credit: the investigation document is the app team engineer's, with the network infrastructure owner running the CDN and WAF changes — *I did not write the root-cause doc or make the network changes*. What I tell as mine is the program side: the monthly review that surfaced the gap, the cut by region, the dated plan with owners, the change-control discipline and the insistence on a like-for-like measurement. **My seat** (my own workbook, Apple v11 B3, and the document): the investigation is authored by the app engineer, and the network-infrastructure owner made the CDN and DNS changes (the plan's owners are named in the document); I drove the follow-ups from the monthly review and kept the plan, the sequencing and the China-host smoke tests honest. I say "I drove the follow-ups", not "I fixed the CDN". **Before-and-after flag:** the investigation window (Apr 24 – May 10) and the review series differ in window and region definition, so the 81% figure I can compute is indicative only — I do not quote it as the result; the like-for-like record is the review series. The document's Asia table and its China-specific plan rows belong to one investigation, with China the sharpest tail; the later review series splits Asia (non-China) from China, so the 21.19 s and the later 4.0 s may not be the same population — I never subtract across them. Overlap: this story is also on the bench in another interviewer's packet.

WHAT THE RECORD SAYS (FROM THE INVESTIGATION DOCUMENT AND THE WEEKLY REVIEW SHEET; RECORD-FROM-DOCS)

Context. The Freight Forwarding App (FFA) launched its import-side app in April 2022 to 11 partners across 10 countries; through May, 48 of 53 import partners were migrated. The monthly review showed Asia countries with much higher latency than other regions. RECORD-FROM-DOCS

The numbers. Page-load P95 for Apr 24 – May 10 (Datadog view loading time, which covers static resources, the GraphQL request and the render): US 3.3 s on 2.19k views; Asia 21.19 s on 6.31k views; Others 4.45 s on 5.42k views. Static JavaScript P95: China 5 s against 500 ms in the US. RECORD-FROM-DOCS

Two named causes. (1) Static resources are served from a CDN host that performs poorly in China; (2) the dynamic host has no geo-based DNS or regional reverse proxy for Asia, so GraphQL requests cross the public internet instead of the data-center backbone. RECORD-FROM-DOCS

The plan. Eight rows, owners and ETAs: three security (WAF proposal, WAF for China, WAF globally — follow-ups to an incident) and five latency (remove the double CDN in China — done, ETA June 24; static-CDN proposal for China and Asia, July 15; tuning for China, July 30; tuning for Asia, August 15; CDN for the dynamic host globally — "decide based on the metrics", Q3 or later). A written change process: override DNS locally, verify resolution, smoke-test

internal single-sign-on and partner users, repeat on the China instance, change, then remove the override and re-test. RECORD-FROM-DOCS

The review series. The weekly review sheet carries page-load P95 overall and by region (weeks 33 to 52 and week 1) and a monthly view (April to September). The sheet's action items include adding GraphQL latency at P50, P90 and P95 by region (done October 11, 2022). RECORD-FROM-DOCS

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. We had launched the import-side freight-forwarder app in April 2022 and moved 48 of 53 import partners onto it by May. At the monthly review the latency chart had an outlier: Asia. **Task.** Find out whether it was one problem or several, and where the time was going, before anyone bought anything. **Action.** I cut the 13.92 thousand views by region over the 17-day window: Asia was 45% of the views and waited 21.19 seconds at the 95th percentile against 3.3 in the US — 6.4 times. Then I asked where the time goes: static JavaScript took 5 seconds in China against half a second in the US, and the dynamic requests from Asia went over the public internet because there was no regional edge. Two causes, two owners. I ordered the plan by reversibility — remove the double CDN, turn on the firewall — held the expensive global CDN until the metrics asked for it, and put every network change behind a written DNS-override and smoke-test checklist. **Result.** Primary metric: the Asia-to-US gap of 6.42× — and in the review series that followed, Asia read 4.0 seconds against 4.09 in North America by the last week of the year, with overall monthly P95 down from 8.39 to 4.68 seconds. I tell that as a gap that closed, not as a single-cause saving.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “Asia waited about twenty-one seconds at the ninety-fifth percentile against about three and a third in the US — roughly six and a half times — over a seventeen-day window of about fourteen thousand page views.”

“What's your source for that?” “It is the latency document's own table; the six-point-four is my division of its two printed figures, and the later four seconds against four-oh-nine is from the monthly review series — I describe it as a gap that closed, not a single-cause saving.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Flexport's freight-forwarder app was in its first months: the import-side app live to 11 partners in 10 countries in April 2022, then 48 of 53 import partners (about 90%) by the end of May. The monthly review flagged that Asian users saw far higher page-load latency than anyone else — 21.19 seconds at P95 against 3.3 in the US on the same two-and-a-half-week window — and Asia was the single largest slice of traffic. The risk: a freight partner who waits twenty seconds for a shipment page goes back to email, and the cost-to-serve case for the whole migration depends on partners using the app.

T — Turn "Asia is slow" into named causes with owners and a dated plan, sequence the changes so the cheap and reversible ones run first, keep the production network and security changes safe, and prove the improvement with a measurement that can be trusted.

Baselines on the table: Page-load P95: US 3.3 s · Asia 21.19 s · Others 4.45 s · views 2.19k / 6.31k / 5.42k · window Apr 24 – May 10 (17 days) · static JS P95 China 5 s vs US 0.5 s · 48 of 53 partners migrated · 8 plan rows (3 security, 5 latency).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — cut P95 and views by region over the window; read the resource-level data; mapped the request path for static content and for GraphQL; wrote the two-cause hypothesis and what each predicts. *Exit bar:* a region table with views, and a named hypothesis per cause.
- **Plan & Design** — sequenced eight rows by reversibility and dependency: security first where an incident was open, then the double-CDN removal, then proposals, then tuning; wrote the decision rule to defer the global CDN until the

metrics asked for it; listed the open measurement question — how to test the China CDN in a consistent, reproducible way. *Exit bar:* owners and ETAs on every row; one conditional row with its trigger.

- **Build & Test** — the change process: override DNS locally, verify resolution, smoke-test internal single-sign-on and partner log-in, workview, log-out and re-log-in, repeat on the China instance; infra performs the change; then remove the override, clear the cache and test again. *Exit bar:* 32 checks per change (16 before, 16 after).
- **Cutover & Go-Live** — WAF on for China and globally and the double CDN removed in China, each behind the checklist, ETAs June 24 – June 30; static-asset tuning for China and Asia dated July 30 and August 15. *Exit bar:* four of eight rows marked done in the document; dated owners on the rest.
- **Hypercare / Close** — the monthly and weekly reviews carry page-load P95 by region; the action item to add P50 / P90 / P95 GraphQL latency by region closed October 11, 2022; the week-47 spike was triaged against the region split rather than assumed to be the network. *Exit bar:* a regional series the review reads every week.

Influence and ownership. Mine (as I would tell it): the review that surfaced the gap, the region cut, the sequencing rule, the change checklist discipline and the insistence on a like-for-like measure. Not mine: the root-cause write-up, the CDN and WAF changes, the security sign-offs and the app optimizations — the app engineer and the network-infra owner own those, with the security reviewer signing the firewall proposal.

> **Designed the process; others own it now.** The change process, the region cut in the weekly review and the P50 / P90 / P95 action item live in the document and the sheet, not in my head. **Trade-off put to leadership.** Buy a global CDN now (certain cost, unknown benefit) against two cheaper reversible fixes first and a metrics trigger — the schedule risk was that the global CDN row slipped to Q3 or later.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — No dollar exists in the record. I model exposure, labeled illustrative: Asia's 45.3% of views against the S12 run-rate ≈ $0.69M a year (modeled upper bound, not savings); the direct tail time is ≈ $1.6K a year, which is why the case is adoption, not seconds.
- **#** — Asia 21.19 s vs US 3.3 s = 6.42× on 6.31k of 13.92k views; 32 smoke checks per change; 8 plan rows, 4 marked done.
- **%** — overall page-load P95, review monthly series, April to September: 8.39 s → 4.68 s, −44.2% (record-from-docs); Asia (non-China) 5.5 → 4.0 s, −27.3%, North America 5.62 → 4.09 s, −27.2% over weeks 39 to 52 (record-from-docs, my arithmetic).
- **Time** — plan span June 24 → August 15 = 52 days; the P50 / P90 / P95 by-region action item closed October 11, 2022.
- **Customer** — Asia partners were the largest slice of views; by the last week of the year the review series put Asia (non-China) within 2% of North America, and China at 1.57× North America.
- **Quality** — zero unreviewed network changes — every WAF or CDN change had its pre-check and post-check — and a before-and-after claim limited to what the like-for-like series supports.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	One complaint, no known cause. The work was to split it into two testable causes (static content path, dynamic routing path) and to name what would disprove each.
Scope & influence	Engineering, network infrastructure and security each owned a piece; the plan was a one-page sequence with owners and dates that none of them could do alone.
Execution trade-off + mechanism others now own	Deferred the global CDN on a metrics trigger — a deliberate schedule risk. The mechanism others own: the change checklist, the by-region series and the P50 / P90 / P95 view.
Impact (long-term)	The review series that followed shows the Asia gap gone by week 39 and China as the remaining outlier — a clear next investigation instead of a vague complaint.
Process improvement (standard work institutionalized)	DNS-override + smoke-test + rollback as the standard for WAF and CDN changes; regional percentiles on the weekly review.
Complexity & consensus	App team, network infrastructure and security with different incentives — a written plan with a conditional row resolved the "just buy the CDN" push without a standoff.
Root cause at scale	The 6.42× gap was a path problem, not an app problem — week 47 proved it from the other side, when North America spiked 2.09× as well.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	P95 and views by region; Asia 45.3% of views; static JS 10× slower in China; P95 versus P99 sample sizes (63 observations against 22); the review series by region and week.
2 · Solution design	Static path versus dynamic path; six options weighed; reversible-first sequence; metrics trigger for the global CDN.
3 · Math & data logic	Gap ratios 6.42× and 1.35×; tail views 315.5; 52-day plan; 32 checks; −44.2% overall; Asia / NA 0.98×; exposure $0.69M (illustrative).
4 · Implementation	Eight plan rows with owners and ETAs; the DNS-override change process; WAF first, double CDN removed next; China tuning, then Asia.
5 · Probe questions	Seven numeric probes plus three trap drills: the 81% temptation, P95s that do not add, P99 on thin data.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. Three sources, all RECORD-FROM-DOCS: (1) the investigation document's region table — page-load P95 and views for Apr 24 – May 10; (2) its resource-level cut — static JS P95 China 5 s vs US 0.5 s; (3) the weekly and monthly review sheet by region (weeks 33–52, week 1; April–September monthly). I added the shares, the ratios and the tail-sample arithmetic.

REGION	PAGE-LOAD P95	VIEWS	% OF VIEWS	VS US	TAIL VIEWS (5%)
US	3.30 s	2,190	15.7%	1.00×	109.5
Asia	21.19 s	6,310	45.3%	6.42×	315.5
Others	4.45 s	5,420	38.9%	1.35×	271.0
Total	—	13,920	100%	—	696

What it revealed. (1) *The largest region was the slowest:* Asia carried 45.3% of views at 6.42× the US P95. (2) *Two paths, two causes:* static JS in China at 10× the US P95; dynamic GraphQL from Asia over the public internet with no regional edge. (3) *Why P95 and not P99:* in 17 days Asia has 6,310 views, so P99 rests on 63 observations and the US P99 on 22 — P95 rests on 316 and 110, which is stable enough to steer by. (4) *P95s do*

not add: the 5 s static-JS P95 is not 5 s of the 21.19 s page-load P95; the right test is to re-measure page load after every change. (5) *The later review series is a different population* — it splits Asia (non-China) from China — so I never subtract across the two sources. (6) *Week 47:* North America went from 4.52 to 9.44 s (2.09×) in the same week Asia went from 7.16 to 16.4 s (2.29×) while China fell — an application-wide regression, not a regional network one. The review sheet's own 12/12 callout explains a GraphQL-latency spike for the Core and Client Apps as a change in the metrics code that made the dashboard numbers wrong, fixed retroactively — a measurement error, not a network one; the October regression is a separate cause (work_view 2.0 un-optimized queries, action item 11). I do not quote week 47 as an Asia result.

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Static content in China	Increase CDN cache time · remove the double CDN (direct connect to the object store) · replicate assets into China / Asia (needs security approval) · another CDN or a global accelerator	Remove the double CDN first; replicate assets after a proposal and sign-off	Removal is reversible and needs no new vendor; replication needs a security approval first.
Dynamic requests from Asia	CDN for the dynamic host · geo-based DNS with a regional reverse proxy over the backbone · leave it	Regional edge proxy via geo DNS is the target; global CDN deferred on a metrics trigger	The document's own wording: finish the other items, then decide on the metrics.
Security vs latency order	Latency first · WAF first	WAF proposal and rollout first for China and globally	An open incident follow-up; the WAF and the CDN share a change path, so one checklist covers both.
How to measure the China CDN	Real-user views only · synthetic probes · both	Open question in the document	Raised as "consistent and reproducible"; resolved partly by adding P50 / P90 / P95 by region to the review (done October 11, 2022) VERIFY-WITH-RAJESH · V4
Priority	By worst P95 · by traffic share	By traffic share, reversibility and dependency	Asia = 45.3% of views; China is the sharpest tail but a smaller share.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Views in the window	`2.19k + 6.31k + 5.42k = 13.92k; Apr 24–30 = 7 days + May 1–10 = 10 days = 17 days`	record-from-docs inputs	13.92k views over 17 days
2	Asia's share of views	`6.31 / 13.92 = 45.3%; per day 6,310 / 17 = 371 of 13,920 / 17 = 819`	my arithmetic on record inputs	45.3% · 371 of 819 a day
3	P95 gaps	`21.19 / 3.30 = 6.42×; 4.45 / 3.30 = 1.35×; 21.19 / 4.45 = 4.76×`	my arithmetic on record inputs	Asia 6.42× US
4	Static JS	`China P95 5 s / US P95 0.5 s = 10×`	record-from-docs inputs	10×
5	Tail sample: why P95	`5% × 6,310 = 315.5 Asia views above P95; 1% × 6,310 = 63.1 above P99; US 1% × 2,190 = 21.9`	my arithmetic; the document gives no P99	P95 on 316 views; P99 on 63 (US 22)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
6	Adoption context	`48 / 53 import partners migrated — about 90% (the document prints 90.5%)`	record-from-docs inputs; rounding mine	about 90%

Table 2 — The fix: the plan, the change control and the measurement rule, values substituted

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Plan composition	`3 security rows + 5 latency rows = 8; marked done: 3 security + 1 latency = 4`	record-from-docs plan rows; count mine	8 rows, 4 done
2	Plan span	`June 24 → August 15: 6 days left in June + 31 in July + 15 in August = 52 days`	record-from-docs ETAs; arithmetic mine	52 days
3	Smoke checks per change	`2 user types (internal SSO, partner) × 4 steps (log-in, workview, log-out, re-log-in) × 2 instances (main, China) = 16; before + after = 16 + 16 = 32`	record-from-docs change process; count mine	32 checks per change
4	Decision rule for the global CDN	`do the reversible items first; re-read the regional P95; buy the global CDN only if Asia and China stay above the others after tuning`	record-from-docs wording; rule stated as I would run it	a trigger, not a date
5	Weekly sample for one P95 read	`6,310 / 17 × 7 = 2,598 Asia views a week; 5% = 130 tail views`	my arithmetic at the window rate; volume has grown since	≈ 130 tail views a week

Table 3 — After: the review series (record-from-docs), before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Overall P95, monthly	`April 8.39 s → September 4.68 s: (8.39 − 4.68) / 8.39 = 44.2%`	record-from-docs; arithmetic mine	−44.2%
2	Asia, weekly	`week 39 5.5 s → week 52 4.0 s: (5.5 − 4.0) / 5.5 = 27.3%`	record-from-docs (Asia non-China); arithmetic mine	−27.3%
3	North America, weekly	`week 39 5.62 s → week 52 4.09 s: (5.62 − 4.09) / 5.62 = 27.2%`	record-from-docs; arithmetic mine	−27.2%
4	China, weekly	`week 39 9.9 s → week 52 6.43 s: (9.9 − 6.43) / 9.9 = 35.1%`	record-from-docs; arithmetic mine	−35.1%
5	Gap to North America	`Asia 5.5 / 5.62 = 0.98× (week 39); 4.0 / 4.09 = 0.98× (week 52); China 6.43 / 4.09 = 1.57×`	my arithmetic on record-from-docs series	Asia at parity; China 1.57×
6	Week 47 spike	`NA 4.52 → 9.44 s = 2.09×; Asia 7.16 → 16.4 s = 2.29×; China 13.8 → 9.36 s (fell)`	record-from-docs; arithmetic mine	not regional
7	Indicative only — do not quote	`(21.19 − 4.0) / 21.19 = 81.1% mixes a 17-day investigation window with a week-52 review figure and possibly different region definitions`	my arithmetic; NOT a result	indicative

Table 4 — The dollars: none in the record; two labeled models

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Record dollars	`the investigation prices nothing; the plan rows carry no cost`	record-from-docs	$0 in the record
2	Excess seconds per tail view	`21.19 − 4.0 = 17.19 s (P95 to P95; indicative)`	illustrative: uses the indicative pair	17.19 s
3	Tail hours in the window	`315.5 views × 17.19 s = 5,423 s; 5,423 / 3,600 = 1.51 h`	illustrative arithmetic	1.51 h

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
4	Direct tail time per year	`1.5065 h × 365 / 17 = 32.3 h; 32.3 h × $50 = $1,617`	illustrative: $50 an hour assumed	≈ $1.6K / yr
5	Exposure on the S12 run-rate	`$76 × 20,000 = $1,520,000; × 6.31 / 13.92 = $689,023`	modeled on a modeled run-rate (volume illustrative); views are not shipments — an upper-bound exposure, not savings	≈ $0.69M / yr at risk

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

PLAN ROW (RECORD)	REGION	ETA	OWNER	STATUS IN THE RECORD
WAF proposal for FF App with the process	All	06/30	app engineer / network infra	done
WAF for the main host in China (incident follow-up)	CN	06/30	network infra / app engineer	done
WAF globally (incident follow-up)	All	06/30	network infra / app engineer	done
Remove the double CDN for static content in China — direct connect to the object store	CN	06/24	network infra	done
Proposal to resolve static CDN performance in China and Asia	CN, Asia	07/15	app engineer + network infra	open
Performance tuning for static content — China	CN	07/30	network infra / app engineer	open; depends on proposal
Performance tuning for static content — Asia	Asia	08/15	network infra / app engineer	open; depends on proposal
CDN for the dynamic host globally	All	Q3 or later	network infra	conditional on the metrics

Change process (record) and my controls. Before: local DNS override, verify resolution, smoke test for internal and partner users, repeat on the China instance. Change: network infra only. After: remove the override on both hosts, clear the cache, re-test. **Risks named:** security approval before replicating assets in China; no consistent way yet to benchmark the China CDN; dependency of the tuning rows on the proposal. **Lean / Kaizen mapping.** Going to see where the seconds are is genchi genbutsu; the two causes are root-cause analysis (5 Whys by path); the DNS-override checklist is poka-yoke and standard work; each change followed by a re-measure is PDCA; removing the double CDN is eliminating a waste step; every tail view is waiting waste.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why P95 and not the average or P99?**
>
> The average hides the tail users feel and P99 is too thin: in 17 days Asia has 6,310 views, so P99 rests on 63 observations and the US on 22, while P95 rests on 316 and 110. P95 is stable enough to steer by and sensitive enough to show 21.19 seconds against 3.3.

> **How do you know it was a network problem and not the app?**
>
> Two checks. Resource-level: static JS P95 was 5 s in China against 0.5 s in the US. And by region over time: in week 47 North America doubled (4.52 → 9.44, 2.09×) with Asia (7.16 → 16.4, 2.29×) while China fell — that is an application regression, a different problem, which I did not blame on the network.

Did the plan actually close the gap?

The like-for-like review series says the Asia (non-China) gap to North America was closed by week 39 — 5.5 against 5.62 — and stayed there at week 52 (4.0 against 4.09); China remained at 1.57×. What I cannot show is which change moved it between May and September — the record does not attribute.

Why defer the global CDN?

Because two reversible changes were ahead of it and the document itself said to decide on the metrics. A global CDN is a certain cost and an unknown benefit; the right time to buy it is when the regional P95 stays above the others after tuning.

What did the investigation cost and save in dollars?

The record prices nothing. I model it as adoption protection: Asia was 45.3% of views, and 45.3% of the S12 run-rate is ≈ $0.69M a year of exposure — an upper bound, not savings. The direct tail time is about $1.6K a year, which is why the argument was never seconds times dollars.

What would you do differently?

Start with P50, P90 and P95 by region from day one — the action item to add them closed only on October 11, 2022 — and decide up front how to benchmark the China CDN consistently, which the document listed as an open question.

Your 81% — where is it from?

It is (21.19 − 4.0) / 21.19, which mixes the 17-day investigation window with a week-52 review figure — an indicative number, not a result. The like-for-like numbers are −27.3% for Asia over weeks 39 to 52 and −44.2% overall from April to September.

BUSINESS IMPACT MODEL — KPI TREE

```
Asia partners can use the app without waiting: gap to the home region closed, tail visible, changes safe
+- Gap: Asia 21.19 s vs US 3.30 s = 6.42x on 45.3% of views (record)
|    +- static path: China JS 5 s vs 0.5 s = 10x -> remove double CDN (done) -> proposal -> tuning
|    +- dynamic path: no geo DNS / regional proxy -> GraphQL over the public internet -> edge proxy (CDN
deferred on metrics)
+- Result (review series, record): overall P95 8.39 -> 4.68 s (-44.2%); Asia 5.5 -> 4.0 s vs NA 5.62 ->
4.09 s; China 6.43 s = 1.57x NA
+- Control: 32 smoke checks per WAF / CDN change; P50 / P90 / P95 by region on the weekly review
+- $ (illustrative / modeled): tail time ~ $1.6K/yr; exposure ~ $0.69M/yr (upper bound, not savings)
(loop)  complaint -> split by region -> split by path -> reversible fix first -> re-measure -> next fix
```

SOLUTION FLOW

```
monthly review: Asia is the outlier
   -> cut by region: views, P95  (17 days; Asia 45.3% of views, 6.42x US)
   -> cut by path: static (CDN) vs dynamic (GraphQL route)
   -> two causes, two owners, one dated plan (8 rows)
   -> every change: DNS override -> smoke test x16 -> change -> remove override -> smoke test x16
   -> re-read regional P95 -> decide: next tuning row, or the conditional CDN row
```

SENSITIVITY — LOW / BASE / HIGH

ASIA SHARE OF VIEWS (ILLUSTRATIVE RANGE)	EXPOSURE ON THE S12 RUN-RATE	READ
30%	$1,520,000 × 30% = $456,000	≈ $456K a year at risk
45.3% (observed)	$1,520,000 × 45.3% = $689,023	≈ $689K a year at risk
60%	$1,520,000 × 60% = $912,000	≈ $912K a year at risk

Both the $76 × 20,000 run-rate and the view-to-shipment mapping are modeled; the three rows are a range of the share, not a forecast. None of it is claimed as savings.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

The record number here is a gap, not a dollar. At the monthly review Asia showed a twenty-one second page-load P95 against three-point-three in the US — six and a half times — on forty-five percent of the views. I split it into two paths: static content in China, where the JavaScript took five seconds against half a second, and dynamic requests that crossed the public internet with no regional edge. We sequenced the cheap, reversible changes first, put every network change behind a thirty-two-check smoke test, and held the expensive global CDN until the metrics asked for it. In the review series that followed, overall monthly P95 fell from eight-point-four to four-point-seven seconds, and Asia read four seconds against four-point-one in North America by the last week of the year. I tell that as a gap that closed, not as one change that saved something — the record doesn't attribute it, and any dollars I attach, about seven-tenths of a million of exposure, are modeled and not claimed.

IF CHALLENGED — THE FULL ANSWERS

"You are claiming credit for an infrastructure fix you didn't build."

No. The root-cause document and the CDN and WAF changes were the app engineer's and the network-infra owner's. What I own is the region cut, the sequence, the change checklist discipline and the measurement honesty — and I say so in the first sentence.

"Latency is a technical metric. Where is the business?"

Asia was 45.3% of views, and the cost-to-serve case for the whole migration depends on partners using the app. The exposure on the S12 run-rate is ≈ $0.69M a year as a modeled upper bound; the direct tail time is ≈ $1.6K — which is the point: the case is adoption.

"Why not just throw a global CDN at it?"

Because the reversible items came first and the document tied the global CDN to the metrics. A CDN would add cost and ops without telling us which hop is slow.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport the monthly review showed Asian users waiting 21 seconds at P95 on the freight-forwarder app — 6.4 times the US, on 45% of the views. I split it into a static-content problem and a routing problem, sequenced the reversible fixes first behind a 32-check smoke test, and the review series later showed Asia level with North America.

NACF BRIDGE — ONE SENTENCE

A geo-routed edge near the user with a backbone between sites is the cross-dock placement idea in software: put a node near demand, run the linehaul between hubs, and measure the tail — a P95 on inbound transit or dock wait, not the average, is what sizes appointment buffers and capacity.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Walk me through a big problem or issue in your organization that you helped to solve. How did you become aware of it? What information did you gather? What information was missing and how did you fill the gaps?" AIQB: Dive Deep Q4 **Dive Deep** FACET: DATA ANALYSIS

Open (first two sentences): "At a monthly review, Asia showed a twenty-one second page-load P95 against three seconds in the US. The average would have hidden that; I split the tail by region and then by request path."

90-second STAR. S Freight-forwarder app launched in April 2022, 48 of 53 partners on it by May; one region far slower than the rest. **T** Find out where the time goes before anyone buys a fix. **A** I cut 13.92 thousand views by region — Asia 45.3% of views at 6.42× the US P95 — then by path: static JS P95 5 s in China against 0.5 s in the US, and GraphQL from Asia over the public internet with no regional edge. I checked sample sizes — P99 would have rested on 63 observations — and used P95. **R** Two named causes with owners, a dated eight-row plan, and a regional series on the weekly review.

Land: 6.42× on 45.3% of views · **Pivot to NACF:** inbound transit-time by origin lane: the tail by region, not the average, is what sizes buffers and dock-appointment slack.

Do NOT say: that I found the root cause alone, or "I cut Asia latency 81%".

B2 · "Tell me about a time when you evaluated the customer experience of your product or service. What did you do? What was the result?" AIQB: Customer Obsession Q5 **Customer Obsession** FACET: SOLUTION DESIGN

Open (first two sentences): "The slowest region was also the largest: Asia was forty-five percent of the views. So the plan was ordered by where the partners actually were, not by what was easiest to tune."

90-second STAR. S Freight partners across 10 countries using the new app; the review showed one region waiting far longer. **T** Prioritize the work by who is hurt most. **A** I weighed six options — cache time, double-CDN removal, asset replication, regional edge, global CDN, an accelerator — and ordered them by traffic share, reversibility and dependency; the global CDN waited on a metrics trigger. **R** A plan where the first changes were the reversible ones, and by week 39 the review series put Asia (non-China) within 2% of North America.

Land: 45.3% of views · **Pivot to NACF:** siting a node: weight the lanes by volume first, then the tail; the largest flow decides the sequence.

Do NOT say: any dollar figure as savings, or customer complaints that the record does not document — the document says the slowness was noticed in the monthly business review.

B3 · "Give me an example of a calculated risk that you have taken where speed was critical. What was the situation and how did you handle it? What steps did you take to mitigate the risk? What was the outcome?" AIQB: Bias for Action Q1 **Bias for Action** FACET: IMPLEMENTATION

Open (first two sentences): "We started with the changes that could be undone — removing a double CDN, turning on the firewall — and held the expensive global CDN until the metrics asked for it."

90-second STAR. S A latency gap and an open security incident follow-up on the same network path. **T** Move fast on the reversible, slow on the expensive. **A** I sequenced eight plan rows with owners and ETAs from June 24 to August 15 — 52 days — put the firewall and the double-CDN removal first, and wrote the DNS-override smoke-test routine so each change could be verified and backed out. **R** Four of eight rows marked done in the document by June 30; the conditional row stayed conditional.

Land: 52-day plan; 4 of 8 rows done · **Pivot to NACF:** peak prep: do the reversible changes first and buy capacity last — the same logic as flexing labor before adding a door.

Do NOT say: that every row finished on time — only four were marked done; the tuning rows had ETAs, not results.

B4 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 **Insist on the Highest Standards**

FACET: PROBE QUESTIONS

Open (first two sentences): "Every WAF or CDN change ran thirty-two smoke checks, and I refused to quote a before-and-after the data couldn't support — including a flattering eighty-one percent I could have computed."

90-second STAR. S Production network changes in two regions, plus a tempting headline number. **T** Keep production safe and the claims honest. **A** The change process: DNS override, resolution check, internal and partner log-in, workview, log-out and re-log-in, repeated on the China instance, before and after. For the result I used the like-for-like review series, not the investigation window against a week-52 figure. **R** No unreviewed network change; claims limited to −44.2% overall and Asia 5.5 → 4.0 s over weeks 39 to 52.

Land: 32 checks per change; 81.1% not quoted · **Pivot to NACF:** a standard pre-check and post-check on every dock or yard change, and a rule not to compare windows that are not like for like.

Do NOT say: that the 81% is a result; that I attributed the drop to one change.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: "The average looked fine; the ninety-fifth percentile in Asia was twenty-one seconds, on forty-five percent of the views."

Headline metric: 6.42× on 6.31k of 13.92k views

- Region cut first; path cut second (static vs dynamic).
- Sample-size logic: P99 on 63 observations vs 22 in the US; P95 on 316 / 110.
- P95s do not add: 5 s of JS is not 5 s of 21.19.
- Week 47: NA 2.09×, Asia 2.29× — application, not network.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: "The loud ask was "just turn on a global CDN"; I held it until the metrics asked for it, and that cost us a schedule row."

Headline metric: one conditional row (Q3 or later); 4 of 8 rows done

- Network infra, the app team and security each owned a piece; none could close it alone.
- The plan's conditional row resolved the push without a standoff.
- Security approval before replicating assets in China — I kept that gate.
- My seat: I drove the follow-ups and kept the plan and the smoke-test checklist honest; the app engineer and network owner made the changes.

Honest failure-and-learning angle **Learn and Be Curious**

Opener: "We measured P95 only; P50 and P90 by region did not exist until October, and we never had a like-for-like before-and-after."

Headline metric: window-mismatched 81.1% — not quoted

- The action item to add P50 / P90 / P95 by region closed on October 11, 2022.
- The China CDN had no consistent way to be benchmarked — an open question in the document.
- What I would do differently: baseline by region on the weekly series from day one.
- What held: no unreviewed change and no overclaimed result.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"So you cut Asia latency by 81%."

Numeric answer. No. (21.19 − 4.0) / 21.19 = 81.1% subtracts a 17-day investigation figure from a week-52 review figure that may not be the same region population. Like-for-like: Asia (non-China) 5.5 → 4.0 s = −27.3% over weeks 39 to 52; overall monthly 8.39 → 4.68 s = −44.2%.

Say exactly: "I wouldn't quote eighty-one percent — it compares two different windows. The like-for-like review series shows overall P95 down forty-four percent from April to September, and Asia level with North America by week 39."

"Static JS was 5 seconds of your 21 — so fixing it saves 5 seconds?"

Numeric answer. No. A P95 of one resource is not a slice of a page-load P95, and P95s do not add. The 5 s China JS P95 against 0.5 s in the US (10×) points at the path; the number to steer by is the page-load P95, re-measured after each change.

Say exactly: "Percentiles don't add — five seconds at P95 for one resource isn't five seconds off the page. I re-measured page-load P95 after each change instead."

"Why P95? Why not P99?"

Numeric answer. Sample size. In the 17-day window Asia had 6,310 views, so P99 rests on 1% × 6,310 = 63.1 observations and the US on 21.9; P95 rests on 315.5 and 109.5. At roughly 2,600 Asia views a week, a weekly P95 still has about 130 tail views.

Say exactly: "In seventeen days the US had about twenty-two views past its P99 — too thin to steer by. P95 had over a hundred in the US and over three hundred in Asia."

S16-A — Flexport — The data model under the Monday review: a container-week grain, a ticket-to-container bridge, a 1% parity gate and a sync-drop alarm that turned four hand-stitched systems into one Snowflake pipeline NEW

LPs this story can serve here: Dive Deep (primary) · Invent and Simplify · Frugality · Insist on the Highest Standards
Process family: Improve — data pipeline and reconciliation design under an operating metric; build-vs-bridge and cutover gating

LEADERSHIP PRINCIPLES THIS STORY CARRIES — ONE-LINE PROOF SENTENCE EACH

Dive Deep The cost-to-serve ratio only works if tickets and containers are counted at the same grain; a join at the wrong grain would have made it look 44.4% better than it was (illustrative), so the grain was fixed before the pipeline.

Invent and Simplify A ticket-to-container cross-reference table shipped the join now, while the real key was pushed upstream to Support — ship with the bridge, fix the source — and seven frozen metrics replaced a pile of ad-hoc pulls.

Frugality The status quo was priced in the BRD (over $100K a year, priced not saved) so the infrastructure team could prioritize the build themselves; scope stayed at seven metrics and the existing BI tool — no new system bought.

Insist on the Highest Standards Nothing switched until every frozen metric matched the manual pull within 1% for consecutive weeks, every mismatch explained by cause, and a failed feed raised an alert instead of silence.

Honesty label. I wrote the business requirements document and owned the acceptance gate, the parallel-run procedure and the cutover call; *the infrastructure team built the pipeline* — I did not lay pipe, I made the pipe buildable (Apple A3). The ">$100K a year" is a **priced baseline** — priced, not claimed as savings: the BRD's own arithmetic is $103,680 (30 → 6 hours a week across two weekly reviews at an assumed $90 an hour) and my split of it into hours and decision risk is a model. The BRD names **no table or column** — it names the Zendesk Ticket ID, the Crux Container Number, Flexport core data and Looker SQL Runner views — so I describe the join, the grain and the artifacts, not a production schema; the container-week grain is my design logic, anchored to the weekly-stock grain the review already used (Apple A2). **Four systems:** the BRD names the ticket system, the Crux container pull (by hand, in Google Sheets) and core data; Apple A3's whiteboard lists ticketing, scraper registry, product-event DB and CRM — say "four systems" and name the ticket system and the container pull with confidence. **Overlap:** the failure angle here is the data-access security review (Apple G3), told as the one miss it was; the data-model, grain and reconciliation angle is this story's own.

WHAT THE RECORD SAYS (FROM THE BUSINESS REQUIREMENTS DOCUMENT AND THE PROGRAM NOTES; RECORD-FROM-DOCS)

Problem. Operator intervention for Shipment Visibility is managed in the ticketing system: a ticket is created when the visibility service fails to start tracking a submitted container automatically, and Data Services closes it. To track key metrics automatically, ticket data must be joined to container numbers from the visibility vendor's data, and then to core shipment data. RECORD-FROM-DOCS

As-is. Manual pulls from the ticketing system matched to a manual pull of containers into spreadsheets: 1.5–2 hours on average to produce one week-over-week snapshot of seven metrics (coverage automated, coverage with human input, tickets generated, tickets per active container, 4PL % within SLA, 3PL % within SLA, and the 4PL-vs-3PL delta). RECORD-FROM-DOCS

Requirements. BR-1 customer data protected; BR-2 interoperates with other capabilities; BR-3 feed daily, preferably hourly; BR-4 an alert to internal support users when the feed fails. Objectives: automated reporting through the BI tool's SQL runner; easy joins to core data. RECORD-FROM-DOCS

The pricing in the document. "On Average 3 resources are spending 5 hrs per week for 2 WBR's" at an *assumed* $90 an hour — printed as a $10,880 monthly cost amounting to $129,600 a year; reduced to 1 hr a week, $25,920; "Savings here would be $103,680 a year" (BRD p1–2; 30 → 6 hours a week). The printed monthly figure does not match its own annual (3 × 5 × 2 × 4 × $90 = $10,800). RECORD-FROM-DOCS **Say:** "over $100K a year — the BRD

priced it at $103,680 on thirty hours a week at an assumed ninety dollars an hour; my split into hours and decision risk is a model; priced, not a claimed saving."

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The Monday review ran on four systems that did not talk, stitched together in spreadsheets by me — system accuracy 60%, 1.5 to 2 hours a snapshot, and tickets that carried no container number, so the program's cost-to-serve could not be computed from the data. **Task.** Write the requirements and the acceptance test for a pipeline that produces the same seven numbers without me. **Action.** I fixed the grain first — container by week, with tickets and containers counted at the same grain, because a join at ticket grain would have made cost-to-serve look about 44% better than it was. I shipped a ticket-to-container cross-reference table so the join could go live and pushed a real container ID upstream to Support. I priced the status quo in the BRD so the infrastructure team could prioritize the build, then set the gate: every frozen metric within 1% of the manual pull for consecutive weeks, mismatches explained by cause — mostly late tickets on the manual side — and an alert whenever the feed fails. **Result.** Primary metric: system accuracy 60% to 90% on one pipeline instead of four systems — and the first automated catch was a ticket-sync drop of about 4% that nobody could have seen by hand. The dollar I carry is a price in the BRD — over $100K a year, $103,680 on its own arithmetic — priced, not a saving.

SAFE-SAY — TWO SENTENCES TO SAY EXACTLY

Most attackable number. “The review was priced at over a hundred thousand dollars a year — the requirements document works it to a hundred and three thousand, six hundred and eighty on thirty hours a week at an assumed ninety dollars an hour — and that is a price, not a saving I claim.”

“What's your source for that?” “It is my own requirements document; the arithmetic and its monthly typo are printed on the page, my split into hours and decision risk is a model, and the accuracy, the parity gate and the four-percent catch are in my interview workbook.”

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The program's weekly review needed seven operating metrics, and two of them — tickets generated and tickets per active container — required joining the support ticketing system to container data. The tickets did not carry a container ID. So the numbers were assembled each Monday by hand from four systems into spreadsheets: system accuracy 60%, one person able to produce them, and silent data loss nobody could see. A VP was steering the program by a number that only one person could produce.

T — Define the data contract and the acceptance test: one grain, a join that survives missing keys, definitions in code, a reconciliation gate to cut over, and a failure alert — and get the infrastructure team to build it by showing them the price of the status quo.

Baselines on the table: Four systems · 1 person · system accuracy 60% · 1.5–2 h per weekly snapshot (BRD) · 7 frozen metrics · tickets with no container ID · no failure alarm · priced baseline: BRD $129,600 a year → $103,680 difference (30 → 6 h/wk at an assumed $90/h), over $100K priced, not saved (my model: $100,360).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is — which systems, which pulls, which join — named the key gap (tickets without a container ID), froze seven metric definitions with their owners, and priced the status quo in the BRD (over $100K a year) so the infrastructure team could prioritize the build themselves. *Exit bar:* infra prioritizes the build off the price; scope is the seven metrics, nothing new.
- **Plan & Design** — wrote the requirements (BR-1 to BR-4), chose the container × week grain, specified the documented ticket-to-container cross-reference, set BR-4 (feed-failure alert) and the stale-table alerts, and sketched the pipeline: scheduled extracts → S3 landing zone → SQS-triggered loads → Snowflake raw → curated metric tables → Looker SQL Runner views. *Exit bar:* requirements accepted by infra; acceptance gate published before build.

- **Build & Test** — extract and load specs per system; the bridge table first; parity checks per metric; the parallel-run procedure, every mismatch explained by cause; the data-access security review that surfaced mid-build and cost two weeks. *Exit bar:* all seven metrics inside 1% of the manual pull for consecutive weeks; review closed.
- **Cutover & Go-Live** — the cutover memo; the Monday deck switched to the pipeline; the manual pull kept running as rollback until parity had held a week beyond the gate. *Exit bar:* first Monday off the pipeline shows the same seven numbers.
- **Hypercare / Close** — watched the failure alerts; triaged the first automated catch (a 4% ticket-sync drop) to a source cause; moved definitions into code; pushed the real container key upstream. *Exit bar:* accuracy 60% → 90%; the pipeline steers the review without me.

Influence and ownership. Mine: the as-is documentation, the grain choice, the bridge-table design, the seven frozen definitions, the pricing, the acceptance gate, the parallel-run procedure and the cutover memo. Not mine: the pipeline build and its alerts (infrastructure team), the real container key (Support), the security review (security team). I influenced by pricing the status quo in the BRD and making the cutover decision unarguable with a published gate.

> **Designed the process; others own it now.** The definitions are in code, the gate is a documented procedure and the alerts page a team, not me. **Trade-off put to leadership.** New metrics for stability (the seven were frozen, with a parking lot), and two weeks of date for a data-access review I should have planned.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (RECORD VS MODELED LABELED)

- **$** — Over $100K a year, priced in the BRD, not saved: the BRD route gives $103,680 (30 → 6 h a week at an assumed $90/h); my model gives $100,360 ($30,160 hours + $70,200 decision risk). Both are priced, not claimed as savings. Realized candidate: 208 hours retired × $90 = $18,720 (modeled).
- **#** — Four systems → one pipeline; seven definitions in code; no longer a single point of failure (one engineer maintains the pipeline instead of one PM rebuilding it, Apple A3); zero WBR deliveries skipped through the build (record).
- **%** — System accuracy 60% → 90% (record); parity gate within 1% per metric for consecutive weeks (record); first automated catch a ticket-sync drop of about 4% (record).
- **Time** — the weekly snapshot (1.5–2 h by hand per the BRD) → automated; the data-access review cost two weeks (record).
- **Customer** — the VP and the core team steered on the same seven numbers the first Monday off the pipeline; Data Services and Support got tickets joined to containers for the first time.
- **Quality** — every parallel-run mismatch explained by cause — mostly late-arriving tickets on the manual side; silence on a failed feed replaced by an alert.

L6 LEVEL EVIDENCE — this story against the Amazon L6 functional rubric; every line is this story's record or labeled model

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No one had written requirements for a program's operating metrics, and the cost-to-serve ratio had no join key. I defined grain, bridge and gate before anyone built.
Scope & influence	A pipeline owned by another organization, a key owned by Support and a review owned by security; I moved all three by pricing and publishing the gate, not by authority.
Execution trade-off + mechanism others now own	Seven metrics frozen for stability; two weeks lost to a review I should have planned. The mechanism others own: definitions in code, the parity procedure, the alerts.
Impact (long-term)	The weekly review no longer depends on one person, and the pipeline sees what the sheets could not — the first catch was a 4% drop.
Process improvement (standard work institutionalized)	Parallel-run-until-parity with a written gate; a bridge-then-fix-the-source rule for missing keys; a kickoff question about data-access approvals and their lead time.
Complexity & consensus	Four source systems, infra, Support and security with different priorities — one price in the BRD and one published gate aligned them.
Root cause at scale	The reason the number could be wrong was structural: the unit of cost (a ticket) and the unit of volume (a container) had no shared key or grain.

FIVE FACETS AT A GLANCE — the labeled structure the probing section below follows

1 · Data analysis	Ticket grain vs container grain; the key gap; late-arriving tickets; which mismatches were the manual side's fault; sample of what silence hid.
2 · Solution design	Sheets automation vs warehouse pipeline; container × week grain; bridge table vs waiting for the key; hourly vs daily feed; alarm design.
3 · Math & data logic	Join fan-out (0.667 vs 1.20); parity 0.42% pass and 1.25% investigate; alarm 4% vs 1%; BRD $129,600 → $103,680 vs model $100,360.
4 · Implementation	BR-1 to BR-4; parallel run to rollback; cutover memo; alert triage; security review as a dependency with a lead time.
5 · Probe questions	Seven numeric probes plus three trap drills: priced vs saved, 1.25% investigate-or-fail, sync-drop threshold.

PROBING — FIVE FACETS, THEN THE FULL DRILL

Facet 1 — Data analysis (what data, grain, joins, cuts, what it revealed)

What I read. Three data sets named in the BRD **RECORD-FROM-DOCS**: (1) Zendesk tickets — one row per ticket, created when the visibility service fails to start tracking a submitted container automatically, closed by Data Services; (2) Crux container data — pulled by hand into Google Sheets, keyed by container number; (3) Flexport core data in the warehouse. The unit of *cost* is a ticket; the unit of *volume* is a container; the ratio needs both, so I analyzed the grain at which each could be counted.

DATA SET	UNIT (GRAIN)	KEY IT CARRIES	WHAT IT IS USED FOR	WEAKNESS FOUND
Zendesk tickets	one row per ticket	Ticket ID; *no container ID*	numerator: tickets generated	cannot be joined to a container
Crux container data	one row per container	Container Number	denominator: active containers; coverage	a manual pull into Google Sheets, taken once a week
Flexport core data	one row per shipment / container	container number, client, mode	cuts: 4PL vs 3PL; SLA	not joined to tickets until a bridge exists

The seven metrics the BRD froze, and the grain each lives at

#	METRIC (BRD WORDING)	NUMERATOR	DENOMINATOR / CUT	GRAIN
1	Coverage — % of containers tracked (automated)	containers tracked with no human input	active containers	container-week stock
2	Coverage — % of containers tracked (with human inputs)	containers tracked after a human touch	active containers	container-week stock
3	Cost to serve — tickets generated	tickets	—	ticket, counted per week
4	Cost to serve — tickets generated / active container	tickets	active containers	needs the ticket → container join
5	Timeliness — 4PL % within SLA	containers inside SLA	4PL containers	container-week, 4PL cut
6	Timeliness — 3PL % within SLA	containers inside SLA	3PL containers	container-week, 3PL cut
7	Timeliness — delta (4PL vs 3PL)	metric 5 less metric 6	—	derived

Metric names are the BRD's. The numerator, denominator and grain columns are my design logic — the BRD does not define them. "Active container" is the review's own definition: cumulative inflow less containers 15 days past departed_terminal (Apple A2, grain 2).

What it revealed. (1) *No key, no ratio:* without a container ID on the ticket, tickets per active container could only be assembled by hand RECORD-FROM-DOCS. (2) *Grain decides the headline — in the real instrument:* the review workbook (Apple A2, SV_28.xlsx) shows 80.6% automated at event grain (1,991 of 10,289 events human-touched = 19.4%), 54.1% at ticket grain (538 auto, 456 human) and 40.0% on 17 Jan falling to 21.9% on 31 Jan at weekly stock — three grains, three answers, so the grain label travels with every number (record). My worked join example makes the same point for the cost ratio: 12 tickets over 4 of 10 active containers give 18 rows at ticket grain, 12 / 18 = 0.667 instead of 12 / 10 = 1.20, a ratio 44.4% too good (illustrative — my design example, not program data). (3) *Late-arriving tickets:* the manual snapshot missed tickets that arrived after the pull — nearly every parallel-run mismatch was the manual side (record, Apple A3). (4) *Silence was invisible:* a ticket-sync drop of about 4% after a routine update looked like a quiet week in the manual era (record).

Facet 2 — Solution design (options considered, chosen design, why)

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Where the logic lives	Automate the spreadsheets · BI on a ticket export · warehouse pipeline + BI views	Warehouse feed + BI SQL views (the objectives in the document)	Joins to core data are one query; the manual pull stays a rollback, not the source.
Grain	Ticket · container · container × week	Container × week (Monday–Sunday, as the review reads it)	Numerator and denominator at one grain prevents fan-out; matches the weekly-stock grain the review already used (design logic; Apple A2 grain 2).
The missing key	Wait for Support to add a container ID · parse it from ticket text · cross-reference table now + real key upstream	Bridge now, key upstream	Ship with the bridge, fix the source: the program did not wait for another team's roadmap.

DESIGN QUESTION	OPTIONS WEIGHED	CHOSEN	WHY (RECORD UNLESS MARKED)
Feed frequency	Daily · hourly	Daily, preferably hourly (BR-3)	BR-3 says daily, preferably hourly; my whiteboard sketch has daily with hourly on the fast tables (Apple A3). The review is weekly, so daily is sufficient; hourly helps alarm latency.
Failure handling	Dashboard only · alert to support users · both	Alert when the feed fails (BR-4) and when sync rate drops	The first catch — a 4% drop — only exists because an alert does.
Cutover	Big-bang · parallel run to a gate · phased by metric	Parallel run to a 1% gate for consecutive weeks, manual pull as rollback	Makes the decision unarguable — the VP sees the same seven numbers.

The data elements, from the record (the BRD names no tables or columns)

ELEMENT	GRAIN / KEY	WHAT THE RECORD SAYS	SOURCE
Zendesk tickets (feed)	one row per ticket · Ticket ID	created when auto-tracking fails to start; closed by Data Services; the queue's UPDATED_BY_ID splits human from auto	BRD p1; Apple A2 grain 3
Crux container data	one row per container · Container Number	was a manual pull into Google Sheets; statuses include departed_terminal, in_review, out_of_network	BRD p2; Apple A2
Flexport core data	shipment / container	joined in Looker SQL Runner; carries the 4PL / 3PL and SLA cuts the metrics need	BRD objectives; metrics 5–7
Cross-reference table	one row per ticket–container pair	documented bridge so the join ships before the real key exists; data classes named, who can read what	Apple A3, G3
Feed and alerts	Zendesk → Snowflake, daily or preferably hourly	BR-3 feed frequency; BR-4 alert to SV Support users when the feed fails	BRD p4
Looker views (SQL Runner)	one row per metric × week	the seven definitions in SQL, no manual reconciliation	BRD use cases
Pipeline sketch (as I'd whiteboard it)	sources → scheduled extracts → S3 landing zone → SQS-triggered loads → Snowflake raw → curated metric tables	daily, hourly on the fast tables; SQL-runner queries behind the same WBR deck; stale-table alerts to the ops channel; one engineer maintains it	Apple A3 design sketch

The names above are the record's own (Ticket ID, Container Number, Looker SQL Runner, UPDATED_BY_ID, the three statuses). Table names, column names and the container-week fact are *design logic* I describe — not a production schema I recall.

Facet 3 — Math & data logic (four tables: Before · Fix substituted · After · Dollars)

Table 1 — Before: the problem in numbers (the three-grain row is record, Apple A2; the join illustration is illustrative; BRD pricing is record-from-docs)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Manual cost of a snapshot	`1.5–2 hours to arrive at one week-over-week snapshot of seven metrics`	record-from-docs (BRD As-Is); a different measure from the about-four-hours-a-week-at-its-worst in Apple A1/A2 — do not stack them	1.5–2 h / snapshot
2	System accuracy	`system accuracy 60% (record); read as a weekly hit rate, (1 − 0.60) × 5 = 2 weeks in 5 off`	record (60%); the weeks-in-5 reading is my arithmetic, not a record	2 weeks in 5
3	Join fan-out at the wrong grain	`10 active containers; 12 tickets over 4 of them; rows after a ticket-grain join = 12 + (10 − 4) = 18`	illustrative example	18 rows for 10 containers
4	Ratio, wrong vs right	`wrong 12 / 18 = 0.667; right 12 / 10 = 1.20; understatement = 1 − 0.667 / 1.20 = 44.4%`	illustrative example	cost-to-serve 44.4% too good
5	BRD labor baseline	`3 people × 5 h × 2 reviews = 30 h a week; × 4 × 12 = 1,440 h; × $90 = $129,600 a year`	record-from-docs; arithmetic mine	$129,600 / yr
6	The printed monthly figure	`30 h × 4 × $90 = $10,800, not the printed $10,880; gap $80`	record-from-docs typo; arithmetic mine	$10,800
7	Three grains, one workbook (real)	`event grain: 1 − 1,991 / 10,289 = 80.6% automated; ticket grain: 538 / (538 + 456) = 54.1% automated`	record (Apple A2, SV_28.xlsx: 1,991 / 10,289 events; 538 auto / 456 human tickets)	80.6% vs 54.1%

Table 2 — The fix: the gates and alarms, values substituted (example values illustrative)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Grain	`container × Monday-week; numerator (tickets) and denominator (active containers, touched in the last 15 days) counted at the same grain`	design; the 15-day active rule is record (Apple A2)	one row per container-week
2	Unmatched-ticket alarm	`unmatched share = 3 / 405 = 0.74% ≤ 1% → pass`	illustrative: 405 tickets, 3 unmatched, threshold 1%	0.74%
3	Parity — pass	`|70.9 − 71.2| / 71.2 = 0.42% ≤ 1% → pass (coverage: manual 71.2%, pipeline 70.9%)`	illustrative values; the 1% gate is the BRD acceptance rule (record)	0.42% → pass
4	Parity — investigate	`|405 − 400| / 400 = 1.25% > 1% → investigate: 5 late tickets, the manual side wrong, the pipeline right`	illustrative values (tickets: manual 400, pipeline 405 — not the real 538 / 456 counts)	1.25% → investigate
5	Sync-drop alarm	`alert if 1 − sync rate > 1%; first catch 1 − 0.96 = 4% = 4× the threshold`	catch is record; the 1% threshold is illustrative	4% → alert
6	Parity checks needed to cut over	`7 metrics × 2 consecutive weeks = 14 checks minimum`	the BRD says "consecutive weeks"; two is the minimum, the actual number is illustrative	≥ 14 checks

Table 3 — After: before → after

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	System accuracy	`60% → 90%; read as a weekly hit rate, (1 − 0.90) × 5 = 0.5 in 5 off`	record (60% → 90%); the weeks-in-5 reading is my arithmetic	2 in 5 → 0.5 in 5
2	Systems and key-person risk	`4 systems → 1 pipeline; one person stitching → one engineer maintaining it`	record (Apple A3); no bus-factor count is claimed	4 → 1 · no single point of failure
3	Manual hours	`~4 h a week × 52 = 208 h a year → ~0`	modeled	208 h retired
4	Silent data loss	`invisible → alert; first catch a 4% ticket-sync drop`	record	seen for the first time
5	Ticket-to-container join	`none → documented cross-reference → real key pushed upstream`	record (Apple A3); the join rate is not stated in any source — I would measure it	join key now upstream

Table 4 — The dollars: two priced baselines, one realized candidate — none claimed as savings

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	BRD route: after and difference	`$129,600 × (1 / 5) = $25,920; $129,600 − $25,920 = $103,680`	record-from-docs (BRD); priced, not claimed	$103,680
2	Model route: hours	`4 h × 52 = 208 h; 208 × ≈ $145 = $30,160`	modeled (rate illustrative; my model, not a source figure)	$30,160
3	Model route: decision risk	`52 weeks × 10% = 5.2 wrong-number weeks; 5.2 × $13,500 = $70,200`	modeled (both inputs illustrative; range $12K–$117K)	$70,200
4	Model route: total	`$30,160 + $70,200 = $100,360`	modeled; the record states "over $100K" as the priced baseline	$100,360
5	Do the two routes agree?	`($103,680 − $100,360) / $100,360 = 3.3%; both clear $100K by different bases (30 h a week vs ~4 h a week plus risk)`	my comparison; the two bases differ (30 h a week in the BRD vs ~4 h in my model)	3.3% apart
6	Realized candidate	`208 h × $90 = $18,720 (hours actually retired at the BRD rate)`	modeled; the honest version of "savings" — measure hours retired and restated Monday numbers	$18,720

Facet 4 — Implementation (phases, owners, rollout, cutover, controls, risks)

STEP	WHAT HAPPENED (RECORD UNLESS MARKED)	OWNER	CONTROL
Requirements	BRD with BR-1 to BR-4, seven definitions, as-is and pricing	me	accepted by infra; gate published before build
Bridge	documented ticket-to-container cross-reference so the join ships; real container ID pushed upstream	infra builds; Support owns the key	unmatched tickets counted and listed, never dropped
Build	scheduled extracts → S3 → SQS loads → Snowflake raw → curated → Looker SQL Runner (the whiteboard sketch); failure alerts	infrastructure team	BR-4 alert; BR-1 data protection
Security review	data-access review for ticket data carrying customer content surfaced mid-build; two weeks; re-sequenced; Monday review kept alive on the manual pull (Apple G3)	me (owned it upward immediately); security team	no review delivery skipped

STEP	WHAT HAPPENED (RECORD UNLESS MARKED)	OWNER	CONTROL
Parallel run	manual vs pipeline per frozen metric, within 1% for consecutive weeks, mismatches explained by cause	me	gate published; late-arriving tickets the usual cause
Cutover	the Monday deck switches to the pipeline; manual pull kept as rollback a week beyond parity	me (memo)	rollback = the manual pull
Hypercare	first alert triaged: 4% ticket-sync drop; definitions into code	infra + me	alerts reach a person, not silence

Risks named. The key gap, parity failure, one person as the single point of failure — and, as it turned out, the data-access review, which was not on the list. **Lean / Kaizen mapping.** The manual pull is motion and waiting waste; the seven frozen definitions are standard work; the parity gate is jidoka — stop the line when the pipeline and the reference disagree; the bridge table is poka-yoke for a missing key; "ship with the bridge, fix the source" is root-cause countermeasure at the right layer; every PDCA starts with the first alert.

Facet 5 — Probe questions (hardest follow-ups, numeric answers)

> **Why container × week?**
>
> Because the review reads weeks and the ratio needs numerator and denominator at one grain: tickets per active container. At ticket grain the container repeats once per ticket — my worked example gives 18 rows for 10 containers and a ratio of 0.667 instead of 1.20, 44.4% too good. At container-week every metric is a sum or a ratio of one table.

> **How did you join tickets that had no container ID?**
>
> A documented cross-reference table so the pipeline could ship, then a real join key pushed upstream so the fix outlived the workaround (record, Apple A3). The design I would add: count and list unmatched tickets, never silently drop them, with an unmatched-share alarm at 1% (threshold illustrative). I do not have a measured join rate to quote.

> **Why daily or hourly?**
>
> BR-3 says daily, preferably hourly; my whiteboard sketch is daily with hourly on the fast tables. The review is weekly, so daily is sufficient for the numbers; hourly matters for the failure alarm — a drop is caught the same day instead of the next.

> **What if parity had failed?**
>
> Then we would not have cut over: the manual pull stays the source of truth. Pass is within 1% on every frozen metric for consecutive weeks — 0.42% passes, 1.25% is investigate not fail (illustrative values), and in the cases I saw the explanation was late-arriving tickets on the manual side.

> **You wrote $103,680 in the BRD — what did you save?**
>
> Nothing I claim. It is a price: 3 × 5 × 2 × 4 × 12 × $90 (an assumed rate) = $129,600, less $25,920 after, $103,680 — and my own model prices it at $100,360 including decision risk. The realized candidate is the hours retired, 208 × $90 = $18,720, modeled, and I would measure it.

> **Why not wait for Support to add the container ID?**
>
> Because a program metric cannot wait on another team's roadmap; the bridge gave us the join now and the upstream fix made it cheap to retire the bridge. Ship with the bridge, fix the source.

> **What did the pipeline catch that you could not?**
>
> A ticket-sync drop of about 4% after a routine update (record, Apple A3). With my illustrative 1% alarm threshold that is 4× the threshold; in the manual era, silence read as a quiet week.

BUSINESS IMPACT MODEL — KPI TREE

```
The weekly review steers on numbers that are right, owned by more than one person, and alarmed when they
break
+- System accuracy 60% -> 90% (record)
+- Data model: container x week grain; bridge table (ticket <-> container); definitions in code
|    +- grain logic: 12 tickets / 10 containers = 1.20, not 12 / 18 = 0.667 (illustrative); real: 80.6%
event vs 54.1% ticket grain (Apple A2)
+- Gate: parity within 1% per metric, consecutive weeks (record; 0.42% pass / 1.25% investigate are
illustrative) | manual pull = rollback
+- Alarm: ~4% ticket-sync drop caught (record); alert threshold 1% (illustrative)
+- $ (priced, not saved): over $100K | BRD $103,680 | model $100,360 | realized candidate $18,720
(modeled)
(loop)  four systems by hand -> one person stitches -> one pipeline -> parity gate -> alarms
```

SOLUTION FLOW

```
Zendesk tickets (no container ID) --+                                          +--> Looker SQL Runner
views --> Monday deck
Crux container data ---------------+--> scheduled extracts -> S3 -> SQS loads -> Snowflake raw -> curated
metric tables
core data (4PL / 3PL, SLA) --------+             ^                                    |
          cross-reference table (documented; ticket <-> container)                     +--> alerts: feed
failure (BR-4), stale tables
parallel run: manual vs pipeline per metric, within 1%, consecutive weeks -> cut over -> manual pull kept
one more week as rollback
(pipeline stages: the whiteboard sketch, Apple A3; the BRD fixes the feed, the alert and the Looker SQL
Runner views)
```

SENSITIVITY — LOW / BASE / HIGH

LABOR BASIS FOR THE HOURS TERM	HOURS × WEEKS × RATE	BASIS
canon model: ~4 h a week	4 h × 52 wk × $145 = $30,160	modeled
one person, 5 h × 2 reviews	10 h × 48 wk × $90 = $43,200	illustrative
BRD: 3 people, 5 h × 2 reviews	30 h × 48 wk × $90 = $129,600	record-from-docs

The hours term alone moves 4.30× across the three bases (129,600 / 30,160), before the decision-risk term. That spread is why the number is a *price*, never a saving. The decision-risk term ($70,200) has its own range, $12K–$117K.

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

> The record number here is a price, not a saving. The manual way of producing the weekly review was priced in my requirements document at over a hundred thousand dollars a year — about a hundred and four thousand on its own arithmetic, thirty hours a week at an assumed ninety dollars an hour; my own model gets to about a hundred thousand with decision risk — and that price let the infrastructure team prioritize the build themselves. What I own is the design under it: a container-by-week grain so tickets and containers are counted together, a bridge table so the join could ship before the source was fixed, and a gate that said nothing switches until all seven metrics match the manual pull within one percent for consecutive weeks. System accuracy went from sixty to ninety percent, and the first automated catch was a ticket-sync drop of about four percent that nobody could have seen by hand. The realized version of the dollars is hours actually retired — about two hundred hours, roughly nineteen thousand at the requirements document's rate, modeled — and I won't call any of it savings.

IF CHALLENGED — THE FULL ANSWERS

"The requirements document says $103,680 saved. That's a lot for a spreadsheet."

It is a price, not a saving. The document's own arithmetic is 3 people × 5 hours × 2 reviews × $90, and its printed monthly figure does not match its annual. I say what the price was for — the BRD priced doing nothing so the infrastructure team could prioritize the build themselves — and I lead with system accuracy and the first catch, not dollars.

"Why build a pipeline for seven metrics?"

Because the ratio the program steered on — tickets per active container — had no join key and the number rested on one person stitching four systems by hand. Seven was the scope I held on purpose; new metrics went to a parking lot until the pipeline was stable.

"What's your actual technical contribution?"

The infrastructure team built the pipeline. Mine were the grain, the bridge-table design, the seven definitions, the pricing, the acceptance gate, the parallel-run procedure and the cutover memo — the part that decides whether a build is trusted.

20-SECOND VERSION AND NACF BRIDGE

20-SECOND VERSION

At Flexport the weekly review ran on four systems stitched by hand by one person. I wrote the requirements and the acceptance gate: a container-by-week grain, a cross-reference table for the missing ticket key, and a one-percent parity gate. System accuracy went from 60% to 90%, and the first automated catch was a ticket-sync drop of about 4%.

NACF BRIDGE — ONE SENTENCE

Capacity planning for inbound sits on the same data discipline: appointments, POs and containers must join at one grain (container × week or door × hour), unmatched records must be counted rather than dropped, and a feed that goes silent must raise an alarm — otherwise a quiet dock schedule looks like a healthy one.

BEHAVIORAL PACK — FOUR QUESTIONS THIS STORY ANSWERS, EACH A SEPARATE 90-SECOND STAR FROM A DIFFERENT FACET

B1 · "Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?" AIQB: Invent and Simplify Q1

Invent and Simplify FACET: SOLUTION DESIGN

Open (first two sentences): "The Monday review depended on four systems and one person assembling them by hand. I simplified it into one pipeline with seven frozen definitions and a bridge table for the one thing that was missing."

90-second STAR. S Four systems, one person, accuracy 60%, tickets with no container ID. **T** Produce the same seven numbers without that person. **A** I wrote the requirements, chose a container-by-week grain, specified a ticket-to-container bridge so the join could ship before the source was fixed, froze seven definitions, and set a one-percent parity gate with the manual pull as rollback. **R** Accuracy 60% → 90%; four systems → one pipeline; the review no longer depends on me.

Land: four systems → one pipeline; 60% → 90% · **Pivot to NACF:** a standard data contract for inbound: one grain, one bridge for missing keys, definitions in code.

Do NOT say: that I built the pipeline myself, or that the dollar is saved.

B2 · "Tell me about a situation that required you to dig deep to get to the root cause. How did you know you were focusing on the right things? What was the outcome? Would you have done anything differently?"

AIQB: Dive Deep Q2 | **Dive Deep** | FACET: DATA ANALYSIS

Open (first two sentences): "A join at ticket grain would have repeated each container once per ticket and made cost-to-serve look about forty-four percent better than it was. I caught that in the design, before it was in a deck."

90-second STAR. S Cost-to-serve per active container was the program's headline ratio, and its join key did not exist. **T** Make the ratio right before anyone steered on it. **A** I laid out the grain of each data set, showed with a worked example that 12 tickets over 4 of 10 containers gives 18 rows and a ratio of 0.667 instead of 1.20, and moved the model to container × week with numerator and denominator at the same grain. The parallel run then showed most mismatches were late-arriving tickets on the manual side. **R** A ratio that foots, and a first automated catch — a 4% ticket-sync drop — that the manual era could not have seen.

Land: 0.667 vs 1.20 (illustrative); 4% drop (record) · **Pivot to NACF:** any network ratio — utilization, cost per unit — is only as good as the grain of its numerator and denominator.

Do NOT say: that the 44.4% was a real finding in production — it is my design example; or that the manual numbers were never right.

B3 · "Tell me about a time when you generated a creative solution to a problem or project without requiring additional resources. What was the problem? What was the solution and how did you come up with it? What was the outcome?" AIQB: Frugality Q5 | **Frugality** | FACET: IMPLEMENTATION

Open (first two sentences): "I didn't ask for budget. I priced the manual process in the BRD, so the infrastructure team put the build on their own roadmap, and I held the scope to seven metrics."

90-second STAR. S The pipeline belonged to another organization with its own priorities. **T** Get it built without owning or funding it. **A** I priced the status quo in the BRD — its labor arithmetic gave $103,680 and my model $100,360 including decision risk; I held scope at seven metrics with a parking lot and used the existing BI tool's SQL runner. **R** Infrastructure prioritized the build itself; no new system bought; zero review deliveries skipped through the build.

Land: over $100K priced, not saved · **Pivot to NACF:** a business case for dock or yard automation: price the status quo, scope tight, no new tool unless the numbers demand it.

Do NOT say: the $103,680 as savings; or the hours basis as settled — it is the BRD's 30 hours a week at an assumed $90 an hour, not my ~4 hours.

B4 · "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?" AIQB: Highest Standards Q1 | **Insist on the Highest Standards** | FACET: PROBE QUESTIONS

Open (first two sentences): "Nothing switched until all seven metrics matched the manual pull within one percent for consecutive weeks, with every mismatch explained by cause, and a failed feed raising an alert."

90-second STAR. S A VP steered the program by this number, and the pipeline was new. **T** Earn the cutover instead of announcing it. **A** I published the acceptance gate before build; ran the parallel run; investigated every mismatch — 1.25% is investigate, not fail (my illustrative example); kept the manual pull as rollback a week beyond parity; and required the failure alert (BR-4) before go-live. **R** The first Monday off the pipeline showed the same seven numbers, and silence stopped meaning health.

Land: 1% gate; 0.42% pass (illustrative) · **Pivot to NACF:** cutting over a yard-management or appointment system: parallel run to a numeric gate, rollback kept, alarm on silence.

Do NOT say: that the 1% threshold on the alarm was in the record — it is illustrative; or the number of consecutive weeks.

SAME STORY, THREE ANGLES — THREE CARDS, THREE OPENERS, THREE METRICS, THREE LPS

Data / analytical angle **Dive Deep**

Opener: “A ticket-grain join would have made cost-to-serve look forty-four percent better than reality — so before we built anything, I fixed the grain.”

Headline metric: 12 / 18 = 0.667 vs 12 / 10 = 1.20 (illustrative); 4% drop (record)

- Unit of cost (ticket) vs unit of volume (container); one grain: container × week.
- Unmatched tickets counted and listed, never dropped.
- Parity: 0.42% pass; 1.25% investigate (illustrative); mismatches the manual side's late tickets.
- Say "illustrative" on every example number.

Conflict / influence / backbone angle **Have Backbone; Disagree and Commit**

Opener: “Everyone wanted one more metric during the build; I held the scope at seven with a parking lot, and shipped the bridge instead of waiting for Support.”

Headline metric: seven frozen metrics; one bridge table

- Stability over coverage — a documented trade-off, not an oversight.
- Bridge now, real key upstream: another team's roadmap did not hold the program.
- Priced in the BRD so infra could prioritize it themselves — influence without authority.
- My seat (Apple A3): the BRD, the seven definitions, the parallel-run gate, the acceptance test, the cutover call; infra built the pipeline.

Honest failure-and-learning angle **Earn Trust**

Opener: “I discovered the data-access security review mid-build instead of planning it — two weeks — on the pipeline that was supposed to retire my own manual review.”

Headline metric: two-week security-review miss; zero Monday reviews skipped

- I owned it upward immediately: the slip was scoping, not engineering — the BRD priced the build and never the review (Apple G3).
- Ticket data carries customer content, so the access review is a gate with lead time, not a formality; I had treated it as a formality.
- I re-sequenced instead of stalling: the parallel run kept the Monday review alive on the manual path, so no review week was skipped.
- I closed the review properly — cross-reference table documented, data classes named, who could read what — and now every data program starts with "whose approval does this data need, and how long does that take?"
- Not a dollar claim: the miss cost margin, not delivery. Do not mix in the $103,680 here.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS

"Your document says $103,680 saved. So you saved over a hundred thousand?"

Numeric answer. No. 3 × 5 × 2 × 4 × 12 × $90 (an assumed rate) = $129,600; after = $25,920; difference $103,680 — a priced labor estimate, and the printed monthly figure ($10,880) does not match its annual ($10,800 × 12). My model prices the baseline at $100,360 ($30,160 + $70,200). The realized candidate is the hours actually retired: 208 × $90 = $18,720, modeled.

Say exactly: “It's a price, not a saving — about a hundred and four thousand on the document's labor basis, about a hundred thousand in my model. The realized version is hours retired, about nineteen thousand, modeled.”

"Pipeline 405 tickets, manual 400. That's a 1.25% gap — a failure?"

Numeric answer. Investigate, not fail (illustrative values): |405 − 400| / 400 = 1.25%, just over the 1% gate, and the explanation in the cases I saw was five late-arriving tickets on the manual side. Contrast 70.9% vs 71.2% coverage: 0.42%, pass. (Example values are illustrative.)

Say exactly: “A gap of one-point-two-five percent is over the gate, so we investigate — and the cause was late tickets the manual pull missed. Coverage at point-four-two percent passed.”

"How do you know the 4% drop wasn't noise?"

Numeric answer. The alarm rule is 1 − sync rate > 1% (threshold illustrative); the observed 1 − 0.96 = 4% is 4× that threshold, and the record says it was a ticket-sync drop of about 4% after a routine update, invisible in the manual era; I did not keep the triage detail beyond that. A threshold like this is tuned from weeks of normal variation, which I would pull from the history.

Say exactly: “Four percent is four times the alarm threshold, and in the manual era it would have looked like a quiet week. I triaged it to a source cause, not to noise.”

Question bank — likely questions for this interviewer, with a one-line answer plan

NEW

Behavioral and functional questions a hiring manager for NACF Supply Chain is likely to ask, in rough order of likelihood. Each plan points at a packet story or a named backup in the master packet. Stories allocated to other interviewers are referenced as pointers only. v2 adds rows 17–28 for the five new primaries; the Router at the front maps all 16 Leadership Principles.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Tell me about yourself.**	The 75-second newest-to-oldest script from the master packet: Sephora (one signed "available"; dual-DC WMS migration) → Flexport (the review; $0.66 → $0.03) → Elementum (measured lanes, $1.5M out) → Manhattan (99% fill). End on "design the supply chain, keep it honest in a weekly review — at network scale."	Master TMAY
2	**Why Amazon, why now?**	Three reasons — the forecast (Elementum lanes), the weekly review (Flexport), one version of the truth (Sephora) — and "what I went to Sephora to build is live."	Master Why Amazon
3	**Tell me about your most significant delivery.**	S11 pitch; end on CTS $0.66 → $0.03 across 7 → 68; stop.	S11
4	**Tell me about a time you changed a plan that was already committed.**	S12 pitch; private-first; 66% less effort, value month 6; ~$5M once.	S12
5	**Tell me about a time leadership doubted your program.**	S11 Discovery: accepted the critique as being about evidence; built the review; left the critic steering by the numbers.	S11
6	**How do you make trade-offs between speed and quality?**	S11: board no faster than automation matures; wave 3 held publicly. S12: confidence buffers over a fast optimistic plan.	S11 / S12
7	**Describe how you would run your first 90 days in NACF Supply Chain.**	Book of record and primary metric first; stocks-and-flows map of the process; the invariant broken today; 3 key results with denominators; a weekly input/output review by week 6 (master §3.0, seven steps).	Master §3.0
8	**How would you decompose a complex network problem (e.g., inbound placement cost rising)?**	Open as the master's §3.99 script: book of record + metric → stocks and flows → broken invariant → minimal design → sequence by Reach × Impact × Confidence ÷ Effort → one metric, one labeled dollar.	Master §3.99
9	**Tell me about a time you failed.**	Full failure story here: S11-B (the pilot ran later than the first testing plan — testing under-sized, monitoring late, now sized on hop count) is the primary; the master's UAT-crash bench story (§4.3 — numeric entrance criteria ever since) is the backup. S18's RF miss (Goforth) and S16's security-review miss (Josyula) are held for other interviewers — pointers only; S4 (the in-transit rule I got wrong) is Lan's backup pointer, so keep it clean.	S11-B · Bench §4.3
10	**How do you measure success for a process you designed?**	A KR without a denominator is not a KR: seven frozen metrics with owners, honest denominators (15-day active rule), inputs tagged to outputs.	S11

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
11	**What's a number you refused to report?**	The 99.4% three-touch week — printed as a data miss.	S11
12	**What would your current manager say you need to work on?**	Start the review at kickoff rather than after the critique (S11); back-test a rule before publishing it (master S4 lesson, pointer only).	S11 / S4 pointer
13	**How would you think about inventory-placement strategy for a network like mine? (placement was her 2017–22 world — public-profile inference)**	S11's logic translated: publish the selection bar before the first node (graduation criteria ≈ placement criteria), calibrate per node (~50 containers ≈ a trial volume), hold a node in public when it misses. Then master §3.99: book of record → stocks and flows → broken invariant → minimal design.	S11 / §3.99
14	**How do you decide connectivity in a crossdock network — which nodes, which flows?**	S12's sequencing logic: rank flows by benefit × impact ÷ effort at the work-item grain (interactions eliminated per flow), FCL-first as "highest-volume flow first," a reversal trigger written per node before the swing.	S12
15	**You ran launch readiness for 50+ buildings — what does readiness mean for a process launch? (her 2014–17 world — public-profile inference)**	FC-readiness discipline translated: a written entry bar per wave (the six criteria), a launch gate on real volume (100 containers × 7 checks), a published rollback rule (the reversal trigger), and the hold taken in public when the bar misses.	S11 / S12
16	**Tell me about building a team or capability from scratch.**	S11's mechanism answer: input owners across nine teams recruited to sign the definitions they would be red against; the review run by hand until the pipeline (S16-A, in this packet) carried it; wave teams grown into owners — capability built while delivering, not before.	S11
17	**Walk me through a data analysis that changed a decision.**	S11-A: the 22× cost-to-serve drop split into 6.0× (60% → 10% of containers needing a human) × 3.67× (touches on each); the 3-touch 99.4% week discarded. Backup S12-A (P95 by region, then by path).	S11-A / S12-A
18	**How would you design the data model behind an operating metric?**	S16-A: one grain (container × week), numerator and denominator at that grain, a documented cross-reference for the missing key, a 1% parity gate and a feed-failure alert — the BRD names no tables, so the join, the grain and the artifacts are the story.	S16-A
19	**How do you decide a product or process is ready to launch?**	S11-B: four test gates, a 100-container ledger × 7 checks, six published graduation criteria — and own that the pilot ran later than the first testing plan.	S11-B
20	**How do you retire or migrate a customer base without stranding anyone?**	S11-C: nine segments, three questions each, concentration first (3 brokers = 75.0% of containers), dated actions, the dissent on the page.	S11-C
21	**Tell me about diagnosing a regional performance problem.**	S12-A: P95 by region (Asia 21.19 s vs US 3.3 s = 6.42×), then by path (static vs dynamic), reversible fixes first, 32 smoke checks a change.	S12-A
22	**How do you know a number is right before you steer on it?**	S16-A parity gate (≤ 1% for consecutive weeks, mismatches explained by cause); S11-A footing and honest denominators; S11-C re-footed the doc's own table.	S16-A / S11-A / S11-C

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
23	**How would you apply Kaizen / Lean here? (her trainer background — public-profile inference)**	Waste and standard work in S11-A (3.67 touches; seven frozen definitions); jidoka in S11-B (hold wave 3); genchi genbutsu and PDCA in S12-A; poka-yoke in S16-A (bridge table, parity gate). Say the gate, never the slogan.	S11-A / S11-B / S12-A
24	**How do you decide what to fix first?**	S11-B Pareto: 41 of 59 failing containers first broke at two hops (69.5%) — fix those; S11-C concentration; S12 benefit × impact ÷ dev-weeks.	S11-B / S11-C / S12
25	**Tell me about a metric that looked great and was not.**	S11-A: 99.4% on three touches — printed as a data miss; a 95% interval on N = 3 is wider than the metric.	S11-A
26	**Tell me about something that took longer than planned.**	S11-B: the first testing plan targeted early November 2021 and the pilot launched in December; the documents disagree on whether that was late, so I say "later than the first testing plan" — testing effort was under-estimated and monitoring arrived late; now sized on hop count.	S11-B
27	**How would your container-week grain apply to inbound capacity? (transferability probe)**	S16-A bridge: one grain for appointments, POs and containers (container × week or door × hour), a documented cross-reference for missing keys, an alert on a silent feed. Then the booklet: doors × hours × unload rate; the 85% knee.	S16-A
28	**Why should I believe your dollars?**	Every dollar carries its basis. Record: 7 → 68, $0.66 → $0.03, ~$5M once. Modeled: ≈ $50K direct CTS, $1.52M run-rate. Priced, not saved: S16-A's $103,680 / $100,360. See the NUMBERS page.	S11 / S12 / S16-A

Backup bench — use only if a primary doesn't fit

Three fallbacks for the hiring manager, none of them another interviewer's primary: S9 (Elementum — the 597-lane Pareto and the ≥95% gate, for Backbone / Think Big / Highest Standards if S11 and S12 are spent), S19 (Elementum · Walmart-client — the promise buffer as one governed dial; a design-flavored Frugality / Invent & Simplify alternate, all model, illustrative labels intact) and S13 (Manhattan — Cyber Monday at 4 AM, the Bias for Action / Customer Obsession alternate). Tell S11 and S12 first.

Bench discipline. The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

LPs this backup can serve: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

Basis. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Shipment-data completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr`	illustrative volumes	~90K docs/yr keyed by hand

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% — build them first
2	Cohort gate	`valid-ASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	"the standard was the data, not the transport" (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live in month one	0	597 = 80% of freight spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in six months	record — measured, never annualized
4	Document handling	~90K docs/yr keyed	X12/XML validated, portal floor for the tail	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000/yr`	derived from the record pair, stated as such	≈$20M/yr
2	The record result	`$20M × 10% = $2M in six months`	record	≈$2M — said as measured
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when the ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes; benchmark rate (modeled add-on)	≈$0.4M/yr

TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

Canon numbers (basis labels): 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

BACKUP — ONLY IF ASKED S19 — Elementum (client: Walmart Marketplace) — the delivery-promise buffer rebuilt as priced risk: one governed dial, +32 points of fast-promise share at the same 95% floor (all model, no record dollar)

LPs this backup can serve: Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate) · Frugality (speed bought with zero new spend) — design-flavored probes

Basis. Engagement lead for the promise-policy workstream; the client's promise-engineering and data-science teams owned and built the engine and the transit models — I never configured or deployed any platform. Every quantity is from the engagement's illustrative twelve-cohort model — say "the model" in the same breath; there is no production dollar and none is ever promoted. Labels (metro two-day, mountain rural) are the model's illustrative cohort names.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. A marketplace client printing delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of variability. In the engagement model — twelve lane cohorts, about a million orders a week, transit σ from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise: dense metro lanes over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Own the decomposition, the alternatives, the validation gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought Φ(5.0), near-certainty, on the tightest cohort and only Φ(0.69) ≈ 75% on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: b = 1.645σ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share 7.6% → 39.6%, +32 points at the held 95% floor; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say "the model" in the same breath.

COMPRESSED STAR

- **S —** A flat 1.10-day buffer on every lane: 95.8% headline accuracy hiding over-insurance on tight lanes and under-insurance on loose ones; fast-promise share 7.6% (model).
- **T —** Lead the promise-policy workstream: decompose the waste, put priced alternatives on one table, design the gates, leave a governance instrument — not a tuning habit.
- **A —** Promise identity and accuracy factorization; buffer-waste audit; four policies on one table; proofs (Φ(1.645) = 0.95 per cohort, equimarginal check, Q4 drill at ~1.45× σ); shadow mode for four weeks; the dial k owned by a business owner and printed every Monday.
- **R —** Fast-promise share 7.6 → 39.6% (+32 pts) at exactly the 95% floor; average promised day 3.09 → 2.75; +$255K/wk ≈ $13.3M/yr — model, labeled; the widest cohorts got honestly slower promises (1.10 → 2.63 days) on purpose.

RESULTS LINE

Fast-promise share 7.6% → 39.6% (+32.0 points) at a held 95.0% floor (model) · average promised day 3.09 → 2.75 (model) · +$255K/wk ≈ $13.3M/yr vs uniform (modeled on the illustrative ~1M-order book) · the profit-optimal frontier (97.8% accuracy, ≈ +$20M/yr) handed to Finance as a priced decision — shown, never claimed, never merged with the $13.3M.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (record-from-docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	record-from-docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (record-from-docs design)	k = 1.645, one number for the book
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645 / 1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
7	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1–2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

TWO PROBES

> **The whole story rests on an illustrative cohort set.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers risk-proportional, the last buffer-day buying the same accuracy everywhere, and a floor that disagrees with the breach economics treated as a governance finding.

> **Why one dial instead of per-lane learned buffers?**
>
> Data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

Canon numbers (basis labels): 7.6% → 39.6% fast-promise share (+32 pts, model) · 95.8% → 95.0% accuracy held at the floor (model) · 3.09 → 2.75 promised days (model) · +$255K/wk ≈ $13.3M/yr (modeled) · 12 cohorts, ~1M orders/wk, σ 0.22–1.60 days, k = 1.645 (illustrative model) · breach ≈$12 vs margin ≈$6.50 (model assumptions). Never a production dollar.

BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

LPs this backup can serve: Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

Basis. OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.
- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	4 AM state	`scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal`	record risk pair; backlog illustrative	the bridge call
2	FIFO failure	`FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age`	record rule problem	value-blind clearing

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Naive node loads	`DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days`	capacities 10K / 8K / 6K orders/day (illustrative)	worst node 2.2 days behind while another idles
4	No liveness detection	`a dead scheduler was found by a VP at 4 AM, not by a page`	record problem	detection by seniority

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise-at-risk ranking	`risk = value × urgency(promise − now); clear descending`	on the bridge log in hour 1 (record rule)	at-risk promises first, not first-in
2	Node choice = the promise	`T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d`	record pair	−3.75 days per re-pointed order
3	Load levelling	`rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days`	load = queue / capacity (illustrative)	worst slip 2.2 → 1.25 days
4	Liveness page (post-mortem)	`page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr`	μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr	caught in one interval, not by a VP

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Peak fill rate	95% goal	99% delivered	record
2	Peak sales	~$6M at risk	~$8M protected	record pair — one-time event
3	Worst-case promise slip	2.2 days	1.25 days	illustrative walk
4	Detection	a VP at 4 AM	one interval via the 3σ page	post-mortem design
5	Structural stockouts	—	−20% over the engagement	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Margin protected	`$8,000,000 × 40% = $3,200,000`	benchmark margin 35–45% (modeled)	≈$3.2M one-time
2	Orders at risk	`$6,000,000 / $150 = 40,000 orders`	AOV benchmark	40,000 orders
3	Cancels / CS avoided	`40,000 × $6.50 = $260,000`	CS contact $5–8 (benchmark; modeled)	≈$0.26M
4	Expedite premium avoided	`40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000`	re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative)	≈$0.13M
5	Structural-stockout run-rate	`$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000`	illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin	≈$1.3M/yr modeled

TWO PROBES

Why act before you knew the cause?

Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

Isn't this a hero story?

The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

Canon numbers (basis labels): 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

NUMBERS I MUST NOT GET WRONG — canon values with basis labels NEW

Canon numbers for all seven primaries with their basis labels (S11 and S12 from the master packet's §4.1 ledger; the five new stories' rows re-cut from the program documents, with the footing mine). Say record numbers flat; say modeled numbers with their inputs; say a priced number as priced.

METRIC	VALUE TO SAY	BASIS	NEVER
S11 cost-to-serve	$0.66 → $0.03 per container (about 95%)	record	"$1.00 → $0.03"
S11 customers	7 → 68 = 9.7×; zero rollbacks; no added ops headcount	record	"10×", or any growth multiple other than 9.7×
S11 automation / accuracy	40% → 90% / 60% → 90%	record (this packet's canon; loop packet differs — do not mix)	"70% baseline"
S11 targets / timing	68 customers · 170 MAU · 32K shipments; both targets two months early	record	—
S11 direct CTS dollars	≈$50K/yr (range $30–81K; exact $30,240–$80,640)	modeled on illustrative containers/shipment	presenting it as the value — the value is scale
S11 pre-prod measurement	300 containers, ±1.6% at 95%; rollout proof 1.5 vs 6 months; ~$1.5M saved	user anchor — separate canonical	merging the $1.5M pre-production estimate with CTS
S11 review by hand	about four hours a week at its worst, by hand for months; seven frozen metrics; ±5% WoW; 15-day active rule	record	"I invented the WBR"
S12 time-to-value / effort	month 18 → month 6; 66% less engineering effort	record	—
S12 cost avoidance	~$5M, said once	record	"$5M annual cost reduction" alongside it
S12 CTS at work-item grain	−50% partner-first vs −25% operator-first; 339.5 min/shipment; 2,200 interactions	record	—
S12 partners / burn-up	48 of 53 within a month; import-side app April 2022: 11 partners / 10 countries; 55.8% → 83.2%	record	—
S12 run-rate / decomposition	≈$1.5M/yr run-rate; ≈$2.9M + ≈$1.5M ≈ $4.4M (12% under the ~$5M record)	modeled (12 eng × $20K; 20K shipments illustrative)	presenting the split as the program's figure
S12 employer	Flexport FF App export launch	record	telling it as a Sephora roadmap (loop-packet conflict)

METRIC	VALUE TO SAY	BASIS	NEVER
S11-A decomposition	22× = 6.0× (60% → 10% of containers needing a human) × 3.67× (touches per container); weekly automation 40.0% (17 Jan) → 21.9% (31 Jan 2022) is record	record inputs; decomposition mine	attributing the 22× to automation alone; any growth multiple other than 9.7×
S11-A honest denominator	active = cumulative inflow less containers 15 days past departed_terminal; 99.4% on N = 3 discarded (95% interval 44%–100%)	record decision; interval is my arithmetic	printing the three-touch week
S11-B ledger and gates	100 containers × 7 checks; 59 failing rows, 41 first failed at two hops (69.5%); 500 containers / 20 min / 99.6%	record-from-docs; my arithmetic	claiming the pilot was on time, or a number of weeks late — say "later than the first testing plan"
S11-C Crux book	79 customers · 294,581 containers · printed $185,892 = six of nine segments · nine rows = $254,952 · 3 brokers = 75.0% of containers	record-from-docs; footing mine	quoting $185,892 as "the book"; naming a keep-alive cost
S12-A Asia latency	Asia 21.19 s vs US 3.3 s = 6.42×; 45.3% of 13.92k views; overall monthly P95 8.39 → 4.68 s = −44.2%	record-from-docs; arithmetic mine	quoting 81%; subtracting P95s across sources; a dollar as savings
S12-A dollars	tail time ≈ $1.6K / yr; exposure ≈ $0.69M / yr upper bound	illustrative / modeled	calling the exposure a saving
S16-A pipeline	system accuracy 60% → 90%; parity ≤ 1% (record gate; 0.42% pass / 1.25% investigate are illustrative); about 4% sync drop; four systems → one	record (accuracy, drop, gate); examples illustrative	the alarm threshold or weeks-to-parity as record
S16-A dollars	over $100K a year priced · BRD $103,680 (30 → 6 h/wk at an assumed $90/h) · model $100,360 · realized candidate $18,720 (208 h × $90)	priced / modeled — none claimed as savings	"I saved $100K"; a monthly figure other than $10,800
S12 decomposition range	$2.85M + $1.52M = $4.4M (12% under ~$5M); range $2.9M–$6.7M; effort piece alone $1.8M–$4.5M	modeled	"the effort band contains the $5M"; $6.8M (rounded high case)
S11 pre-production set	~$1.5M pre-production estimate · n = 300 → ±1.6% · 1.5 vs 6 months	user anchor (GFiber v8 only) — kept separate, never added to cost-to-serve	saying it as program record; "E2E 85% → ~98%" (say 86% tracked, 500 containers / 20 min / 99.6%)
S11 FTE equivalent	80,000 × 3.5 touches × 1 min ÷ 1,800 h ≈ 2.6 FTE (2.4 at 1,920 h)	modeled (assumptions named)	presenting it as record headcount

Never-say, one glance before the call.

- Never "10×", or any growth multiple other than 9.7× — say 7 → 68 (9.7×).
- Never merge the ~$1.5M pre-production saving with cost-to-serve; never stretch it into a range.
- Never "$5M avoidance" and "$5M annual reduction" as two results — the avoidance is said once.
- Never "I invented the WBR" — copied from the LSA review, which was the point.
- Never claim platform configuration; program-lead voice throughout. The only personally configured platform in canon is Manhattan DOM.
- S14, S15, S10 are the bar raiser's, and S16 is his primary too — this packet's S16-A is the data-model cut only; S1/S3, S2/S18, S6/S8 belong to Lan, Goforth and Hoerger — pointers only here.

WHAT TO SAY ABOUT MY ROLE — the I-versus-we line for each story NEW

> **The I-versus-we rule.** Say *I* for what I decided or authored — the plan, the gate, the analysis, the document, the read-out. Say *we* only for results other people delivered, and name the owner when the work was theirs. No source documents a direct report, so every people answer is program-lead voice: owners, gates and cadence — never "my team of …".

STORY	"I" — WHAT I OWNED	"WE" / THEIR WORK	NEVER SAY	EVIDENCE
S11 · wave machine	I owned the rollout: I wrote the wave plan as a schedule, published the entry bar (about fifty containers of the customer's own data) and the six graduation criteria before wave one, and ran the weekly review that gated each wave.	Nine teams (~100 people) built and operated the product; onboarding and sales sat with other teams; the VP decided the holds.	I built the platform · I ran the pilots myself · I scaled it alone	AP A1 p6–7; P4PL p1–2
S11-A · the Monday review	I built and ran the review by hand for months, froze the seven definitions with the input owners, and made the denominator honest. The dollars per ticket come from Data Services and FP&A at their loaded rates, never my own.	The input owners ran their lines; the pipeline later carried the formulas.	I set the cost per ticket · the LSA workbook's numbers are mine	AP A2 p8–10
S11-B · launch readiness	One of two technical program managers. I defined the gates and ran the go / no-go read-out; I made the 100-container run the UAT spine with a Jira ID on every missed row.	The QA team ran the container log day to day; business users ran UAT; the test lead designed the hop-by-hop trace.	I ran the test · the pilot was on time · it was N weeks late	AP A5 p13–15; PTA p1–3
S11-C · Crux transition	I owned the mitigation of the decommission risk: top of the risk register, the apples-to-apples comparison, the 79-customer base priced per segment, and the one-canonical-source rule.	The transition document's owners built the per-customer paths with me; the decision was the executive decider's (RAPID).	I decided to retire Crux · I own the revenue number	AP A4 p12–13; RAPID p1
S12 · export roadmap	I drove the sequence call: I priced both orders on shared axes and ranked the backlog by benefit × impact ÷ dev-weeks; my manager saw the math first and pressure-tested it with engineering for two days; the onsite ratified it.	Engineering did the re-estimates and the build; the 339.5-minute baseline is the program's time-and-motion analysis — I did not run the stopwatch.	I built the app · it was my idea alone · $5M a year	AP B2 p17–19
S12-A · Asia latency	I drove the follow-ups from the monthly review and kept the plan, the sequencing and the smoke tests honest.	The app engineer wrote the investigation; the network-infrastructure owner made the CDN and DNS changes (ASIA p3 names the owners).	I fixed the CDN · I found the root cause	ASIA p1–4; AP B3 p19–21
S16-A · Snowflake pipeline	I wrote the BRD, froze the seven definitions, made the Monday review the acceptance test and owned the cutover call; I own the miss — I planned the build and not the security review.	The infrastructure team built the pipeline: I didn't lay pipe, I made the pipe buildable.	I built the pipeline · I saved $100K · bus factor ≥ 2	AP A3 p10–11, G3 p45–46; BRD p1–5

Seat sources: Apple v11 chapters A1–A5, B2–B3 and G3 (the candidate's own account), the BRD, the RAPID table, the investigation document and the testing approach. Anything not in those is not said as "I".

VERIFY BEFORE YOU SAY IT — 4 specifics in no source file NEW

The first pass carried 37 yellow items. Everything the program documents, the Apple v11 workbook, the BRD or the review workbooks state was converted to a fact or a labeled anchor (see SOURCE TRACE). These are the only specifics that are in **none** of the source files. **Rule: do not say the specific — say the safe sentence beside it.**

ROUTER AND FRONT MATTER

☐ V1 a named person I mentored or developed, with an outcome — no source names one. **Safe sentence:** "I led through programs rather than a reporting line: I grew the input owners across nine teams until each signed and defended their own definition. I don't have a one-to-one mentee story I can document, so that is the one I'll give you."

S11-A

☐ V2 whether the 10- and 30-day alternatives were compared before 15 was frozen — not in any source
Safe sentence: “I froze the fifteen days with the input owners; I did not record a ten- and thirty-day comparison, and I would run one now.”

S11-C

☐ V3 how Crux handled exceptions — not in any source
Safe sentence: “I don't know how Crux handled exceptions; for the new product, a failed auto-track opened a ticket to Data Services.”

S12-A

☐ V4 whether synthetic probes were adopted — not in any source
Safe sentence: “The document raised synthetic probes as an open question; I added P50, P90 and P95 by region to the review, and I can't say whether probes were adopted.”

Left as labeled anchors, not VERIFY items. The ~$1.5M pre-production set (n = 300, 1.5 vs 6 months) and the “#3 → #14 of 16” ranking are in the candidate's own prior master only — say them as “my estimate” and “as I recorded it”; Snowflake table and column names are in no source (the BRD names the Ticket ID, the container number and Looker SQL Runner views only), so the data-model story describes the join, the grain and the artifacts and never names a table.

SOURCE TRACE — every quantitative claim in the new and modified stories NEW

Number · story · source file · page / row / sheet · match. **Y** = the source says it; **Y-with-rounding** = the source says it and I rounded or divided; **CONFLICT** = sources disagree — say the safe wording; **UNSUPPORTED-anchor** = only my own prior master or my own model — say it as “my estimate / my model”. Apple page = PDF page of the v11 workbook.

AP Apple_PMIII_Interview_Stories_v11_2026-08-26 (PDF page; chapter in brackets) · **BRD** BRD_-_SV_Zendesk_Data_in_Snowflake · **TPM** Technology_Program_Management_Status_Week_of_November_19th__2021 · **PCRS** SV_-_Pilot_Client_Roll-out_Schedule · **PTA** SV_-_Product_Testing_Approach · **P3PL** Shipment_Visibility_3PL_ingestion-_One_Pager · **P4PL** Path_to_4PL__Visibility__Rajesh · **WBRAI** WBR_Metric_Culture_using_AI · **LSA** LSA_WBR_2022 workbook (the .CSV upload is an .xlsx; sheet and Excel row given) · **C100** 100_Container_Debugging_Session_-_Basic_test_cases (.csv; my recount of the working log) · **RAPID** RAPID_-_Crux_Customer_Transition · **ASIA** FF_App_Asia_Latency_Investigation · **HLSD** High-Level_Solution_Document_FF_App__Schedules · **GF8** GFiber_MASTER_Packet_v8 (the candidate's own prior master — a user anchor, not a program document) · **RECON** numbers-reconciliation.md (CANON v2 ledger and §2 tables)

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
1	7 → 68 customers; 9.7× (68 / 7 = 9.71)	S11	AP [A1]; WBRAI	AP p6–7; WBRAI p1–2	**Y-with-rounding**
2	170 monthly active users — public target, landed **Say: Say "170 monthly active users — the public target, landed." Never the workbook's multiplier.**	S11	AP [A1]	AP p6–7 (WBRAI p1 says "MAU from 400 to 3X")	**CONFLICT**
3	32,000 shipments tracked = a target, not a result	S11	AP [A1]	AP p6–7 ("a number we managed toward all year")	**Y**
4	68 customers and 170 users landed two months early; zero rollbacks	S11	AP [A1]	AP p7	**Y**
5	Cost-to-serve $0.66 → $0.03 per container	S11 · S11-A	AP [A1, A2]; WBRAI	AP p7, p9; WBRAI p2 ("66 cents to 3 cents")	**Y**
6	Cost-to-serve cut: about 95% (1 − 0.03 / 0.66 = 95.45%) **Say: Say "about ninety-five percent" — never "95.5%".**	S11 · S11-A	AP [A1]; RECON S11.06	AP p7–8 ("a 95% cut")	**Y-with-rounding**
7	$0.18 per eliminated interaction; 0.66 / 0.18 ≈ 3.67 touches → 0.03 / 0.18 ≈ 0.17	S11 · S11-A	AP [A2]	AP p9–10 (the cross-walk; the ratios are my division)	**Y-with-rounding**
8	Automation 40% → 90%; 70% no-touch gate	S11 · S11-A	AP [A1, A2]	AP p7, p9	**Y**
9	Six graduation criteria (verification SLA ≥ 90, data quality ≥ 90, ETA accuracy ≥ 80, …)	S11 · S11-B	AP [A1]; PCRS	AP p6–7; PCRS p1–2	**Y**
10	~50-container calibration per new client	S11 · S11-B	AP [A1]; PCRS	AP p6; PCRS p4	**Y**
11	Waves 2 → 4 → 20+ → 40 → 68	S11	AP [A1]; PCRS	AP p6–7; PCRS p1 (40 and 68 are Apple-only milestones)	**Y**
12	Nine teams / ~100 people (input owners across nine teams)	S11 · S11-A	P4PL; AP [A1]	P4PL p1–2; AP p7	**Y**
13	First pilot customers (names) **Say: Say "the first two hand-carried customers" — no names.**	S11 · S11-B	PCRS; TPM; AP [A5]	PCRS p1, p3; TPM p10–11; AP p13–14	**CONFLICT**
14	Manual scorecard effort: about four hours a week at its worst **Say: Say "about four hours a week at its worst"; the BRD hours are a different measure, quoted only inside S16-A and labeled.**	S11 · S11-A · S16-A	AP [A1, A2]; BRD	AP p8–9; BRD p2 (1.5–2 h per snapshot; p1–2: 30 h/wk priced)	**CONFLICT**
15	99.4% week on three touches, discarded as a data miss	S11 · S11-A	AP [A2]	AP p8–9	**Y**
16	Event grain 1,991 / 10,289 = 19.4% human-touched (80.6% automated)	S11-A · S16-A	AP [A2]	AP p9 (extract SV_28.xlsx, not in the source folder)	**Y**
17	Ticket queue grain: 538 auto / 456 human = 54.1% automated	S16-A	AP [A2]	AP p9 (grain 3)	**Y-with-rounding**
18	Weekly stock: 13 Dec 40.2% · 17 Jan 40.0% · 31 Jan 21.9% (carrier format change) · 21 Feb 69.2%; Active 287 → 546 (+90.2%)	S11 · S11-A	AP [A2]	AP p9	**Y**
19	40.0% → 21.9% = −45.3% week over week (my division)	S11-A	AP [A2]	AP p9	**Y-with-rounding**
20	15-day active = cumulative inflow less containers 15 days past departed_terminal	S11-A · S16-A	AP [A2]	AP p9	**Y**
21	Pre-production: 500 containers in 20 minutes at 99.6% connectivity	S11 · S11-B	TPM; AP [A2]	TPM p10; AP p9	**Y**
22	End-to-end dry run: 100% processed, 86% tracked; job stories 98.4%	S11 · S11-B	TPM; AP [A2]	TPM p10; AP p9–10	**Y**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
23	"E2E 85% → ~98%" **Say: Say "eighty-six percent of containers trackable in the dry run."**	S11 (original text)	GF8 only	not in TPM p10 (86% tracked, 98.4% job stories)	**CONFLICT**
24	Pre-production sample n = 300, ±1.6% at 95% (1.96 × √(0.98 × 0.02 / 300) = 1.58%) **Say: Label it "user anchor": "about three hundred containers, plus-or-minus one-point-six at ninety-five — my design."**	S11	GF8	GF8 only — my design, arithmetic checks	**UNSUPPORTED-anchor**
25	~$1.5M pre-production saving; rollout in 1.5 vs 6 months **Say: Label it "user anchor", keep it separate from cost-to-serve: "my estimate of a month and a half versus six."**	S11	GF8	GF8 only (PCRS p2 supports only a ~6-month pilot)	**UNSUPPORTED-anchor**
26	Direct cost-to-serve ≈ $50K a year ($30,240–$80,640) — modeled **Say: Say "modeled on illustrative container counts — the point is scale without headcount."**	S11	RECON S11.25	arithmetic on illustrative container counts	**UNSUPPORTED-anchor**
27	80,000 containers × 3.5 touches × 1 min ÷ 1,800 h ≈ 2.6 FTE — modeled **Say: Say "about two and a half people never hired — modeled, not record headcount."**	S11	RECON	my assumptions named in the dollars table	**UNSUPPORTED-anchor**
28	Wave 3 held once on ETA accuracy	S11	AP [A1]	AP p7	**Y**
29	Critique: initiative #3 → #14 of 16 **Say: Say "leadership said the program wasn't showing value" and the ranking only as "as I recorded it in my master".**	S11	GF8	GF8 only (not in AP or any program document)	**UNSUPPORTED-anchor**
30	LSA review format: Monday–Sunday week, five weeks in view, (B − A)/A, red / green beyond ±5%	S11-A	LSA	sheet "Instructions", rows 6–9	**Y**
31	Real FFA tab, 19-value excluded series: −28.7%, −33.3%, 1 red / 4 green, −36.8% / +74.5%, 47.5% / 49.9%, 0.216, 7 / 52, −31.0%	S11-A (Table 5)	LSA	sheet "FFA" (the FFA tab is another team's slide — used for the mechanics only); formulas in Table 5	**Y**
32	99.4% on N = 3 → 95% Wilson interval 44% to 100%	S11-A	AP [A2]	AP p9 (the week); the interval is my arithmetic	**Y-with-rounding**
33	22× = 6.0× (60% → 10% needing a human) × 3.67× (touches) **Say: Say "the split is my model on two record endpoints."**	S11-A	AP [A1, A2]	AP p7–9 endpoints; the split is my model and a convention	**Y-with-rounding**
34	UAT entry gate: 80% of shipments minor-to-no issues	S11-B	PTA	PTA p3	**Y**
35	100-container test; seven hops (created, subscribed, data back, route, milestone, exception, webhook)	S11-B	PTA; C100	PTA p1–2; C100 columns "Shipment created" to "Webhook Created"	**Y**
36	Ledger: 474 rows, 142 tested, 105 tested rows with hop results, 59 failing **Say: Say "my recount of a partial working log — it tells you where to look, not a defect rate."**	S11-B	C100	C100 rows 1–474 (106 rows have any hop result; one is untested)	**Y-with-rounding**
37	First failing hop: route 23 + milestone 18 = 41 of 59 (69.5%); ingestion 12 (20.3%)	S11-B	C100	C100 columns "Route created?", "Milestone shown in UI?"	**Y-with-rounding**
38	Scraper error rates: Air 5.09%, Ocean 8.99% against a < 10% goal	S11-B	TPM	TPM p11	**Y**
39	Trace-ID / MTTR improvement of 50%	S11-B	P3PL	P3PL p1	**Y**
40	Pilot later than the first testing plan (early November 2021 → December 2021) **Say: Say "later than the first testing plan" — never "on time", never a number of weeks.**	S11-B	PTA; TPM; PCRS	PTA p1; PCRS p1 (Wave 1 in Q4 2021); year-end summary — the documents disagree on "late"	**CONFLICT**
41	Localization-labor dollar **Say: Say the hours, not this dollar.**	S11-B	my model	no source	**UNSUPPORTED-anchor**
42	79 customers · 294,581 containers · $185,892 printed ARR	S11-C	RAPID; AP [A4]	RAPID p5 (total row); AP p12–13	**Y**
43	Printed total equals six of nine segment rows; nine rows = $21,246 MRR / $254,952 ARR **Say: Say the printed 79 / $185,892 / 294,581 and add "the printed total equals six of its nine rows" — lead with containers.**	S11-C	RAPID	RAPID p2–5 (segment rows; the footing is my arithmetic)	**CONFLICT**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
44	77 names in the appendix vs 79 counted (37 SMB vs 39) **Say: Say "the appendix lists 77 names against 79 counted — I would reconcile before the cutover list."**	S11-C	RAPID	RAPID p12 (appendix; my count)	**CONFLICT**
45	Three brokers = 75.0% of containers; 3 of 79 customers = 3.8%	S11-C	RAPID	RAPID p3–5; the concentration is my arithmetic on the table	**Y-with-rounding**
46	25% schedule reliability against a competitor's 63% — unlike data	S11-C	AP [A4]	AP p12	**Y**
47	Decider is an executive (RAPID: Recommend, Agree, Perform, Input, Decide)	S11-C	RAPID	RAPID p1	**Y**
48	18 months → value in month 6; 66% less effort	S12	AP [B2]	AP p17–19	**Y**
49	Cost-to-serve −25% operator-first vs −50% partner-first; 9 vs 6 months (work-item grain)	S12	AP [B2]	AP p17–19	**Y**
50	339.5-minute baseline; ~2,200 operator interactions per departed shipment; ≈ $153 per shipment	S12	AP [B2]	AP p18–19 (339.5 × $0.45 = $152.78)	**Y-with-rounding**
51	About $5M avoided — said once; never annual; no "15% lower overhead" **Say: Say "about five million of avoided cost," once.**	S12	AP [B2]; GF8	AP p18–19 carries both "avoidance" and "annual cost reduction"	**CONFLICT**
52	$2.85M effort + $1.52M run-rate ≈ $4.4M, 12% under the ~$5M record **Say: Say "my bottom-up model lands within about twelve percent of the record."**	S12	RECON S12.13–14	my bottom-up model	**UNSUPPORTED-anchor**
53	Import-side app live April 2022 with 11 partners in 10 countries **Say: Say "the import-side app went live April 2022 with eleven partners in ten countries"; the export launch is the later Q1 2023 milestone.**	S12 · S12-A	ASIA; AP [B1]; HLSD	ASIA p1; AP p16 (Apple also prints "MVP Q1'23" at p17; HLSD p4: export launch March 2023)	**CONFLICT**
54	48 of 53 import partners migrated within a month (about 90%) **Say: Say "about ninety percent" and always "import partners."**	S12 · S12-A	ASIA; AP [B1]	ASIA p1 ("90.5%"); AP p17	**Y-with-rounding**
55	Burn-up 55.8% → 83.2%; 15 domains × 6 modules; 8-dev-week reuse rule	S12	AP [B1, B2]	AP p16–19	**Y**
56	P95 page load: Asia 21.19 s vs US 3.3 s = 6.42×; Others 4.45 s	S12-A	ASIA	ASIA p1 (table, 04/24–05/10)	**Y**
57	Page views 2.19k / 6.31k / 5.42k = 13.92k; Asia 45.3%	S12-A	ASIA	ASIA p1; the sum and share are my arithmetic	**Y-with-rounding**
58	17-day window (04/24 – 05/10)	S12-A	ASIA	ASIA p1	**Y**
59	Static JS P95: China 5 s vs US 0.5 s	S12-A	ASIA	ASIA p2	**Y**
60	P99 rests on 63 Asia views (1% of 6,310), P95 on 316	S12-A	ASIA	ASIA p1 views; the percentile counts are my arithmetic (the document gives no P99)	**Y-with-rounding**
61	Plan: 8 rows, 4 done, 52 days (24 Jun → 15 Aug 2022); remove double CDN 2022/06/24	S12-A	ASIA	ASIA p3	**Y**
62	32 smoke checks per change (4 steps × 2 instances × before / after, 2 environments)	S12-A	ASIA	ASIA p4 (the workflow); the count is mine	**Y-with-rounding**
63	Overall monthly P95 8.39 s → 4.68 s = −44.2% (Apr → Sep)	S12-A	LSA	sheet "OE Page 0", Excel row 50, columns 26 and 31	**Y-with-rounding**
64	Asia 4.0 s vs North America 4.09 s in the last week of the year **Say: Say "the gap closed" — never subtract the 21.19 s from the 4.0 s.**	S12-A	LSA	sheet "OE Page 0", Excel rows 53 (Asia) and 51 (NA); the Asia (non-China) series is not the same population as the 21.19 s	**CONFLICT**
65	Week-47 spike: metrics-code error (12/12 callout) plus the October regression (action item 11)	S12-A	LSA	sheet "OE Page 0", Excel row 94; sheet "Action Items", item #11 (Excel row 12)	**Y**
66	P50 / P90 / P95 by region added to the review (11 Oct 2022)	S12-A	LSA	sheet "Action Items", item #1 (Excel row 2), target 2022-10-11	**Y**
67	Exposure ≈ $0.69M a year; tail time ≈ $1.6K a year **Say: Say "illustrative exposure, not a saving."**	S12-A	my model	illustrative / modeled; no source figure	**UNSUPPORTED-anchor**
68	Zendesk tickets joined to Crux container numbers and Flexport core data; Looker SQL Runner views	S16-A	BRD	BRD p1–2, p5	**Y**
69	Seven metrics frozen	S16-A	BRD; AP [A3]	BRD p2–3; AP p10	**Y**

#	NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW / SHEET	MATCH
70	BR-3 feed daily or preferably hourly; BR-4 alert when the feed fails	S16-A	BRD	BRD p4	**Y**
71	As-is snapshot 1.5–2 hours	S16-A	BRD	BRD p2	**Y**
72	BRD pricing: 3 × 5 h × 2 reviews at an assumed $90 → $129,600 → $25,920 → $103,680 (30 → 6 h/wk) **Say: Say "over $100K a year — the BRD priced it at $103,680 on thirty hours a week at an assumed ninety dollars an hour; my split is a model; priced, not a claimed saving."**	S16-A	BRD; RECON S16.01	BRD p1–2	**CONFLICT**
73	BRD monthly figure $10,880 vs $129,600 / 12 = $10,800 **Say: If you quote a monthly figure, say $10,800.**	S16-A	BRD; RECON S16.02	BRD p2	**CONFLICT**
74	Model split $30,160 (4 h × 52 × $145) + $70,200 (52 × 10% × $13,500) = $100,360 **Say: Say "about a hundred thousand — my model" and never the split as record.**	S16-A	RECON S16.11	my model; rates and probabilities illustrative	**UNSUPPORTED-anchor**
75	Realized candidate 208 h × $90 = $18,720 **Say: Say "about two hundred hours retired — modeled."**	S16-A	my model	modeled; not claimed	**UNSUPPORTED-anchor**
76	System accuracy 60% → 90% **Say: Say "system accuracy" — never ETA accuracy.**	S16-A	AP [A3, A4]; WBRAI	AP p10–11, p13; WBRAI p2	**Y**
77	Four systems → one pipeline	S16-A	AP [A3]; WBRAI	AP p10; WBRAI p1	**Y**
78	Parity gate: manual vs pipeline within 1% for consecutive weeks; mismatches mostly the manual side	S16-A	AP [A3]	AP p10–11	**Y**
79	First automated catch: ticket-sync drop of about 4% **Say: Say "about four percent."**	S16-A	AP [A3]	AP p10–11	**Y-with-rounding**
80	Data-access security review: two weeks, zero skipped reviews	S16-A	AP [A3, G3]	AP p42, p45–46	**Y**
81	"Bus factor 1 → ≥ 2" **Say: Say "no longer a single point of failure."**	S16-A (removed)	AP [A3]; WBRAI	AP: single point of failure (qualitative); WBRAI p2	**UNSUPPORTED-anchor**
82	Illustrative parity values 0.42% / 1.25%; unmatched 0.74%; alarm 1%; join fan-out 12 / 18 vs 12 / 10 (44.4%) **Say: Say "illustrative — my design example, not program data."**	S16-A	my examples	illustrative — labeled in every table	**UNSUPPORTED-anchor**

Tally. 82 claims: 41 Y · 17 Y-with-rounding · 12 CONFLICT (safe wording given) · 12 UNSUPPORTED-anchor (labeled).